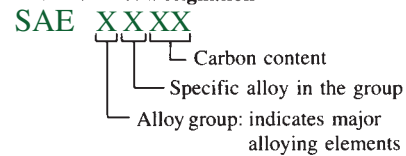


Furthermore, SAE, in cooperation with ASTM, is promoting an expanded system for designating wrought steels and the new system is placed within the Unified Numbering System (UNS). Refer back to Table 2–5. See SAE Standard J402 (R) *New Steel Designation System for Wrought or Rolled Steel*. Three UNS series for steels, G, H, and K, are being changed to conform to the new system that provides for specified modifications of the chemistry of the alloys and may also define special processing methods or limits on material properties.

The system starts with the familiar four-digit designation system that has been used for several decades under both the SAE and AISI systems, as shown in Figure 2–12. The first two digits indicate the specific alloy group that identifies the primary alloying elements other than carbon in the steel as shown in Table 2–8. The last two digits indicate the amount of carbon in the steel in hundredths of a percent. For example, when the last two digits are 20,

General Form of Designation



Examples

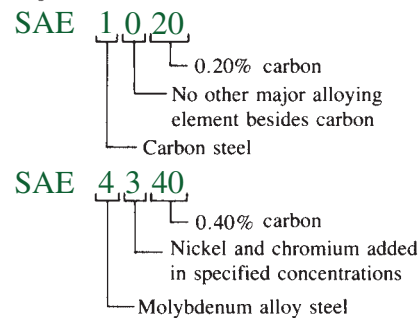


FIGURE 2–12 Steel designation system

TABLE 2–8 Alloy Groups in the SAE Numbering System

10xx	Plain carbon steel: No significant alloying element except carbon and manganese; less than 1.0% manganese. Also called <i>nonresulfurized</i> .
11xx	Free-cutting steel: Resulfurized. Sulfur content (typically 0.10%) improves machinability.
12xx	Free-cutting steel: Resulfurized and rephosphorized. Presence of increased sulfur and phosphorus improves machinability and surface finish.
12Lxx	Free-cutting steel: Lead added to 12xx steel further improves machinability.
13xx	Manganese steel: Nonresulfurized. Presence of approximately 1.75% manganese increases hardenability.
15xx	Carbon steel: Nonresulfurized; greater than 1.0% manganese.
23xx	Nickel steel: Nominally 3.5% nickel.
25xx	Nickel steel: Nominally 5.0% nickel.
31xx	Nickel-chromium steel: Nominally 1.25% Ni; 0.65% Cr.
33xx	Nickel-chromium steel: Nominally 3.5% Ni; 1.5% Cr.
40xx	Molybdenum steel: 0.25% Mo.
41xx	Chromium-molybdenum steel: 0.95% Cr; 0.2% Mo.
43xx	Nickel-chromium-molybdenum steel: 1.8% Ni; 0.5% or 0.8% Cr; 0.25% Mo.
44xx	Molybdenum steel: 0.5% Mo.
46xx	Nickel-molybdenum steel: 1.8% Ni; 0.25% Mo.
48xx	Nickel-molybdenum steel: 3.5% Ni; 0.25% Mo.
5xxx	Chromium steel: 0.4% Cr.
51xx	Chromium steel: Nominally 0.8% Cr.
51100	Chromium steel: Nominally 1.0% Cr; bearing steel, 1.0% C.
52100	Chromium steel: Nominally 1.45% Cr; bearing steel, 1.0% C.
61xx	Chromium-vanadium steel: 0.50–1.10% Cr; 0.15% V.
86xx	Nickel-chromium-molybdenum steel: 0.55% Ni; 0.5% Cr; 0.20% Mo.
87xx	Nickel-chromium-molybdenum steel: 0.55% Ni; 0.5% Cr; 0.25% Mo.
92xx	Silicon steel: 2.0% silicon.
93xx	Nickel-chromium-molybdenum steel: 3.25% Ni; 1.2% Cr; 0.12% Mo.

the alloy includes approximately 0.20% carbon, and it is often said that the steel has *20 points of carbon*. Small variations are allowed, say from 0.18% to 0.23%.

Carbon content is part of the designation system because the primary mechanical properties of strength, hardness, and ductility are strongly dependent on it. As carbon content increases, strength and hardness also increase during processing and heat treatment. Ductility typically decreases so designers must balance strength and ductility when specifying a steel and its condition. It is useful to categorize steels into four broad classes, *low carbon*, *medium carbon*, *high carbon*, and *bearing steels*.

- *Low carbon*: Less than 30 points of carbon. Relatively low strength but good formability. Used where high strength is not required.
- *Medium carbon*: From 30 to 50 points of carbon. Higher strength and hardness with moderate ductility. A large percentage of machine elements are made from steels in this class.
- *High carbon*: 50 to 95 points of carbon. Even higher strength and hardness but often with lower ductility. The high hardness provides good wear properties leading to uses such as knives requiring a durable cutting edge, parts subject to abrasion, tools, chisels, and agricultural implements.
- *Bearing steels*: typically 100 points of carbon or higher ($\geq 1.0\%$). Often specified for balls, rollers, and races of rolling contact bearings because of the very high contact stresses developed in operation.

In this book, when standard specifications for a steel alloy are intended, without modification, we will use the traditional designation as shown in Figure 2–12: for example, SAE 1020 or SAE 4340 steel. See also Appendix 3. Designers, manufacturing professionals, and engineers sometimes refer to only the number in conversation, such as 1020 steel and 4340 steel, or describe them colloquially as *plain carbon steel* and *chrome-moly steel*. Of course, when specifying a steel on design drawings or purchase orders, the complete, formal designations should be used.

The New UNS Designations for G, H, and K Steels. SAE Standard J402 defines a 12-character designation using the format:

GXXXX-XXX-XXXX

where,

- The lead letter may be G for carbon and alloy steels, H for steels with specific hardenability, or K for miscellaneous steels and ferrous alloys as previously listed in the UNS.
- The first set of four Xs are the traditional four-digit SAE designation described earlier.
- The second set of three Xs denote modifications to the standard chemistry.

- The final set of four Xs refer to special production methods, quality standards, or processing.

Details of the chemistry and other requirements of various alloys are contained in other SAE standards, including:

- J403 *Chemical Compositions of SAE Carbon Steels*; for carbon steels in the 10XX, 11XX, 12XX, and 15XX series (see Table 2–8).
- J404 *Chemical Compositions of SAE Alloy Steels*; for alloy steels in the remaining series.
- J1086 *Numbering Metals and Alloys*.
- J1249 *Former SAE Standard and Former SAE-Ex Steels*.
- J1268 *Hardenability Bands for Carbon and Alloy H Steels*.
- J1868 *Restricted Hardenability Bands for Selected Alloy Steels*.

The place of steels in the overall UNS is documented in the standard jointly published by SAE and ASTM: SAE HS-1086/ASTM DS 56H *Metals and Alloys in the Unified Numbering System*. See Internet site 4 and 5 for additional information and to acquire these standards.

The new designation system permits steel manufacturers to modify the standard chemistry for commonly used steels to meet specific customer needs without creating an entirely new four-digit designation number. Performance requirements for press-forming, roll-forming, abrasion resistance, environmental corrosion resistance, cleanliness of the steel, and similar special needs can be assured more easily and reliably with such a system. These situations often arise in production operations in the automotive, aerospace, appliance, construction equipment, agricultural equipment, manufacturing equipment, energy production equipment, and similar industries.

A *bearing steel* nominally contains 1.0% carbon. Common grades are 50100, 51100, and 52100; the usual four-digit designation is replaced by five digits, indicating 100 points of carbon.

Alloy Groups

As indicated in Table 2–8, sulfur, phosphorus, and lead improve the machinability of steels and are added in significant amounts to the 11xx, 12xx, and 12Lxx grades. These grades are used for screw machine parts requiring high production rates where the resulting parts are not subjected to high stresses or wear conditions. In the other alloys, these elements are controlled to a very low level because of their adverse effects, such as increased brittleness.

Nickel improves the toughness, hardenability, and corrosion resistance of steel and is included in most of the alloy steels. Chromium improves hardenability, wear and abrasion resistance, and strength at elevated temperatures. In high concentrations, chromium provides significant corrosion resistance, as discussed in the section on

TABLE 2-9 Uses of Some Steels

SAE number	Applications
1015	Formed sheet-metal parts; machined parts (may be carburized)
1030	General-purpose, bar-shaped parts, levers, links, keys
1040	Shafts, gears
1080	Springs; agricultural equipment parts subjected to abrasion (rake teeth, disks, plowshares, mower teeth)
1112	Screw machine parts
4140	Gears, shafts, forgings
4340	Gears, shafts, parts requiring good through-hardening
4640	Gears, shafts, cams
5150	Heavy-duty shafts, springs, gears
6150	Gears, forgings, shafts, springs
8650	Gears, shafts
9260	Springs

stainless steels. Molybdenum also improves hardenability and high-temperature strength.

The steel selected for a particular application must be economical and must provide optimum properties of strength, ductility, toughness, machinability, and formability. Frequently, metallurgists, manufacturing engineers, and heat-treatment specialists are consulted. (See also References 3, 8, and 25.) See also Section 2-18.

Table 2-9 lists some common steels used for machine parts, with typical applications listed for the alloys. You should benefit from the decisions of experienced designers when specifying materials.

AHSS—Advanced High-Strength Steels

Significant developments are being made in the field of steels used in automotive and other transportation-related fields that depend on high strength with good ductility. Many emerging products are called *advanced high-strength steels* (AHSS). Goals for their development include safety enhancement, cost reduction, durability, reduced environmental impact, and increased fuel economy for vehicles. AHSS steels are characterized by unique microstructural phases other than ferrite and pearlite, including martensite, austenite, and/or retained austenite in quantities sufficient to produce unique mechanical properties with high strength, enhanced formability, and additional stretchability as compared with more traditional high strength-low alloy (HSLA) steels. See Reference 37 and Internet site 40. Types of AHSS steels include:

- Dual phase (DP).
- Transformation induced plasticity (TRIP).
- Complex phase (CP).
- Martensitic (MS).

Examples of grades are (numbers give yield strength/ultimate tensile strength in MPa):

DP 300/500 (30–34% el.)	DP 350/600 (24–30% el.)
TRIP 450/800 (26–32% el.)	DP 500/800 (14–20% el.)
CP 700/800 (10–15% el.)	DP 700/1000 (12–17% el.)
MS 950/1200 (5–7% el.)	MS 1250/1520 (4–6% el.)
HSLA 350/450 (23–27% el.) ¹	

¹Listed for comparison.

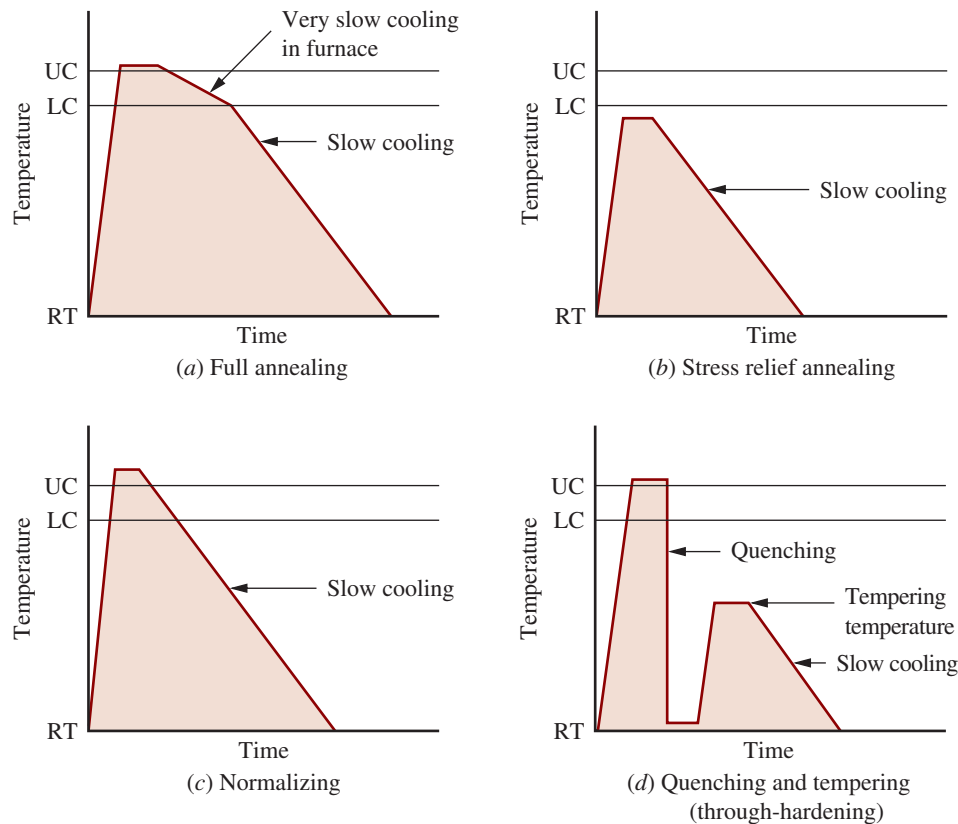
2-6 CONDITIONS FOR STEELS AND HEAT TREATMENT

The final properties of steels are dramatically affected by the way the steels are produced. Some processes involve mechanical working, such as rolling to a particular shape or drawing through dies. In machine design, many bar-shaped parts, shafts, wire, and structural members are produced in these ways. But most machine parts, particularly those carrying heavy loads, are heat-treated to produce high strength with acceptable toughness and ductility.

Carbon steel bar and sheet forms are usually delivered in the *as-rolled condition*; that is, they are rolled at an elevated temperature that eases the rolling process. The rolling can also be done cold to improve strength and surface finish. Cold-drawn bar and wire have the highest strength of the worked forms, along with a very good surface finish. However, when a material is designated to be *as-rolled*, it should be assumed that it was hot-rolled.

Heat Treating

Heat treating is any process in which steel is subjected to elevated temperatures to modify its properties. Of the several processes available, those most used for machine steels are annealing, normalizing, through-hardening



Note:

RT = room temperature

LC = lower critical temperature

UC = upper critical temperature

FIGURE 2-13 Heat treatments for steel

(quench and temper), and case hardening. (See References 6 and 16–18.)

Figure 2-13 shows the temperature–time cycles for these heat-treatment processes. The symbol RT indicates normal room temperature, and LC refers to the lower critical temperature at which the transformation of ferrite to austenite begins during the heating of the steel. At the upper critical temperature (UC), the transformation is complete. These temperatures vary with the composition of the steel. For most medium-carbon (0.30–0.50% carbon) steels, UC is approximately 1500°F (822°C). References giving detailed heat-treatment process data should be consulted.

Annealing. *Full annealing* [Figure 2-13(a)] is performed by heating the steel above the upper critical temperature and holding it until the composition is uniform. Then the steel is cooled very slowly in the furnace to below the lower critical temperature. Slow cooling to room temperature outside the furnace completes the process. This treatment produces a soft, low-strength form of the material, free of significant internal stresses. Parts are frequently cold-formed or machined in the annealed condition.

Stress relief annealing [Figure 2-13(b)] is often used following welding, machining, or cold-forming to relieve

residual stresses and thereby minimize subsequent distortion. The steel is heated to approximately 1000°F to 1200°F (540°C–650°C), held to achieve uniformity, and then slowly cooled in still air to room temperature.

Normalizing. *Normalizing* [Figure 2-13(c)] is performed in a similar manner to annealing, but at a higher temperature above the transformation range where austenite is formed, approximately 1600°F (870°C). The result is a uniform internal structure in the steel and somewhat higher strength than annealing produces. Machinability and toughness are usually improved over the as-rolled condition.

Through-Hardening and Quenching and Tempering.

Through-hardening [Figure 2-13(d)] is accomplished by heating the steel to above the transformation range where austenite forms and then rapidly cooling it in a *quenching* medium. The rapid cooling causes the formation of martensite, the hard, strong form of steel. The degree to which martensite forms depends on the alloy's composition. An alloy containing a minimum of 80% of its structure in the martensite form over the entire cross section has *high hardenability*. This is an important property to look for when selecting a steel requiring high strength and hardness. The common quenching liquid media are

water, brine, and special mineral oils. Formulations of water-soluble polyalkylene glycol liquid organic polymer (PAG) with corrosion-resisting additives often replace other quenchants to provide tailored control of the rate of quenching to reduce distortion and residual stresses. They can be used either in immersion systems or applied as a spray for either steels or aluminum alloys. One brand name is UCON™ (a trademark of The Dow Chemical Company) that offers ten different formulations.

Air or other gases may also be used for quenching of some materials. Inert gases such as helium, nitrogen, hydrogen, and argon can be used in an enclosed tank, typically under low pressure (vacuum), to inhibit oxidation. Careful specification of the quenchants is critical to ensure compatibility with the particular material being heat-treated. The selection of a quenching medium depends on the rate at which cooling should proceed. Most machine steels use either oil or water quenching.

Tempering is usually performed immediately after quenching and involves reheating the steel to a temperature of 400°F to 1300°F (200°C–700°C) and then slowly cooling it in air back to room temperature. This process modifies the steel's properties: Tensile strength and yield strength decrease with increasing tempering temperature, whereas ductility improves, as indicated by an increase in the percent elongation. Thus, the designer can tailor the properties of the steel to meet specific requirements. Furthermore, the steel in its as-quenched condition has high internal stresses and is usually quite brittle. Machine parts should normally be tempered at 700°F (370°C) or higher after quenching.

To illustrate the effects of tempering on the properties of steels, several charts in Appendix 4 show graphs of strength versus tempering temperature. Included in these charts that show graphs of strength versus tempering temperature are tensile strength, yield point, percent elongation, percent reduction of area, and hardness number HB, all plotted in relation to tempering temperature. Note the difference in the shape of the curves and the absolute values of the strength and hardness when comparing the plain carbon SAE 1040 steel with the alloy steel SAE 4340. Although both have the same nominal carbon content, the alloy steel reaches a much higher strength and hardness. Note also the as-quenched hardness in the upper right part of the heading of the charts; it indicates the degree to which a given alloy can be hardened. When the case-hardening processes (described next) are used, the as-quenched hardness becomes very important.

Appendix 3 lists the range of properties that can be expected for several grades of carbon and alloy steels. The alloys are listed with their SAE numbers and conditions. For the heat-treated conditions, the designation reads, for example, SAE 4340 OQT 1000, which indicates that the alloy was oil-quenched and tempered at 1000°F. Expressing the properties at the 400°F and 1300°F tempering temperatures indicates the end-points of the possible range of properties that can be expected for that alloy. To specify a strength between these limits, you could refer to graphs such as those shown in Appendix 4, or you could determine the required

heat-treatment process from a specialist. For the purposes of material specification in this book, a rough interpolation between given values will be satisfactory. As noted before, you should seek more specific data for critical designs.

Case Hardening. In many cases, the bulk of the part requires only moderate strength although the surface must have a very high hardness. In gear teeth, for example, high surface hardness is necessary to resist wear as the mating teeth come into contact several million times during the expected life of the gears. At each contact, a high stress develops at the surface of the teeth. For applications such as this, *case hardening* is used; the surface (or *case*) of the part is given a high hardness to a depth of perhaps 0.010 to 0.040 in (0.25–1.00 mm), although the interior of the part (the *core*) is affected only slightly, if at all. The advantage of surface hardening is that as the surface receives the required wear-resisting hardness, the core of the part remains in a more ductile form, resistant to impact and fatigue. The processes used most often for case hardening are flame hardening, induction hardening, carburizing, nitriding, cyaniding, and carbo-nitriding. (See References 16–18.)

Figure 2–14 shows a drawing of a typical case-hardened gear-tooth section, clearly showing the hard case surrounding the softer, more ductile core. Case hardening is used in applications requiring high wear and abrasion resistance in normal service (gear teeth, crane wheels, wire-rope sheaves, and heavy-duty shafts).

The most commonly used processes for case hardening are described in the following list.

1. **Flame hardening and induction hardening:** The processes of flame hardening and induction hardening involve the rapid heating of the surface of the part for a limited time so that a small, controlled depth of the material reaches the transformation range. Upon immediate quenching, only that part above the transformation range produces the high level of martensite required for high hardness.

Flame hardening uses a concentrated flame impinging on a localized area for a controlled amount of time to heat the part, followed by quenching in a bath or by a stream of water or oil. *Induction hardening* is a process in which the part is surrounded by a coil through which high-frequency electric current is passed. Because of the electrical conductivity of the steel, current is *induced* primarily near the surface of the part. The resistance of the material to the flow of current results in a heating effect.

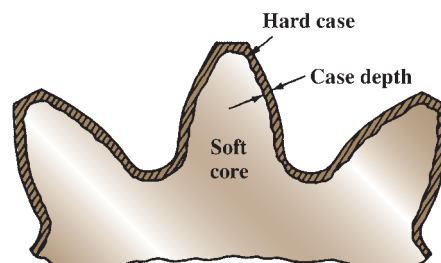


FIGURE 2–14 Typical case-hardened gear-tooth section

Controlling the electrical power and the frequency of the induction system, and the time of exposure, determines the depth to which the material reaches the transformation temperature. Rapid quenching after heating hardens the surface. (See Reference 17.)

The design of the devices for heating the part is critical, especially when localized hardening is desired, as for gear teeth and selective hardening of an area subjected to abrasive wear. For gears of relatively small overall size and small teeth, the impinging flame or the induction heating coil may encircle the entire gear, allowing all teeth to be hardened at the same time. For larger gears with larger teeth, this may be impractical and heating may be tooth-by-tooth or over a small segment of the gear. A system for immediate quenching after heating must be provided. The most critical area for hardening of gear teeth is typically the flanks where the two meshing teeth make contact and transmit the driving forces. High contact stresses where the two curved surfaces meet must be resisted. Pitting resistance of the teeth, as discussed in Chapter 9, is directly related to the hardness of the flanks and that is the most prominent failure mode for gears that are to be used for many thousands of hours of expected life. The area where the root of the tooth blends with the involute tooth form of the flank experiences the highest bending stress. For gears that are used intermittently, bending fatigue is often the predominant mode of failure. Ensuring that the fillet area is adequately hardened will optimize the fatigue life.

Note that for flame or induction hardening to be effective, the material must have a good hardenability. Usually the goal of case hardening is to produce a case hardness in the range of Rockwell C hardness HRC 55 to 60 (Brinell hardness approximately HB 550 to 650). Therefore, the material must be capable of being hardened to the desired level. Carbon and alloy steels with fewer than 30 points of carbon typically cannot meet this requirement. Thus, the alloy steels with 40 points or more of carbon are the usual types given flame- or induction-hardening treatments.

Typical steel materials specified are SAE 1045, 1552, 4140, 4150, 4340, and 5150.

2. *Carburizing, nitriding, cyaniding, and carbo-nitriding:*

The remaining case-hardening processes—carburizing, nitriding, cyaniding, and carbo-nitriding—actually alter the composition of the surface of the material by exposing it to carbon-bearing gases, liquids, or solids at high temperatures that produce carbon and diffuse it into the surface of the part. The concentration and the depth of penetration of carbon depend on the nature of the carbon-bearing material and the time of exposure. Nitriding and cyaniding typically result in very hard, thin cases that are good for general wear resistance. Where high load-carrying capability in addition to wear resistance is required, as with gear teeth, carburizing is preferred because of the thicker case.

Several steels are produced as carburizing grades. Among these are 1015, 1020, 1022, 1117, 1118, 4118, 4320, 4620, 4820, and 8620. Appendix 5 lists the expected properties of these carburized steels. Note when evaluating a material for use that the core properties determine its ability to withstand prevailing stresses, and the case hardness indicates its wear resistance. Carburizing, properly done, will virtually always produce a case hardness from HRC 55 to 64 (Rockwell C hardness) or from HB 550 to 700 (Brinell hardness).

Carburizing has several variations that allow the designer to tailor the properties to meet specific requirements. The exposure to the carbon atmosphere takes place at a temperature of approximately 1700°F (920°C) and usually takes eight hours. Immediate quenching achieves the highest strength, although the case is somewhat brittle. Normally, a part is allowed to cool slowly after carburizing. It is then reheated to approximately 1500°F (815°C) and then quenched. A tempering at the relatively low temperature of either 300°F or 450°F (150°C or 230°C) follows, to relieve stresses induced by quenching. As shown in Appendix 5, the higher tempering temperature lowers the core strength and the case hardness by a small amount, but in general it improves the part's toughness. The process just described is *single quenching and tempering*.

When a part is quenched in oil and tempered at 450°F, for example, the condition is *case hardening by carburizing, SOQT 450*. Reheating after the first quench and quenching again further refines the case and core properties; this process is *case hardening by carburizing, DOQT 450*. These conditions are listed in Appendix 5.

Cautions for Heat-Treating Operations. Several methods of performing heat-treating operations exist and the choice can affect the final quality and performance of the finished part. Consultation with heat-treating specialists is recommended and careful specifications should be agreed upon. Issues that arise include:

- Final range of acceptable hardness
- Distribution of hardness over critical surfaces of the parts
- Case depth from case-hardening processes
- Microstructure of heat-treated zones in either the case or the core of the part
- Distortion of the part as a result of heat-treating
- Residual stresses developed during heat-treating, particularly residual tensile stresses that reduce fatigue life of the part
- Cracking in critical areas that may affect fatigue life
- Appearance of the part after heat-treating

A wide range of equipment is used for heat-treating; a few are described briefly here. All include heating in some kind of furnace or by flame or induction heating at temperatures up to 1250°C (2300°F), controlled for

the individual types of metals being treated. Heating by immersion in molten salts is also sometimes used. Then, parts are quenched by any of several means:

1. Transfer to liquid quenching tanks manually, by gravity dropping, or by conveyor or robotic handling. The quenching liquid may be at room temperature or elevated and it may be still or agitated.
2. Air quenching either in ambient air or with high-velocity blowers or enclosed chambers
3. Impinging of the quenchant on the part by flow from a nozzle
4. High-temperature vacuum carburizing in enclosed tanks with high-pressure gas quenching using inert gases such as helium, argon, hydrogen, or neon.

Careful support of the part during heating and quenching is often needed to reduce distortion. Special racks, shelving, and material-handling equipment may be needed.

Shot Peening for Favorable Residual Stresses. As mentioned earlier in the discussion of machining, grinding, forming, and heat-treating steels, unfavorable tensile residual stresses are often created that exacerbate problems of crack formation where high applied tensile stress is also present in operation. The combination hastens the onset of fatigue failure. *Shot peening* is a secondary process that can mitigate this problem by producing favorable residual *compressive stresses* near the surface of a part.

Shot peening is a process in which fine steel or cast iron shot is projected at high velocity on critical surfaces of a

part. The multiple impacts plastically deform the surface and after completing the process, residual compressive stresses remain. Thus, when operating under conditions where applied tensile stresses are produced, the net result is a lower tensile stress which would reduce the likelihood of initiating fatigue cracks. Cases of increasing fatigue life by 100% by shot peening have been reported. Many variables are involved and testing is recommended to quantify the benefits in any given situation. Products for which shot peening have been used include gears, shafts, helical compression and tension springs, leaf springs, and turbine blades.

Steel Alloys for Castings. Often, the composition of steel alloys for casting is similar to some already discussed here for wrought forms of steel produced primarily by rolling processes. However, because casting results in different internal structures from those of rolled forms, special designations are made by ASTM International. General classes, in increasing level of performance, are *carbon*, *low alloy*, *high alloy*, and *special alloy*. Material selection must consider operational conditions for a given application, particularly:

- Structural strength in tension, compression, and shear
- Toughness required to resist impact loading
- High temperature operation
- Need for wear resistance
- Need for pressure containment

Table 2–10 gives some of the pertinent standards for cast steel alloys. You should refer to the indicated ASTM

TABLE 2–10 Cast Carbon and Alloy Steels

ASTM designation	Description
General applications	
A27/A27M-13	Carbon steel; Heat-treated to a range of tensile strengths Grades: 60–30 (415–205), 65–35 (450–240), 70–36 (485–250), and 70–40 (485–275) Each set of numbers is tensile strength–yield strength in ksi (MPa)
A915/A915M-08(2013)	Carbon and alloy steels; similar composition to standard wrought steels Grades: SC1020, SC1025, SC1040, SC1045, SC4130, SC4140, SC4330, SC4340, SC8620, SC8625, SC8630 Grade numbers match similarly named steel in wrought form
A128/A128M-93(2007)	Hadfield manganese steel castings
A148/A148M-15A	High strength carbon, alloy, and martensitic stainless steels for castings
Pressure-containing parts	
A757/A757M-15 and A352/352M-06(2012)	Carbon and alloy, ferritic and martensitic, for low temperature
A351/A351M-15	General pressure-containing service Some grades are suitable for high temperatures or corrosive environments
A216/A216M-14e1	Carbon steel for high temperature; weldable
A389/A389M-13	Alloy steel for high temperatures

specification for details of strength limits and other performance factors for the indicated steel alloys. Note that the compositions of steels in ASTM A915/915M-08(2013) are designed to be similar to commonly used wrought steels designated in the SAE system. However, properties may be different because of the production processes.

2-7 STAINLESS STEELS

The term *stainless steel* characterizes the high level of corrosion resistance offered by alloys in this group. To be classified as a stainless steel, the alloy must have a chromium content of at least 10%. Most have 12% to 18% chromium. (See Reference 9.)

The SAE designates most stainless steels by its 200, 300, and 400 series.

The three main groups of stainless steels are austenitic, ferritic, and martensitic. *Austenitic* stainless steels fall into the SAE 200 and 300 series. They are general-purpose grades with moderate strength. Most are not heat-treatable, and their final properties are determined by the amount of working, with the resulting temper referred to as 1/4 hard, 1/2 hard, 3/4 hard, and full hard. These alloys are nonmagnetic and are typically used in food processing equipment.

SAE 304 and 316 are most often used in commercially available bars, hexagons, squares rectangles, and tubing as listed in Table 2-6.

Ferritic stainless steels belong to the SAE 400 series, designated as 405, 409, 430, 446, and so on. They are magnetic and perform well at elevated temperatures, from 1300°F to 1900°F (700°C–1040°C), depending on the alloy. They are not heat-treatable, but they can be cold-worked to improve properties. Typical applications include heat exchanger tubing, petroleum refining equipment, automotive trim, furnace parts, and chemical equipment.

Martensitic stainless steels are also members of the SAE 400 series, including 403, 410, 414, 416, 420, 431, and 440 types. They are magnetic, can be heat-treated, and have higher strength than the 200 and 300 series, while retaining good toughness. Typical uses include turbine engine parts, cutlery, scissors, pump parts, valve parts, surgical instruments, aircraft fittings, and marine hardware.

There are many other grades of stainless steels, many of which are proprietary to particular manufacturers. A group used for high-strength applications in aerospace, marine, and vehicular applications is of the precipitation-hardening type. They develop very high strengths with heat treatments at relatively low temperatures, from 900°F to 1150°F (480°C–620°C). This characteristic helps to minimize distortion during treatment. Some examples are 17-4PH, 15-5PH, 17-7PH, PH15-7Mo, and AMS362 stainless steels.

2-8 STRUCTURAL STEEL

Most structural steels are designated by ASTM numbers established by the American Society for Testing and Materials. One common grade is ASTM A36, which has

a minimum yield point of 36 000 psi (248 MPa) and is very ductile. It is basically a low-carbon, hot-rolled steel available in sheet, plate, bar, and structural shapes such as some wide-flange beams, American Standard beams, channels, and angles. The geometric properties of some of each of these sections are listed in Appendix 15.

Most wide-flange beams (W-shapes) are currently made using ASTM A992 structural steel, which has a yield point of 50 to 65 ksi (345 to 448 MPa) and a minimum tensile strength of 65 ksi (448 MPa). An additional requirement is that the maximum ratio of the yield point to the tensile strength is 0.85. This is a highly ductile steel, having a minimum of 21% elongation in a 2.00-inch gage length. Using this steel instead of the lower strength ASTM A36 steel typically allows smaller, lighter structural members at little or no additional cost.

Hollow structural sections (HSS) are typically made from ASTM A500 steel that is cold-formed and either welded or made seamless. Included are round tubes and square and rectangular shapes. Note in Appendix 7 that there are different strength values for round tubes as compared with the shaped forms. Also, several strength grades can be specified. Some of these HSS products are made from ASTM A501 hot-formed steel having properties similar to the ASTM A36 hot-rolled steel shapes.

Many higher-strength grades of structural steel are available for use in construction, vehicular, and machine applications. They provide yield points in the range from 42 000 to 100 000 psi (290–700 MPa). Some of these grades, referred to as *high-strength, low-alloy (HSLA) steels*, are ASTM A242, A440, A514, A572, and A913.

Appendix 7 lists the properties of several structural steels.

2-9 TOOL STEELS

The term *tool steels* refers to a group of steels typically used for cutting tools, punches, dies, shearing blades, chisels, and similar uses. The numerous varieties of tool steel materials have been classified into seven general types as shown in Table 2-11. Whereas most uses of tool steels are related to the field of manufacturing engineering, they are also pertinent to machine design where the ability to maintain a keen edge under abrasive conditions is required (Type H and F). Also, some tool steels have rather high shock resistance which may be desirable in machine components such as parts for mechanical clutches, pawls, blades, guides for moving materials, wrench sockets, screw driver bits, and clamps (Types S, L, F, and W). (See Reference 10 for a more extensive discussion of tool steels.)

2-10 CAST IRON

Large gears, machine structures, brackets, linkage parts, and other important machine parts are made from cast iron. The several types of grades available span wide ranges of strength, ductility, machinability, wear resistance, and cost. These features are attractive in many applications.

TABLE 2-11 Examples of Tool Steel Types

General type	Type symbol	Specific types	Examples		Typical uses (and other common alloys)
		Major alloying elements	AISI No.	UNS No.	
High-speed	M	Molybdenum	M2	T11302	General-purpose tool steels for cutting tools and dies for forging, extrusion, bending, drawing, and piercing (M1, M3, M4–M7, M30, M34, M36, M41–M47)
			M10	T11310	
			M42	T11342	
	T	Tungsten	T1 T15	T12001 T12015	Similar to uses for M-types (T2, T4, T5, T6, T8)
Hot-worked	H	Chromium	H10	T20810	Cold-heading dies, shearing knives, aircraft parts, low-temperature extrusion and die-casting dies (H1–H19)
		Tungsten	H21	T20821	Higher-temperature dies, hot shearing knives (H20–H39)
		Molybdenum	H42	T20842	Applications that tend to produce high wear (H40–H59)
Cold-worked	D	High-carbon, high-chromium	D2	T30402	Stamping dies, punches, gages (D3–D5, D7)
	A	Medium-alloy, air-hardening	A2	T30102	Punches, thread-rolling dies, die-casting dies (A3–A10)
	O	Oil-hardening	O1	T31501	Taps, reamers, broaches, gages, jigs and fixtures, bushings, machine tool arbors, tool shanks (O2, O6, O7)
Shock-resisting	S		S1	T41901	Chisels, pneumatic tools, heavy-duty punches, machine parts subject to shocks (S2, S4–S7)
Molded steels	P		P2	T51602	Plastic molding dies, zinc die-casting dies (P3–P6, P20, P21)
Special-purpose	L	Low-alloy types	L2	T61202	Tooling and machine parts requiring high toughness (L3, L6)
	F	Carbon-tungsten types	F1	T60601	Similar to L-types but with higher abrasion resistance (F2)
Water-hardened	W		W1	T72301	General-purpose tool and die uses, vise and chuck jaws, hand tools, jigs and fixtures, punches (W2, W5)

The three most commonly used types of cast iron are gray iron, ductile iron, and malleable iron. Appendix 8 (U.S. units) and 8A (SI Units) list the properties of several types and grades of cast iron. See also Reference 13.

Gray iron is available in grades having tensile strengths ranging from 20 000 to 60 000 psi (138–414 MPa). Its ultimate compressive strength is much higher, three to five times as high as the tensile strength. One disadvantage of gray iron is that it is brittle and therefore should not be used in applications where impact loading is likely. But it has excellent wear resistance, is relatively easy to machine, has good vibration damping ability, and can be surface-hardened. Applications include engine blocks, gears, brake parts, and machine bases. The gray irons are rated by the ASTM specification A48-03 (2012) and A48M in classes 20, 25, 30, 40, 50, and 60, where the number refers to the minimum tensile strength in kips/in²(ksi). For example, class 40 gray iron has a minimum tensile strength of 40 ksi or 40 000 psi (276 MPa). Because it is brittle, gray iron does not exhibit the property of yield strength.

Malleable iron is a group of heat-treatable cast irons with moderate to high strength, high modulus of elasticity (stiffness), good machinability, and good wear resistance.

The five-digit designation roughly indicates the yield strength and the expected percent elongation of the iron. For example, Grade 40010 has a yield strength of 40 ksi (276 MPa) and a 10% elongation. The strength properties listed in Appendix 8 are for the non-heat-treated condition. Higher strengths would result from heat treating. See ASTM specifications A 47-99 (2014) and A 220-99 (2014).

Ductile irons have higher strengths than the gray irons and, as the name implies, are more ductile. However, their ductility is still much lower than that of typical steels. A three-part grade designation is used for ductile iron in the ASTM A536-84 (2014) specification. The first number refers to the tensile strength in ksi, the second is the yield strength in ksi, and the third is the approximate percent elongation. For example, the grade 80-55-06 has a tensile strength of 80 ksi (552 MPa), a yield strength of 55 ksi (379 MPa), and a 6% elongation in 2.00 in. Higher-strength cast parts, such as crankshafts and gears, are made from ductile iron.

Austempered ductile iron (ADI) is an alloyed and heat-treated ductile iron. (See Reference 13.) It has attractive properties that lead to its use in transportation equipment, industrial machinery, and other applications where

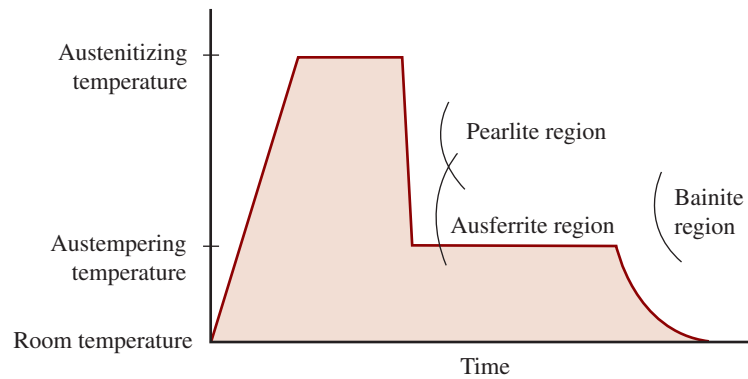


FIGURE 2-15 Heat treatment cycle for austempered ductile iron (ADI)

the low cost, good machinability, high damping characteristics, good wear resistance, and near-net-shape advantages of casting offer special benefits. Examples are drive train gears, parts for constant velocity joints, and suspension components. ASTM Standard A897/A897M-15 lists five grades of ADI ranging in tensile strength from 125 ksi (850 MPa) to 230 ksi (1600 MPa). Yield strengths range from 80 ksi (550 MPa) to 185 ksi (1300 MPa). Ductility decreases with increasing strength and hardness with percent elongation values in the range from approximately 10% to less than 1%. ADI begins as a conventional ductile iron with careful control of composition and the casting process to produce a sound, void-free casting. Small amounts of copper, nickel, and molybdenum are added to enhance the metal's response to the special heat treatment cycle shown in Figure 2-15. It is heated to the austenitizing temperature (1550°F–1750°F, or 843°C–954°C) depending on the composition. It is held at that temperature for one to three hours as the material becomes fully austenitic. A rapid quench follows in a medium at 460°F to 750°F (238°C–400°C), and the casting is held at this temperature for one-half to four hours. This is the *austempering* part of the cycle during which all of the material is converted to a mixture of mostly austenite and ferrite, sometimes called *ausferrite*. It is important that neither pearlite nor bainite form during this cycle. The casting is then allowed to cool to room temperature.

An emerging form of ADI, called *carbidaustempered ductile iron (CADI)*, is produced by alloying conventional ADI with carbide stabilizers such as chromium, molybdenum, or titanium. CADI exhibits significantly higher wear and abrasion resistance while retaining a good level of toughness. Applications are found in railroad rolling stock, earthmoving equipment, agricultural machinery, materials processing equipment, crushers, pump components, and bulk material conveyors. In these applications, it competes well with *white iron*, described next. Another good application is piping that transports aerated dry bulk materials or slurries containing suspended abrasive constituents. Wear is particularly a problem in elbows for such systems.

White iron is produced by rapidly chilling a casting of either gray iron or ductile iron during the solidification process. Chilling is typically applied to selected areas

that will experience heavy abrasion because chilled areas become very hard and have high wear resistance. Chilling does not allow the carbon in the iron to precipitate out during solidification, resulting in the white appearance. Areas away from the chilling medium solidify more slowly and acquire the normal properties of the base iron. One disadvantage of the chilling process is that the white iron is somewhat brittle. White iron can be alloyed with chromium and molybdenum to achieve high hardness ranging from 65 to 70 HRC. ASTM Standard A532/A532M-10(2014) covers the composition and heat-treatment for white iron. Typical applications include liners of ball or roll mills used to crush aggregates or other materials into powder form, the balls and rolls themselves, pulveriser components, clay-mixing equipment, brick-molding processes, and wear parts of crushers and material-handling devices.

2-11 POWDERED METALS

Making parts with intricate shapes by powder metallurgy can sometimes eliminate the need for extensive machining. Metal powders are available in many formulations whose properties approach those of the wrought form of the metal. The processing involves preparing a preform by compacting the powder in a die under high pressure. Sintering at a high temperature to fuse the powder into a uniform mass is the next step. Re-pressing is sometimes done to improve properties or dimensional accuracy of the part. Typical parts made by the powder metallurgy (PM) process are gears, gear segments, cams, eccentrics, and various machine parts having oddly shaped holes or projections. Dimensional tolerances of 0.001 in to 0.005 in (0.025–0.125 mm) are typical.

One disadvantage of PM parts is that they are sometimes brittle and should not be used in applications where high-impact loading is expected. Another important application is in sintered bearings, which are made to a relatively low density with consequent high porosity. The bearing is impregnated with a lubricant that may be sufficient for the life of the part. This type of material is discussed further in Chapter 16.

Manufacturers of metal powders have many proprietary formulations and grades. However, the Metal Powder Industries Federation (MPIF) is



FIGURE 2-16 Example of powder metal components. (Courtesy of the Metal Powder Industries Federation, Princeton, NJ)

promoting standardization of materials. Figure 2-16 shows photographs of some powder metal parts. (See Reference 6 and Internet site 9.)

A variety of metals can be used for PM parts, samples of which are listed in Table 2-12. Several processes are applied to achieve particular properties and to meet production goals. Brief descriptions are given here.

■ Press and Sinter

- Constituent powders in the proper percentage for the desired material are blended uniformly.
- Blended metal powders are fed into a die.
- A press compacts the powders into the desired shape, producing a *green part* having relatively low strength; the part is ejected from the die.
- A heating process, called *sintering*, metallurgically bonds the grains of powder into a solid, strong form.
- Optional manufacturing steps may be performed as needed, such as:

- Repressing to achieve refined properties and greater dimensional accuracy
- Machining of features that could not be made in the die
- Heat-treating to achieve higher mechanical strength
- Plating or other surface treatments

■ Powder Forging

- The steps described for the press and sinter process are completed as before. However, the geometry is not yet in final form or size.
- As the part is withdrawn from the sintering furnace, it is coated with a lubricant, transferred to a forging press in a close-form die where it is hot worked, causing plastic flow of the material to its final shape.
- The forging process refines the strength of the part, nearly equaling that of wrought forms of similar metals, while tending to orient the internal structure in favorable directions according to the anticipated loads.

TABLE 2-12 Materials Used for Powder Metallurgy (PM) Parts

Material type	Designation and composition	Tensile strength		Density	
		(ksi)	(MPa)	(lb _m /in ³)	(kgm ³)
Ferrous metals					
Carbon steel	F-0008 0.8% C	54	370	0.249	6900
	F-0008-HT 0.8% C	85	590	0.249	6900
Copper steel	FC-0208 2% Cu, 0.8% C	60	410	0.242	6700
	FC-0208-HT 2% Cu, 0.8% C	85	590	0.242	6700
Nickel steel	FN-0205 2% Ni, 0.5% C	57	390	0.257	7100
	FN-0205-HT 2% Ni, 0.5% C	145	1000	0.257	7100
Low-alloy steel	FL-4405 0.85% Mo, 0.5% C	66	460	0.257	7100
	FL-4405-HT 0.85% Mo, 0.5% C	160	1100	0.257	7100
Low-alloy steel hybrid	FLN-4405 2% Ni, 0.85% Mo, 0.5% C	80	550	0.255	7050
	FLN-4405-HT 2% Ni, 0.85% Mo, 0.5% C	170	1170	0.255	7050
Diffusion-alloyed steel	FD-0205 1.75% Ni, 1.5% Cu, 0.5% Mo, 0.5% C	78	540	0.251	6950
	FD-0205-HT 1.75% Ni, 1.5% Cu, 0.5% Mo, 0.5% C	130	900	0.251	6950
Sinter-hardened steel	FLC-4608-HT 1.75% Ni, 1.5% Cu, 0.5% Mo, 0.5% C	100	690	0.253	7000
Stainless steel	SS-316N2 17% Cr, 12% Ni, 2.5% M	60	410	0.235	6500
	SS-430N2 17% Cr	60	410	0.257	7100
Copper	C-0000 0.2% maximum other	Electrical applications		0.289	8000
Bronze	CTG-1001 10% Sn, 1% Graphite	Oil-impregnated bearings		0.224	6200
Red brass	CZ-1000 10% Zn	11	76	0.285	7900
Yellow brass	CZ-3000 30% Zn	16	110	0.275	7600
Nickel silver	CNZ-1818 18% Ni, 18% Zn	20	138	0.285	7900
Aluminum	Aluminum 4% Cu, 0.5% Mg, 1% Si	32	221	0.094	2600

Note: C = Carbon, Cu = Copper, Ni = Nickel, Mo = Molybdenum, Cr = Chromium, Sn = Tin, Zn = Zinc, Mg = Magnesium, Si = Silicon
See Internet site 9 for additional data.

- **Isostatic Pressing**
 - The blended metal powders are placed into a hermetic chamber and restrained by a flexible membrane.
 - A high pressure is exerted on the membrane compressing the powder uniformly and producing a high density.
 - Compressing at room temperature is called *cold isostatic pressing*.
 - Compressing at elevated temperatures is called *hot isostatic pressing*, which accomplishes the forming to the desired shape and simultaneously sintering the powders, thus eliminating one step in the process.
 - The ability to make larger, more complex parts and the use of a wider range of powder types are advantages of using the isostatic pressing processes.
 - The density of the completed part is typically more uniform as compared with the press and sinter process and the resulting density and mechanical properties are very close to those obtained in wrought forms of metals of similar composition.
- **Metal Injection Molding (MIM)**
 - Blended metal powders are enhanced with binders and additives such as waxes and thermoplastic polymers.
 - The mixture is fed into an injection molding machine, similar to plastics injection molding systems to create the *green part*.
 - The green part is extracted from the mold and sintered as with other powder metal processes.
 - More complex shapes can be made with MIM than with other processes and dimensional accuracy is improved.
 - Mechanical properties of the finished metal part are nearly equal to those of wrought forms of similar composition.

Production of powder metal products often requires expensive tooling, presses, and special equipment, lending its use for high volume or high-value products. Typical industries using powder metal products are:

Aircraft	Space vehicles	Satellites	Military equipment
Medical devices	Automotive	Appliances	Power tools
Hydraulic systems	Agricultural machinery	Weapons	Business machines
Motor components	Recreational equipment	Consumer products	
Computer peripheral components	Chemical processing equipment		

Guidelines for Specifying Powder Metal Parts. While it is not practical to define all possible conditions, the following discussion presents the framework in which powder metal products fall. Consultation with an experienced supplier is recommended.

1. Part size is typically limited to those with a projected surface area of compressed powder of 50 in² (32 000 mm²) based on press capacity.
2. Part length should be less than 6 in (300 mm) but greater than 0.06 in (1.5 mm).
3. The length/diameter ratio should be less than 5.
4. The length/wall thickness ratio should be less than 8.
5. The part shape must be such that it can be formed by linear pressing by a ram.
6. No undercut, threaded, or re-entrant forms can be made.
7. Density variations should be expected from normal press and sinter processing for relatively long parts.
8. Relatively high production volumes are normally required to amortize tooling and equipment costs.
9. Consultation with the supplier is recommended on powder composition, processing method, and performance testing of the finished product.
10. Some PM parts may retain some porosity so the product may not be pressure tight without subsequent impregnation of sealing materials.

2-12 ALUMINUM

Aluminum is widely used for structural and mechanical applications. Chief among its attractive properties are light weight, good corrosion resistance, relative ease of forming and machining, and pleasing appearance. Its density is approximately one-third that of steel. However, its strength is somewhat lower, also. (See References 1, 2, 12, and 23.) Table 2-13 lists the commonly used alloy groups for wrought aluminums.

The standard designations for aluminum alloys listed by the Aluminum Association use a four-digit system. The first digit indicates the alloy type according to the major alloying element. The second digit, if it is other than zero, indicates modifications of another alloy or limits placed on impurities in the alloy. The presence of impurities is particularly important for electrical conductors. Within each group are several specific alloys, indicated by the last two digits in the designation.

TABLE 2-13 Wrought Aluminum Alloy Groups

Alloy designations (by major alloying element)

1xxx	99.00% or greater aluminum content
2xxx	Copper
3xxx	Manganese
4xxx	Silicon
5xxx	Magnesium
6xxx	Magnesium and silicon
7xxx	Zinc

TABLE 2-14 Common Aluminum Alloys and Their Uses

Alloy	Applications	Forms
1060	Chemical equipment and tanks	Sheet, plate, tube
1350	Electrical conductors	Sheet, plate, tube, rod, bar, wire, pipe, shapes
2014	Aircraft structures and vehicle frames	Sheet, plate, tube, rod, bar, wire, shapes, forgings
2024	Aircraft structures, wheels, machine parts	Sheet, plate, tube, rod, bar, wire, shapes, rivets
2219	Parts subjected to high temperatures (to 600°F)	Sheet, plate, tube, rod, bar, shapes, forgings
3003	Chemical equipment, tanks, cooking utensils, architectural parts	Sheet, plate, tube, rod, bar, wire, shapes, pipe, rivets, forgings
5052	Hydraulic tubes, appliances, sheet-metal fabrications	Sheet, plate, tube, rod, bar, wire, rivets
6061	Structures, vehicle frames and parts, marine uses	All forms
6063	Furniture, architectural hardware	Tube, pipe, extruded shapes
7001	High-strength structures	Tube, extruded shapes
7075	Aircraft and heavy-duty structures	All forms except pipe

Table 2-14 lists 11 of the 50 or more available aluminum alloys along with the forms in which they are typically produced and some of their major applications. See also Table 2-6 that lists six aluminum alloys used for commercially available shapes. These tables should aid you in selecting suitable alloys for particular applications.

The mechanical properties of the aluminum alloys are highly dependent on their condition. For this reason, the specification of an alloy is incomplete without a reference to its *temper*. The following list describes the usual tempers given to aluminum alloys. Note that some alloys respond to heat treating, and others are processed by strain hardening. *Strain hardening* is controlled cold working of the alloy, in which increased working increases hardness and strength while reducing ductility. Commonly available tempers are described next.

F (as-fabricated): No special control of properties is provided. Actual limits are unknown. This temper should be accepted only when the part can be thoroughly tested prior to service.

O (annealed): A thermal treatment that results in the softest and lowest strength condition. Sometimes specified to obtain the most workable form of the alloy. The resulting part can be heat-treated for improved properties if it is made from alloys in the 2xxx, 4xxx, 6xxx, or 7xxx series. Also, the working itself may provide some improvement in properties similar to that produced by strain hardening for alloys in the 1xxx, 3xxx, and 5xxx series.

H (strain-hardened): A process of cold working under controlled conditions that produces improved, predictable properties for alloys in the 1xxx, 3xxx, and 5xxx groups. The greater the amount of cold work, the higher the strength and hardness, although the ductility is decreased. The *H* designation is

followed by two or more digits (usually 12, 14, 16, or 18) that indicate progressively higher strength. Temper H18 is called *full hard*, produced by approximately 75% reduction in area during the strain hardening process. H16 is 3/4 hard; H14 is 1/2 hard; and H12 is 1/4 hard. H19 extra-hard temper is available and results in harder and stronger properties than H18. Other designations are used.

T (heat-treated): A series of controlled heating and cooling processes applied to alloys in the 2xxx, 4xxx, 6xxx, and 7xxx groups. The letter *T* is followed by one or more numbers to indicate specific processes. The more common designations for mechanical and structural products are T3, T4, and T6.

Property data for aluminum alloys are included in Appendix 9. Because these data are typical values, not guaranteed values, the supplier should be consulted for data at the time of purchase.

For mechanical design applications, alloy 6061 is one of the most versatile types. Note that it is available in virtually all forms, has good strength and corrosion resistance, and is heat-treatable to obtain a wide variety of properties. It also has good weldability. In its softer forms, it is easily formed and worked. Then, if higher strength is required, it can be heat-treated after forming. However, it has low machinability.

Aluminum Casting Alloys

Aluminum casting alloys are designated by a system of three digits followed by a decimal point and one digit to the right: xxx.x. The first digit represents the alloying element in the base aluminum having the highest concentration as follows:

- 1xx.x—99.00% minimum aluminum
- 2xx.x—Copper

- 3xx.x—Silicon with added copper and/or magnesium
- 4xx.x—Silicon
- 5xx.x—Magnesium
- 7xx.x—Zinc
- 8xx.x—Tin

The second two digits represent a particular composition of alloying elements, typically some combination of those listed above. The last digit after the decimal point indicates the form of the product being described, as follows:

- 0—Casting, with specification of sand, permanent mold, or die cast.
- 1—Ingot form, as supplied to the caster with typical control of nonspecified elements.
- 2—Ingot form, with tighter limits of specified elements.

Temper designations must also be applied to the alloy, typically *F* (*as-cast*), *O* (*annealed*), or *Tx* (*heat-treated*), where *x* is one or more numbers indicating the nature of the heat-treatment similar to those used for wrought alloys. Tempers T4 and T6 are most popular.

Mechanical properties data for three popular alloys, 204.0-T4, 206.0-T6, and 356.0-T6, for permanent mold castings are listed in Appendix 9. At least 75 other alloys are available, such as 201, 222, 242, 319, 360, 380, 413, 443, 444, 512, 518, 535, 712, 713, 771, 850, 851, and 852. The 850 series alloys are used primarily for bearings in applications similar to those for which bronzes are used. For special purposes, standard alloys are modified by slightly different compositions, specifications of tighter

limits on constituents, or specifying minimum or maximum amounts of trace elements. Designations for such alloys are preceded by a letter (A, B, C, D, and so forth).

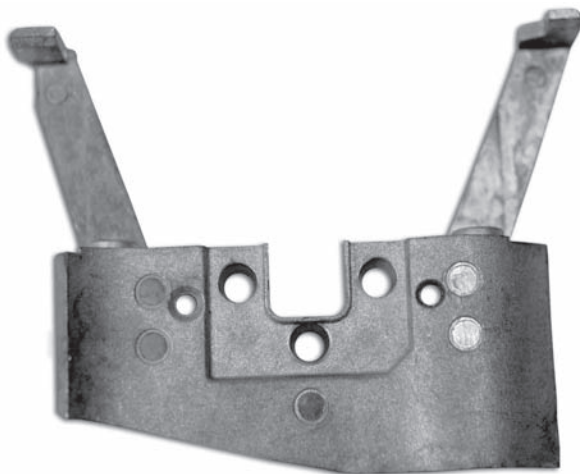
Aluminum Forging Alloys

Many of the same alloys used in wrought form are also used for forging stock. Examples are general-purpose alloys 2014, 2024, 5052, and 6061. Alloys 7075, 7175, and 7145 offer much higher strengths and are often used in aerospace applications. Alloys 2218 and 2025 have high fatigue strengths and are applied in rotating engine parts, aircraft propellers, and machine parts that are cyclically loaded. Alloys 2219, 2618, and 4032 are designed for high-temperature operation as experienced in engines. Class I forging alloys are typically supplied in extruded form with wide tolerances in *F* or *O* tempers as rounds, squares, or rectangles. Class II forging stock is wrought and cold finished with much tighter tolerances, sometimes as high as ten times tighter. See Internet site 7.

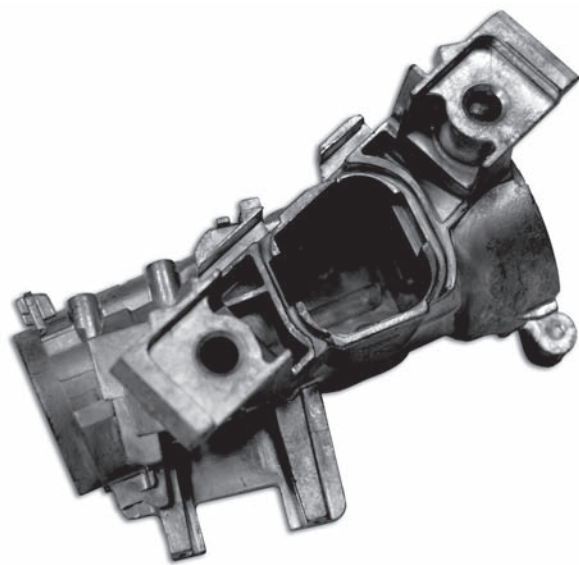
2-13 ZINC ALLOYS AND MAGNESIUM

Zinc Alloys

Zinc is the fourth most commonly used metal in the world. Much of it is in the form of zinc galvanizing used as a corrosion inhibitor for steels, but very large quantities of zinc alloys are used in castings and for bearing materials. See Reference 28 and Internet site 10. Figure 2-17 shows two examples of zinc die-cast parts having complex shapes and many cast-in holes, bosses, and other features, many of which can be used without further machining.



(a) Camera support bracket for an interactive whiteboard



(b) Steering wheel lock housing

FIGURE 2-17 Examples of zinc die-cast parts (Photo courtesy of North American Die Casting Association, Wheeling, IL)

High production quantities are made using zinc pressure die casting, which results in very smooth surfaces and excellent dimensional accuracy. A variety of coating processes can be used to produce desirable finish appearance and to inhibit corrosion. Although the as-cast parts have inherently good corrosion resistance, the performance in some environments can be enhanced with chromate or phosphate treatments or anodizing. Painting and chrome plating are also used to produce a wide variety of attractive surface finishes.

In addition to die casting, zinc products are often made by permanent mold casting, graphite permanent mold casting, sand casting, and shell-mold casting. Other, less frequently used processes are investment casting, low-pressure permanent mold casting, centrifugal casting, continuous casting, and rubber-mold casting. Plaster-mold casting is often used for prototyping. Continuous casting is used to produce standard shapes (rod, bar, tube, and slabs). Prototypes or finished products can then be machined from these shapes.

Zinc alloys typically contain aluminum and a small amount of magnesium. Some alloys include copper or nickel. The performance of the final products can be very sensitive to small amounts of other elements, and maximum limits are placed on the content of iron, lead, cadmium, and tin in some alloys.

The most widely used zinc casting alloy is called *alloy No. 3*, sometimes referred to as *Zamak 3*. It has 4% aluminum and 0.035% magnesium. Another form, called *alloy No. 5* (*Zamak 5*), is often used in Europe and it also has 4% aluminum with 0.055% magnesium and 1.0% copper.

A group of alloys having higher aluminum content are the *ZA-alloys*, with ZA-8, ZA-12, and ZA-27 in wide use. The numbers 8, 12, and 27 indicate the nominal percentage of aluminum in the alloy, far higher than the 4% in the Zamak alloys. This results in higher strength, particularly in die-cast form. Each of the ZA alloys can be sand cast, permanent mold cast, or die cast. Percent elongation of cast zinc materials are relatively low, ranging from 2% to 13% depending on the alloy and casting method. Brittle behavior is to be expected. Appendix 10–1 gives a summary of the typical properties of these alloys.

Two additional alloys, No. 2 and No 7, are available. No. 2 is sometimes called KIRKSITE and is good for gravity casting. No. 7 is modified from No. 3 with lower magnesium content which gives it improved fluidity to fill more complex castings or those requiring modest forming during assembly.

Magnesium Alloys

One of the lightest metals used in common practice, magnesium is applied in many applications in automotive, military vehicles and equipment, motorcycles, construction tools, chainsaws, luggage, business equipment, ladders, consumer products, computers, cell phones,

other electronic devices, and appliances. Its density of approximately 1800 kg/m^3 ($0.065 \text{ lb}_m/\text{in}^3$) is two-thirds that of aluminum and about one-fifth that of steel. Magnesium has high inherent damping capacity to absorb vibration energy providing quieter operation of equipment when used for housings and enclosures. It is readily cast using die casting (the most prevalent choice), permanent mold casting, sand casting, and squeeze casting. However, strengths are relatively low and its modulus of elasticity, typically 45 GPa ($6.53 \times 10^6 \text{ psi}$), is the lowest of common metals.

The most widely used alloy is AZ91, where A refers to aluminum and Z refers to zinc, the two major alloying elements. A series of three AM alloys contain aluminum and magnesium. The AS alloys contain aluminum and silicon that give them better high temperature performance. Appendix 10–2 gives values for mechanical properties of AZ91, AM-60, AM-50, and AM-20 alloys.

2-14 NICKEL-BASED ALLOYS AND TITANIUM

Nickel-Based Alloys

Nickel-based alloys are often specified for applications where high corrosion resistance and/or high strength at elevated temperatures are required. Examples are turbine engine components, rocket motors, furnace parts, chemical processing systems, marine-based systems, aerospace systems, valves, pumps, heat exchangers, food processing equipment, pharmaceutical processing systems, nuclear power systems, and air pollution equipment.

Appendix 11–1 gives typical properties of three nickel-based alloys listed according to their UNS designations in the NXXXXX series. Most nickel-based alloys are not heat-treated; however, cold working is employed to increase strength as is evident for the N04400 alloy in Appendix 11–1.

Alloys N06600 and N06110 retain approximately 70% to 80% of room-temperature strength up to 1200°F (649°C). Some alloys are usable up to 2000°F (1093°C). Alloys are often grouped under the categories of *corrosion resistant* or *heat resistant*. While nickel makes up 50% or more of these alloys, the major alloying elements include chromium, molybdenum, cobalt, iron, copper, and niobium. See Internet sites 28–30 for some producers of these alloys. Note the brand names offered by these companies and the extensive set of property data available on their websites. See Internet sites 28–30.

Creep performance is an important property for the nickel-based alloys because they often operate under significant loads and stresses at high temperatures where creep rupture is a possible failure mode. Also, the modulus of elasticity decreases with increasing temperature, which can result in unacceptable deformations that must be considered in the design phase. Producers publish creep and elastic modulus versus temperature data for their materials in addition to the basic strength data.

Titanium Alloys

The applications of titanium include aerospace structures and components, chemical tanks and processing equipment, fluids-handling devices, and marine hardware. Titanium has very good corrosion resistance and a high strength-to-weight ratio. Its stiffness and density are between those of steel and aluminum; its modulus of elasticity is approximately 16×10^6 psi (110 GPa), and its density 0.160 lb/in^3 (4.429 kg/m^3). Typical yield strengths range from 25 to 175 ksi (172–1210 MPa). Disadvantages of titanium include relatively high cost and difficult machining. See Reference 29.

The classification of titanium alloys usually falls into four types: commercially pure alpha titanium, alpha alloys, alpha–beta alloys, and beta alloys. The term *alpha* refers to the hexagonal, close-packed, metallurgical structure that forms at low temperatures, and *beta* refers to the high-temperature, body-centered, cubic structure.

Generally speaking, the alpha–beta alloys and the beta alloys are stronger forms of titanium. They are heat-treatable for close control of their properties. Since several alloys are available, a designer can tailor the properties to meet special needs for formability, machinability, forgeability, corrosion resistance, high-temperature strength, weldability, and creep resistance, as well as basic room-temperature strength and ductility. Alloy Ti-6Al-4V contains 6% aluminum and 4% vanadium and is used in a variety of aerospace applications.

Appendix 11–2 lists the properties of:

- Three commercially pure alpha alloys in wrought form: Ti-35A, Ti-50A, and Ti-65. For each, the number represents the nominal tensile strength in ksi.
- Alpha alloy Ti-0.2Pd in wrought form
- Beta alloy Ti-3Al-13V-11Cr in two forms of heat-treatment
- Alpha-beta alloy Ti-6Al-4V in one annealed and one heat-treated form.

2-15 COPPER, BRASS, AND BRONZE

Copper is widely used in its nearly pure form for electrical and plumbing applications because of its high electrical conductivity and good corrosion resistance. It is rarely used for machine parts because of its relatively low strength compared with that of its alloys, *brass* and *bronze*. (See Reference 6 and Internet site 8.)

Brass is a family of alloys of copper and zinc, with the content of zinc ranging from about 5% to 40%. Brass is often used in marine applications because of its resistance to corrosion in salt water. Many brass alloys also have excellent machinability and are used as connectors, fittings, and other parts made on screw machines. *Yellow brass* contains about 30% or more of zinc and often contains a significant amount of lead to improve

machinability. *Red brass* contains 5% to 15% zinc. Some alloys also contain tin, lead, nickel, or aluminum.

Bronze is a class of alloys of copper with several different elements, one of which is usually tin. They are useful in gears, bearings, and other applications where good strength and high wear resistance are desirable.

Wrought bronze alloys are available in four types:

Phosphor bronze: Copper-tin-phosphorus alloy

Leaded phosphor bronze: Copper-tin-lead-phosphorus alloy

Aluminum bronze: Copper-aluminum alloy

Silicon bronze: Copper-silicon alloy

Cast bronze alloys have four main types:

Tin bronze: Copper-tin alloy

Leaded tin bronze: Copper-tin-lead alloy

Nickel tin bronze: Copper-tin-nickel alloy

Aluminum bronze: Copper-aluminum alloy

The cast alloy called *manganese bronze* is actually a high-strength form of brass because it contains zinc, the characteristic alloying element of the brass family. Manganese bronze contains copper, zinc, tin, and manganese.

In the UNS, copper alloys are designated by the letter C, followed by a five-digit number. Numbers from 10000 to 79900 refer to wrought alloys; 80000 to 99900 refer to casting alloys. See Appendix 12 for typical properties.

Wrought forms of brass and bronze include circular rods, flat bars, hexagonal bars, sheet, strip, and wire. Over 120 different wrought alloys are available, listed on Internet site 8. Four common types of wrought brasses are:

Extra high-leaded brass—C35600 Copper-lead-tin alloy

Uses include: Gears, gear wheels, and general machined parts

Free-cutting brass—C36000 Copper-lead-tin alloy

Uses include: Screw machine parts, gears, and roll-threaded parts

Free-cutting Muntz metal—C37000 Copper-tin alloy with a small amount of lead

Uses include: Screw machine parts in high-production operations

Naval brass—C46400 Copper-tin-lead-zinc alloy

Uses include: Marine hardware, propeller shafts, valve stems, nuts, bolts, and rivets

Phosphor bronze—C54400 Copper-lead-tin-zinc-phosphorous alloy

Uses include: Gears, bushings, stamped parts, drawn parts, machine components

Silicon bronze—C65500Copper (97%)-
silicon (3%)

Uses include: Electrical fittings, marine hardware, heavy screws, bolts, propeller shafts

Manganese bronze—C67500Copper-zinc-iron-
tin-manganese alloy

Uses include: Clutch discs, pump parts, shafting, aircraft parts, forgings, valve stems

Casting alloys are made using sand casting, permanent mold casting, and centrifugal casting. See Internet site 8 for properties of over 90 casting alloys. A few common types are:

Manganese bronze—C86200 Copper-zinc, aluminum,
iron, manganese alloy

Uses include: Marine castings, rudders, gun mounts, large gears, cams, machine parts

Bearing bronze—C93200 Copper-tin-lead-zinc
alloy

Uses include: Bearings and bushings for heavy equipment, pump impellers, hose fittings

Aluminum bronze—C95400 Copper-aluminum-iron
alloy

Uses include: Valve seats, gears, worm wheels, bearings, landing gear parts, welding tips

**Copper-nickel-zinc alloy—
C96200**

Uses include: Pump housings, valve bodies, marine hardware, and plumbing components

Nickel silver—C97300 Copper-zinc-nickel alloy

Uses include: Camera parts, optical equipment, jewelry, fasteners, drawn shapes

Conditions for Copper Alloys. Copper alloys typically have a wide range of properties, heavily dependent on thermal treatment, strain hardening by cold working, or a combination of both. Thermal treatments include annealing, solution heat-treating, and precipitation hardening.

Strain hardening is categorized by many systems, one of which is called *cold-worked tempers (H)*, in which control of the degree of working produces predictable strength and ductility properties. Working tends to align the grains of the alloy's structure and higher degrees of working create finer grain sizes, a preferred condition. The range of cold-worked tempers is typically given in ten steps with the initial condition being annealed.

H00 1/8 hard	H04 Full hard	H12 Special spring
H01 1/4 hard	H06 Extra hard	H13 Ultra spring
H02 1/2 hard	H08 Spring	H14 Super spring
H03 3/4 hard	H10 Extra spring	

Other conditions or tempers include:

<i>annealed tempers (O)</i>	<i>cold-worked and stress relieved (HR)</i>
<i>average grain size control tempers (OS)</i>	<i>cold-worked and order-strengthened (HT)</i>
<i>mill-hardened tempers (TM)</i>	<i>solution heat-treated (TB)</i>
<i>precipitation hardened tempers (TF)</i>	<i>solution heat-treated and cold-worked (TD)</i>
<i>precipitation hardened and cold-worked (TL)</i>	<i>precipitation hardened and stress relieved (TR)</i>
<i>Cold-worked and precipitation hardened (TH)</i>	

There is a strong relationship between the condition of copper alloys and the ease of which they can be fabricated, especially the tempers involving cold working. In general, as the degree of cold working increases, the strength is greater, but the ductility is lower indicated by the percent elongation. Figure 2–18 illustrates the general nature of this situation. The designer's responsibility is to specify a condition that has adequate strength while retaining sufficient ductility to permit forming and fabricating using such methods as:

machining	thread rolling	press forming	brake forming
roll forming	drawing	forging	blanking
stamping	coining	punching	upsetting
cold heading	knurling	shearing	swaging
spinning	soldering	welding	brazing

2-16 PLASTICS

Plastics include a wide variety of materials formed of large molecules called *polymers*. The thousands of different plastics are created by combining different chemicals to form long molecular chains.

One method of classifying plastics is by the terms *thermoplastic* and *thermosetting*. In general, the *thermoplastic* materials can be formed repeatedly by heating or molding because their basic chemical structure is unchanged from its initial linear form. *Thermosetting* plastics do undergo some change during forming and result in a structure in which the molecules are cross-linked and form a network of interconnected molecules. Some designers recommend the terms *linear* and *cross-linked* in place of the more familiar *thermoplastic* and *thermosetting*.

Listed next are several thermoplastics and several thermosets that are used for load-carrying parts and that are therefore of interest to the designer of machine elements. These listings show the main advantages and uses of a sample of the many plastics available. Appendix 13 lists typical properties.

Thermoplastics

- **Polyamide (PA or Nylon):** Good strength, wear resistance, and toughness; wide range of possible properties depending on fillers and formulations. Used for

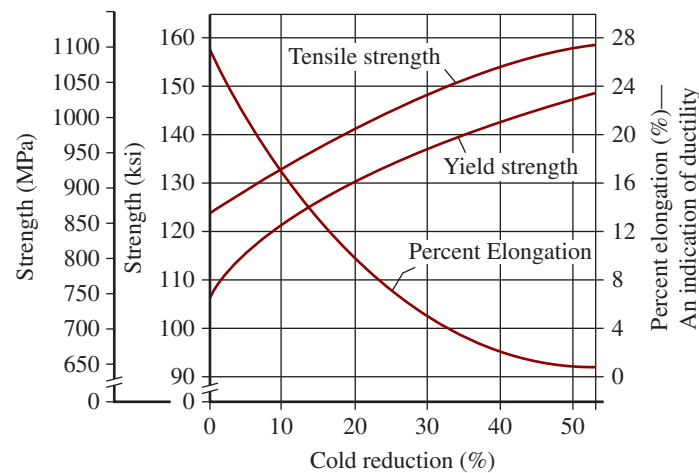


FIGURE 2-18 General characteristics of the variation of strength and ductility of brass alloys as related to the degree of cold working

structural parts, mechanical devices such as gears and bearings, and parts needing wear resistance.

- **Acrylonitrile-butadiene-styrene (ABS):** Good impact resistance, rigidity, moderate strength. Used for housings, helmets, cases, appliance parts, pipe, and pipe fittings.
- **Polycarbonate:** Excellent toughness, impact resistance, and dimensional stability. Used for cams, gears, housings, electrical connectors, food processing products, helmets, and pump and meter parts.
- **Acrylic:** Good weather resistance and impact resistance; can be made with excellent transparency or translucent or opaque with color. Used for glazing, lenses, signs, and housings.
- **Polyvinyl chloride (PVC):** Good strength, weather resistance, and rigidity. Used for pipe, electrical conduit, small housings, ductwork, and moldings.
- **Polyimide (PI):** Good strength and wear resistance; very good retention of properties at elevated temperatures up to 500°F. Used for bearings, seals, rotating vanes, and electrical parts.
- **Acetal:** High strength, stiffness, hardness, and wear resistance; low friction; good weather resistance and chemical resistance. Used for gears, bushings, sprockets, conveyor parts, and plumbing products.
- **Polyurethane elastomer:** A rubberlike material with exceptional toughness and abrasion resistance; good heat resistance and resistance to oils. Used for wheels, rollers, gears, sprockets, conveyor parts, and tubing.
- **Thermoplastic polyester resin (PET):** Polyethylene terephthalate (PET) resin with fibers of glass and/or mineral. Very high strength and stiffness, excellent resistance to chemicals and heat, excellent dimensional stability, and good electrical properties. Used for pump parts, housings, electrical parts, motor parts, auto parts, oven handles, gears, sprockets, and sporting goods.

- **Polyether-ester elastomer:** Flexible plastic with excellent toughness and resilience, high resistance to creep, impact, and fatigue under flexure, good chemical resistance. Remains flexible at low temperatures and retains good properties at moderately elevated temperatures. Used for seals, belts, pump diaphragms, protective boots, tubing, springs, and impact absorbing devices. High modulus grades can be used for gears and sprockets.

Thermosets

- **Phenolic:** High rigidity, good moldability and dimensional stability, very good electrical properties. Used for load-carrying parts in electrical equipment, switchgear, terminal strips, small housings, handles for appliances and cooking utensils, gears, and structural and mechanical parts. Alkyd, allyl, and amino thermosets have properties and uses similar to those of the phenolics.
- **Polyester:** Known as *fiber glass* when reinforced with glass fibers; high strength and stiffness, good weather resistance. Used for housings, structural shapes, and panels.

Special Considerations for Selecting Plastics

A particular plastic is often selected for a combination of properties, such as light weight, flexibility, color, strength, stiffness, chemical resistance, low friction characteristics, or transparency. Table 2-15 lists the primary plastic materials used for six different types of applications. References 14, 19, 27, and 32 provide us extensive comparative studies of the design properties of plastics.

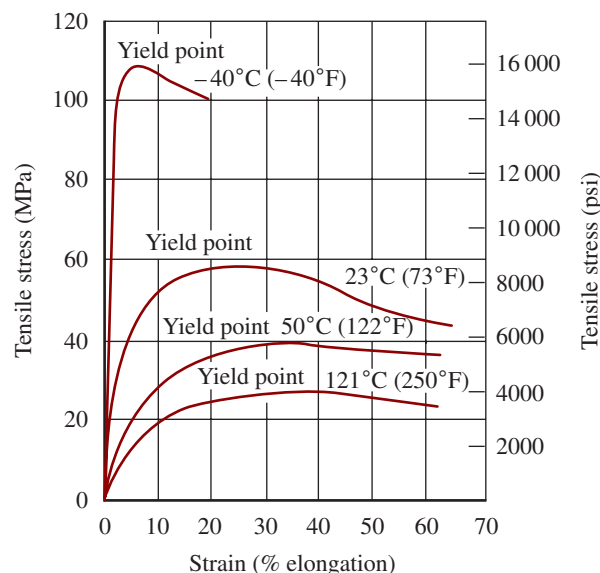
While most of the same definitions of design properties described in Section 2-2 of this chapter can be used for plastics as well as metals, a significant amount of additional information is typically needed to

TABLE 2-15 Applications of Plastic Materials

Applications	Desired properties	Suitable plastics
Housings, containers, ducts	High impact strength, stiffness, low cost, formability, environmental resistance, dimensional stability	ABS, polystyrene, polypropylene, PET, polyethylene, cellulose acetate, acrylics
Low friction—bearings, slides	Low coefficient of friction; resistance to abrasion, heat, corrosion	TFE fluorocarbons, nylon, acetals
High-strength components, gears, cams, rollers	High tensile and impact strength, stability at high temperatures, machinable	Nylon, phenolics, TFE-filled acetals, PET, polycarbonate
Chemical and thermal equipment	Chemical and thermal resistance, good strength, low moisture absorption	Fluorocarbons, polypropylene, polyethylene, epoxies, polyesters, phenolics
Electrostructural parts	Electrical resistance, heat resistance, high impact strength, dimensional stability, stiffness	Allyls, alkyds, amines, epoxies, phenolics, polyesters, silicones, PET
Light-transmission components	Good light transmission in transparent and translucent colors, formability, shatter resistance	Acrylics, polystyrene, cellulose acetate, vinyls

specify a suitable plastic material. Some of the special characteristics of plastics follow. The charts shown in Figures 2-19 to 2-22 are examples only and are not meant to indicate the general nature of the performance of the given type of material. There is a wide range of properties among the many formulations of plastics even within a given class. Consult the extensive amount of design guidance available from vendors of the plastic materials.

1. Most properties of plastics are highly sensitive to temperature. In general, tensile strength, compressive strength, elastic modulus, and impact failure energy decrease significantly as the temperature increases. Figure 2-19 shows the tensile strength of



Sample has been conditioned to 50% relative humidity.

FIGURE 2-19 Stress-strain curves for nylon 66 at four temperatures

nylon 66 at four temperatures. Note also the rather different shapes of the stress-strain curves. The slope of the curve at any point indicates the elastic modulus, and you can see a large variation for each curve.

2. Many plastics absorb a considerable amount of moisture from the environment and exhibit dimensional changes and degradation of strength and stiffness properties as a result. See Figure 2-20 that shows the flexural modulus versus temperature for a nylon in dry air, 50% relative humidity (RH), and 100% RH. A consumer product may well experience a major part of this range. At a temperature of 20°C (68°F, near room temperature), the flexural modulus would decrease dramatically from approximately 2900 MPa to about 500 MPa as humidity changes from dry air to 100% RH. The product may also see a temperature range from 0°C (32°F, freezing point of water) to 40°C (104°F). Over this range, the flexural

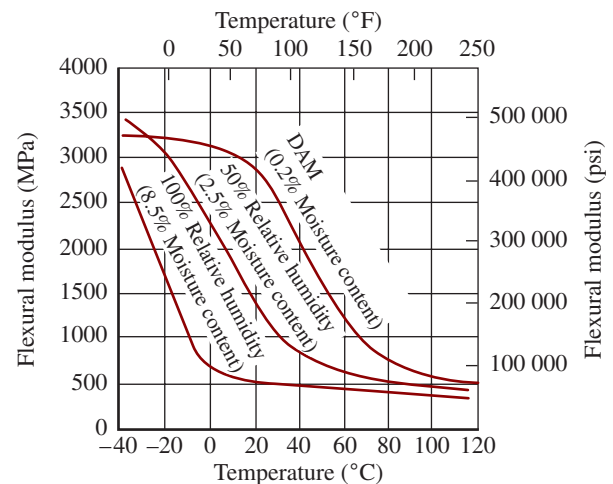


FIGURE 2-20 Effect of temperature and humidity on the flexural modulus of nylon 66

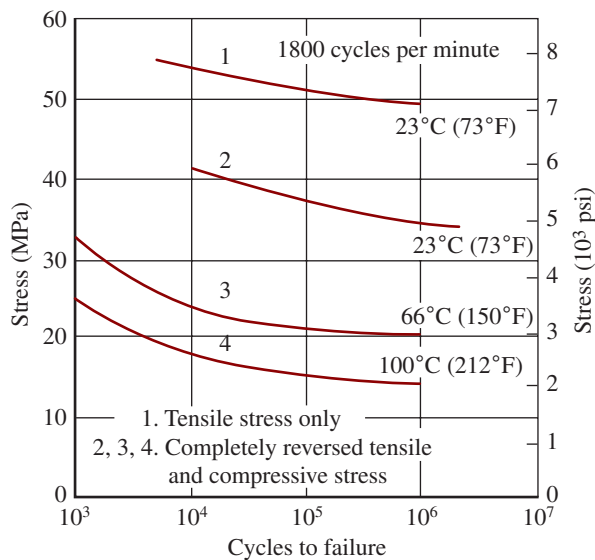


FIGURE 2-21 Fatigue stress versus number of cycles to failure for an acetal resin plastic

modulus for the nylon at 50% RH would decrease from approximately 2300 MPa to 800 MPa.

3. Components that carry loads continuously must be designed to accommodate creep or relaxation. See Figures 2-8 and 2-9 and Example Problem 2-1.
4. Fatigue resistance data of a plastic must be acquired for the specific formulation used and at a representative temperature. Chapter 5 gives more information about fatigue. Figure 2-21 shows the fatigue stress versus number of cycles to failure for an acetal resin plastic. Curve 1 is at 23°C (73°F, near room temperature) with cyclic loading in tension only as when a tensile load is applied and removed many times. Curve 2 is at the same temperature, but the loading is completely reversed tension and compression as would be experienced with a rotating beam or shaft loaded in bending. Curve 3 is the reversed bending load at 66°C (150°F), and Curve 4 is the same loading at 100°C (212°F) to show the effect of temperature on fatigue data.
5. Processing methods can have large effects on the final dimensions and properties of parts made from plastics. Molded plastics shrink significantly during solidification and curing. Parting lines produced where mold halves meet may affect strength. The rate of solidification may be widely different in a given part depending on the section thicknesses, the complexity of the shape and the location of sprues that deliver molten plastic into the mold. The same material can produce different properties depending on whether it is processed by injection molding, extrusion, blow molding, or machining from a solid block or bar. See Reference 27.
6. Resistance to chemicals, weather, and other environmental conditions must be checked.

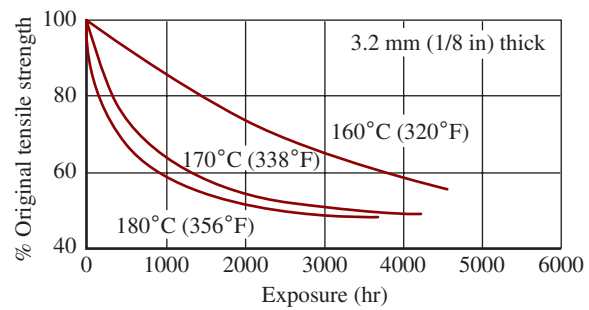


FIGURE 2-22 Effect of exposure to elevated temperature on a thermoplastic polyester resin (PET)

7. Plastics may exhibit a change in properties as they age, particularly when subjected to elevated temperatures. Figure 2-22 shows the reduction in tensile strength for a thermoplastic polyester resin when subjected to temperatures from 160°C (320°F) to 180°C (356°F) for a given number of hours of exposure. The reduction can be as much as 50% in as little time as 2000 hours (12 weeks).
8. Flammability and electrical characteristics must be considered. Some plastics are specially formulated for high flammability ratings as called for by Underwriters Laboratory and other agencies.
9. Plastics used for food storage or processing must meet U.S. Food and Drug Administration standards.

2-17 COMPOSITE MATERIALS

Composites are materials having two or more constituents blended in such a way that results in bonding between the materials. To form a composite, a discrete *reinforcement* material is distributed in a second continuous material called a *matrix* so that it adds strength, stiffness, or other desirable properties. Typically, the reinforcement material is strong and stiff while the matrix has a relatively low density and specific weight. When the two materials bond together, much of the load-carrying ability of the composite is produced by the reinforcement. The matrix serves to hold the reinforcement in a favorable orientation relative to the manner of loading and to distribute the loads to the reinforcement. The result is a somewhat optimized composite that has high strength and high stiffness with low weight. See References 3, 6, 20, 25, 33, 35, and 36 and Internet sites 1, 19-26, 31, 32, and 34-37.

Designers can tailor the properties of composite materials to meet the specific needs of a particular application by the selection of each of several variables that determine the performance of the final product. Among the factors under the designer's control are the following:

1. Matrix resin or metal.
2. Type of reinforcing fibers.
3. Amount of fiber contained in the composite.
4. Orientation of the fibers.

TABLE 2-16 Examples of Composite Materials and Their Uses

Type of composite	Typical applications
Glass/epoxy	Automotive and aircraft parts, tanks, sporting goods, printed wiring boards
Boron/epoxy	Aircraft structures and stabilizers, sporting goods
Graphite/epoxy	Aircraft and spacecraft structures, sporting goods, agricultural equipment, material handling devices, medical devices
Aramid/epoxy	Filament-wound pressure vessels, aerospace structures and equipment, protective clothing, automotive components
Glass/polyester	Sheet-molding compound (SMC), body panels for trucks and cars, large housings

5. Number of individual layers used.
6. Overall thickness of the material.
7. Orientation of the layers relative to each other.
8. Combination of two or more types of composites or other materials into a composite structure.

Table 2-16 lists some of the composites formed by combinations of resins and fibers and their general characteristics and uses. A virtually unlimited variety of composite materials can be produced by combining different matrix materials with different reinforcements in different forms and in different orientations.

Examples of Finished Products Made from Composite Materials

The number and variety of applications for composite materials is large and growing. The following items are but a sampling of these applications.

Consumer Products and Recreation: Sporting goods such as tennis rackets, snow skis, snowboards, water skis, surfboards, baseball bats, hockey sticks, vaulting poles, and golf clubs; numerous products having the familiar fiberglass housings and panels; boat hulls and other onboard equipment; medical systems and prosthetic devices.

Ground Transportation Equipment: Bicycle frames, wheels, and seats; automotive and truck body panels and support structures, air ducts, air bags, drive shafts, springs for high-performance sports cars and trucks, floor pans, pickup truck beds, and bumpers.

Aircraft and Aerospace Systems: Fuselage panels and internal structural elements, wings, control surfaces (ailerons, spoilers, tails, rudders), floor systems, engine cowls and nacelles, landing gear doors, cargo compartment structure and fittings, interior sidewalls, trim, partitions, ceiling panels, dividers, environmental control system ducting, stowage bins, lavatory structure systems and fixtures, airfoils (blades) in the compressor section of turbine engines, rocket nozzles, helicopter rotors, propellers, and onboard tanks for storing water and wastewater.

Industrial Facilities: Storage tanks and pressure vessels for chemical, agricultural, and petroleum processing, piping for chemicals and through corrosive environments, septic systems, wastewater treatment facilities, chemical cleaning and plating systems, pulp and paper making equipment, portable tanks for trucks and railroad applications, environmental treatment equipment, protective clothing and helmets, food processing and storage systems, mining, and material handling systems.

Electrical and Electronic Systems: Printed circuit boards, printed wiring boards, surface mount cards, packaging of electronic components, and switching system components.

Building Construction: Structural shapes, exterior panels, roofing and decking systems, doors, window frames, equipment housings, gutters and downspouts, cooling towers, bridges and walkways, piping systems, and ductwork.

Classifications of Composite Materials by Matrix

One method of classifying composite materials is by the type of matrix material. Three general classifications are used as described next, along with typical matrix materials, uses, and matrix-reinforcement combinations.

Polymer Matrix Composites (PMC)

Thermoplastics: Polyethylenes, Polyamides (nylons), polystyrenes, polypropylenes, polycarbonates, polyetheretherketones (PEEK), polyphenylene sulfides (PPS), polyvinyl chloride (PVC)

Thermosets: Polyesters, epoxies, phenolics, polyimides (PI), vinyl esters, silicones

PMCs are used for their high strength and stiffness, low density, and relatively low cost in aerospace, automotive, marine, chemical, electrical, and sporting applications. Common PMC composites include polyester-glass (conventional fiber-glass), epoxy-glass, polyimide-glass, epoxy-aramid, epoxy-carbon, PEEK-carbon, and PPS-carbon.

Metal Matrix Composites (MMC) Aluminum (Al), titanium (Ti), magnesium (Mg), iron (Fe), copper (Cu), nickel (Ni), and alloys of these metals with themselves and with molybdenum (Mo), cesium (Ce), boron (B).

MMCs are preferred for high strength, high stiffness, abrasion resistance, dimensional stability, electrical and thermal conductivity, ability to operate in high temperatures, and toughness and are applied typically in aerospace and engine applications. Examples of MMC composites include Al-SiC (silicon carbide), Ti-SiC, Al-B, Al-C (carbon), Al-graphite, Mg-SiC, and Al-Al₂O₃ (aluminum oxide).

Ceramic Matrix Composites (CMC) Silicon carbide, silicon nitride, alumina, zirconia, glass-ceramic, glass, carbon, graphite.

CMCs are preferred for high strength, high stiffness, high fracture toughness relative to ceramics alone, ability to operate at high temperatures, and low thermal expansion and are attractive for furnaces, engines, and aerospace applications. Common CMC composites include carbon-carbon (C-C), silicon carbide-carbon (SiC-C), silicon carbide-silicon carbide (SiC-SiC), glass ceramic-silicon carbide, silicon carbide-lithium aluminosilicate (SiC-LAS), and silicon carbide-calcium aluminosilicate (SiC-CAS). Where the same basic material is listed as both the matrix and the reinforcement, the reinforcement is of a different form such as whiskers, chopped fibers, or strands to achieve the preferred properties.

Forms of Reinforcement Materials

Many forms of reinforcement materials are used as listed here.

- Continuous fiber strand consisting of many individual filaments bound together.
- Chopped strands in short lengths (0.75 to 50 mm or 0.03 to 2.00 in).
- Chopped longer strands randomly spread in the form of a mat.
- Chopped longer strands aligned with the principal directions of the load path.
- Roving: A group of parallel strands.
- Woven fabric made from roving or strands.
- Braided sleeves.
- Metal filaments or wires.
- Metal, glass, or mica flakes.
- Single crystal whiskers of materials such as graphite, silicon carbide, and copper.

Types of Reinforcement Materials

Reinforcement fibers come in many types based on both organic and inorganic materials. The following are some of the more popular types:

- Glass fibers in five different types:
 - A-glass: Good chemical resistance because it contains alkalis such as sodium oxide.
 - C-glass: Special formulations for even higher chemical resistance than A-glass.
 - E-glass: Widely used glass with good electrical insulating ability and good strength.
 - S-glass: High-strength, high-temperature glass.
 - D-glass: Better electrical properties than E-glass.
- Quartz fibers and high-silica glass: Good properties at high temperatures up to 2000°F (1095°C).
- Carbon fibers made from PAN-base carbon (PAN is polyacrylonitrile): Approximately 95% carbon with very high modulus of elasticity.
- Graphite fibers: Greater than 99% carbon and an even higher modulus of elasticity than carbon; the stiffest fibers typically used in composites.
- Boron coated onto tungsten fibers: Good strength and a higher modulus of elasticity than glass.
- Silicon carbide coated onto tungsten fibers: Strength and stiffness similar to those of boron/tungsten, but with higher temperature capability.
- Aramid fibers: A member of the polyamide family of polymers; higher strength and stiffness with lower density as compared with glass; very flexible. (Aramid fibers produced by the DuPont Company carry the name *Kevlar*™.)

Processing of Composites

One method that is frequently used to produce composite products is first to place layers of sheet-formed fabrics on a form having the desired shape and then to impregnate the fabric with wet resin. Each layer of fabric can be adjusted in its orientation to produce special properties of the finished article. After the lay-up and resin impregnation are completed, the entire system is subjected to heat and pressure while a curing agent reacts with the base resin to produce cross-linking that binds all of the elements into a three-dimensional, unified structure. The polymer binds to the fibers and holds them in their preferred position and orientation during use.

An alternative method of fabricating composite products starts with a process of preimpregnating the fibers with the resin material to produce strands, tape, braids, or sheets. The resulting form, called a *pregreg*, can then be stacked into layers or wound onto a form to produce the desired shape and thickness. The final step is the curing cycle as described for the wet process. See Internet sites 23, 24, and 31.

Injection molding and extrusion can be used for composites made from thermoplastics in conventional plastics processing equipment, provided the reinforcement fibers are short so the material can flow properly. Glass fiber reinforcement is most commonly used.

Sheet molding compounds (SMC) are used in compression molding of panels and shells used for automotive body panels, industrial and business machine cabinets, watercraft, and electrical parts. Complex three-dimensional products can be made in final form or near-net-shape using SMC materials. Glass reinforcement fibers [typically in lengths from 1/2 in to 1.0 in (12 mm to 25 mm) and from 10% to 60% volume fraction] are randomly mixed with a polymer resin (often polyester), catalysts, colorants, thickeners, release agents, and inert fillers. Plastic films of nylon or polyethylene cover the mixture top and bottom as a sandwich and the combination is rolled to the desired thickness. At the time of use, the sheets are placed into a mold and shaped and cured simultaneously under heat and pressure. The resulting product has good strength and stiffness properties and excellent surface appearance. See Internet site 32. Where higher mechanical properties are required, SMC sheets can be made from an epoxy matrix and carbon fiber reinforcements. (See Internet sites 31 and 32.)

Pultrusion is a process in which the fiber reinforcement is coated with resin as it is pulled through a heated die to produce a continuous form in the desired shape. This process is used to produce rod, tubing, structural shapes (I-beams, channels, angles, and so on), tees, and hat sections used as stiffeners in aircraft structures, truck and trailer bodies, and similar large structures.

Filament winding is used to make pipe, pressure vessels, rocket motor cases, instrument enclosures, and other cylindrical containers. The continuous filament can be placed in a variety of patterns, including helical, axial, and circumferential, to produce desired strength and stiffness characteristics.

Selection of Composite Materials

Selection of which matrix, reinforcement, and processing method to use for a component made from composites is indeed a complex process. However, some generalizations can be made depending on two major design criteria, *cost* and *performance*.

When *low cost* dominates the design criteria, fiberglass is the predominant choice. Common fiberglass made from thermosetting polyester resin and glass fiber reinforcement is most often used in boats, piping, housings, canisters, and structural components.

For *high-volume applications* that can be made from injection molding or extrusion, low-cost designs can be made using thermoplastics such as polyamides and polycarbonates with short glass reinforcements. Higher strength or stiffness properties may result when using PEEK, PPS, or PVC.

When *high performance* is required, carbon fiber in an epoxy matrix is the predominant choice. Typical applications include aircraft structural components; high-end sporting goods such as tennis rackets, golf clubs, and fishing poles; and race car components.

See also the discussion of *Materials Selection* in this chapter.

Advantages of Composites

Designers typically seek to produce products that are safe, strong, stiff, lightweight, and highly tolerant of the environment in which the product will operate. Composites often excel in meeting these objectives when compared with alternative materials such as metals, wood, and unreinforced plastics. Two parameters that are used to compare materials are *specific strength* and *specific modulus*, defined as follows:

Specific strength is the ratio of the tensile strength of a material to its specific weight.

Specific modulus is the ratio of the modulus of elasticity of a material to its specific weight.

Because the modulus of elasticity is a measure of the stiffness of a material, the specific modulus is sometimes called *specific stiffness*.

Although obviously not a length, both of these quantities have the *unit* of length, derived from the ratio of the units for strength or modulus of elasticity and the units for specific weight. In the U.S. Customary System, the units for tensile strength and modulus of elasticity are lb/in², whereas specific weight (weight per unit volume) is in lb/in³. Thus, the unit for specific strength or specific modulus is inches. In SI units, strength and modulus are expressed in N/m² (pascals), whereas specific weight is in N/m³. Then the unit for specific strength or specific modulus is meters.

Table 2–17 gives comparisons of the specific strength and specific stiffness of selected composite materials with certain steel, aluminum, and titanium alloys. Figure 2–23 shows a comparison of these materials using bar charts. Figure 2–24 is a plot of these data with specific strength on the vertical axis and specific modulus on the horizontal axis. When weight is critical, the ideal material will lie in the upper-right part of this chart. Note that data in these charts and figures are for composites having the reinforcement materials aligned in the most favorable direction to withstand the applied loads.

Advantages of composites can be summarized as follows:

1. Specific strengths for composite materials can range as high as five times those of high-strength steel alloys. See Table 2–17 and Figures 2–23 and 2–24.
2. Specific modulus values for composite materials can be as high as eight times those for steel, aluminum, or titanium alloys. See Table 2–17 and Figures 2–23 and 2–24.
3. Composite materials typically perform better than steel or aluminum in applications where cyclic loads can lead to the potential for fatigue failure.
4. Where impact loads and vibrations are expected, composites can be specially formulated with materials that provide high toughness and a high level of damping.

TABLE 2-17 Comparison of Specific Strength and Specific Modulus for Selected Materials

Material	Modulus of elasticity, E		Tensile strength, s_u		Specific weight, γ		Specific strength		Specific modulus	
	(psi)	(GPa)	(ksi)	(MPa)	(lb/in ³)	(kN/m ³)	(in)	(m)	(in)	(m)
Metals										
Steel										
SAE 1020 HR	30.0×10^6	207	55	379	0.283	76.8	0.194×10^6	0.494×10^4	1.06×10^8	2.69×10^6
SAE 5160 OQT 700	30.0×10^6	207	263	1813	0.283	76.8	0.929×10^6	2.36×10^4	1.06×10^8	2.69×10^6
Aluminum										
6061-T6	10.6×10^6	69.0	45	310	0.100	27.1	0.450×10^6	1.14E04	1.00×10^8	2.54E + 06
7075-T6	10.4×10^6	71.7	83	572	0.100	27.1	0.830×10^6	2.11E04	1.04×10^8	2.64E + 06
Titanium										
Ti-6Al-4V	16.5×10^6	114	160	1103	0.160	43.4	1.00×10^6	2.54×10^4	1.03×10^8	2.62×10^6
Composites										
Glass/epoxy	4.00×10^6	28	114	786	0.061	16.6	1.87×10^6	4.75×10^4	0.656×10^8	1.67×10^6
34% fiber content										
Aramid/epoxy	11.0×10^6	76	200	1379	0.050	13.6	4.00×10^6	10.2×10^4	2.20×10^8	5.59×10^6
60% fiber content										
Boron/epoxy	30.0×10^6	207	270	1862	0.075	20.4	3.60×10^6	9.14×10^4	4.00×10^8	10.2×10^6
60% fiber content										
Graphite/epoxy	19.7×10^6	136	278	1917	0.057	15.5	4.88×10^6	12.4×10^4	3.46×10^8	8.78×10^6
62% fiber content										
Graphite/epoxy	48.0×10^6	331	160	1103	0.058	15.7	2.76×10^6	7.01×10^4	8.28×10^8	21.0×10^6
Ultrahigh modulus										

- Some composites have much higher wear resistance than metals.
- Careful selection of the matrix and reinforcement materials can provide superior corrosion resistance.
- Dimensional changes due to changes in temperature are typically much less for composites than for metals.
- Because composite materials have properties that are highly directional, designers can tailor the placement of reinforcing fibers in directions that provide the required strength and stiffness under the particular loading conditions to be encountered.
- Composite structures can often be made in complex shapes in one piece, thus reducing the number of parts in a product and the number of fastening operations required. The elimination of joints typically improves the reliability of such structures as well.
- Composite structures are typically made in their final form directly or in a near-net shape, thus reducing the number of secondary operations required.

Limitations of Composites

Designers must balance many properties of materials in their designs while simultaneously considering manufacturing operations, costs, safety, life, and service of the product. Listed next are some of the major concerns when using composites.

- Material costs for composites are typically higher than for many alternative materials.
- Fabrication techniques are quite different from those used to shape metals. New manufacturing equipment may be required, along with additional training for production operators.
- The performance of products made from some composite production techniques is subject to a wider range of variability than the performance of products made from most metal fabrication techniques.
- The operating temperature limits for composites having a polymeric matrix are approximately 500°F (260°C). (But ceramic or metal matrix composites

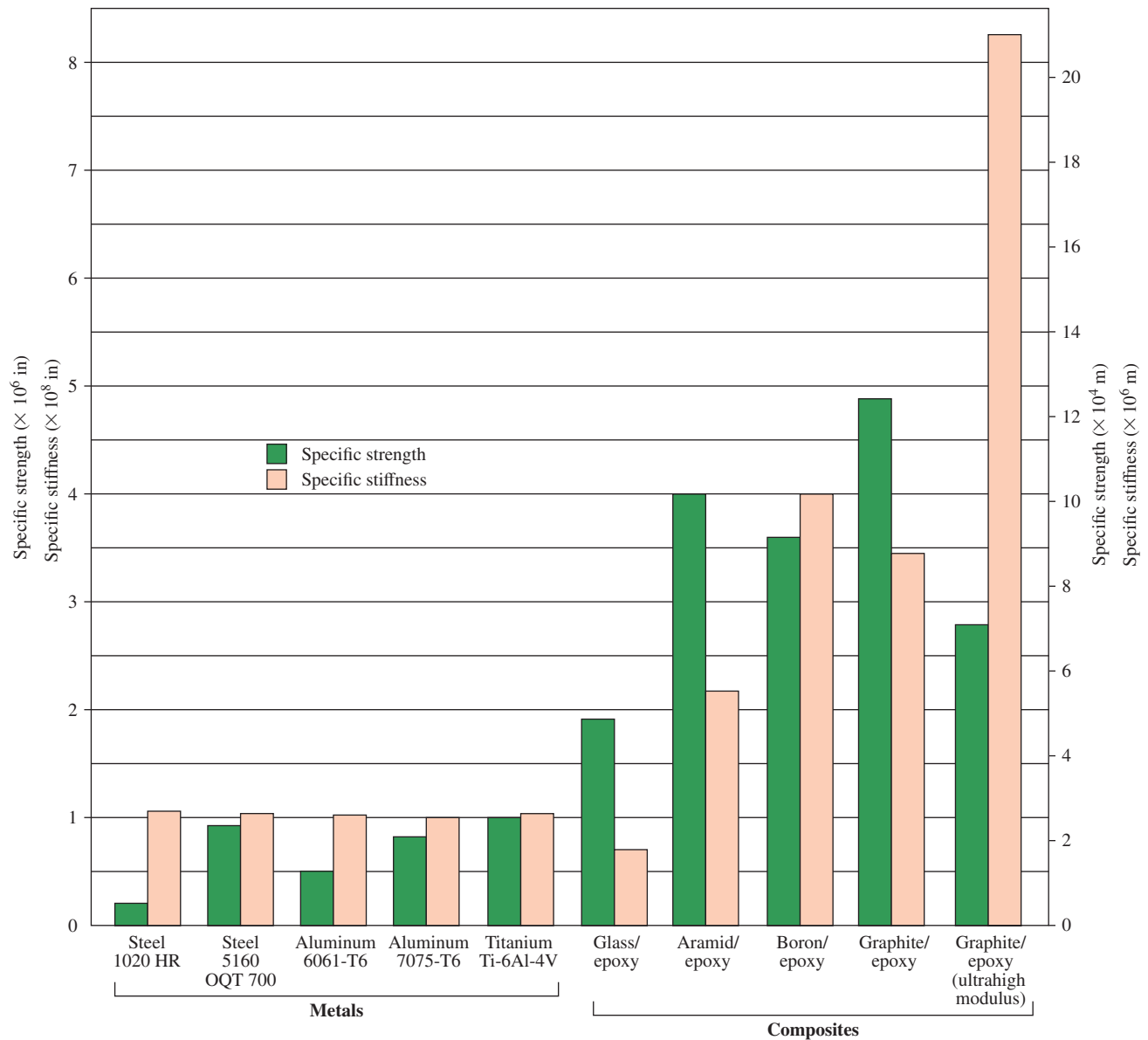


FIGURE 2-23 Comparison of specific strength and specific stiffness for selected metals and composites

can be used at higher temperatures such as those found in engines.)

5. The properties of composite materials are not isotropic: Properties vary dramatically with the direction of the applied loads. Designers must account for these variations to ensure safety and satisfactory operation under all expected types of loading.
6. At this time, many designers lack understanding of the behavior of composite materials and the details of predicting failure modes. Whereas major advancements have been made in certain industries such as the aerospace and recreational equipment fields, there is a need for more general understanding about designing with composite materials.
7. The analysis of composite structures requires detailed knowledge of more properties of the materials than would be required for metals.
8. Inspection and testing of composite structures are typically more complicated and less precise than for metal structures. Special nondestructive techniques may be required to ensure that there are no major voids in the final product that could seriously weaken the structure. Testing of the complete structure may be required rather than testing of simply a sample of the material because of the interaction of different parts on each other and because of the directionality of the material properties.
9. Repair and maintenance of composite structures are serious concerns. Some of the initial production techniques require special environments of temperature and pressure that may be difficult to reproduce in the field when damage repair is required. Bonding of a repaired area to the parent structure may also be difficult.

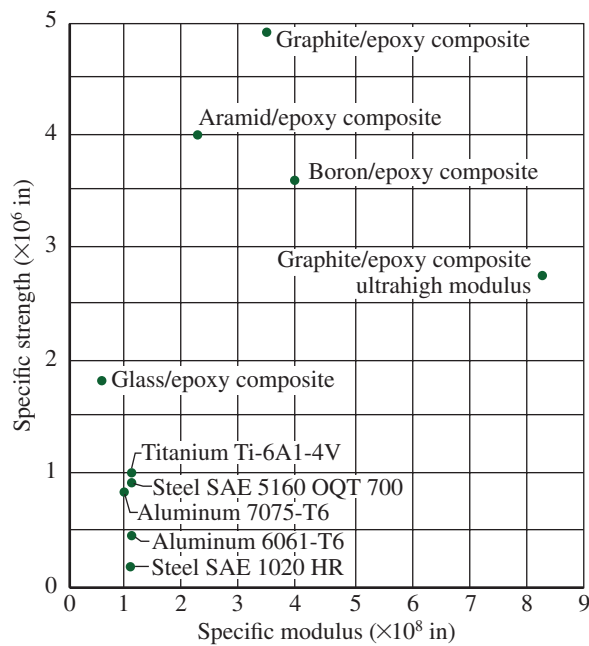


FIGURE 2-24 Specific strength versus specific modulus for selected metals and composites

Laminated Composite Construction

Many structures using composite materials are made from several layers of the basic material containing both the matrix and the reinforcing fibers. The manner in which the layers are oriented relative to one another affects the final properties of the completed structure.

As an illustration, consider that each layer is made from a set of parallel strands of the reinforcing filler material, such as E-glass fibers, embedded in the resin matrix, such as polyester. As mentioned previously, in this form, the material is sometimes called a *prepreg*, indicating that the reinforcement has been preimpregnated with the matrix prior to the forming of the structure and the curing of the assembly. To produce the maximum strength and stiffness in a particular direction, several layers or plies of the prepreg could be laid on top of one another with all of the fibers aligned in the direction of the expected tensile load. This is called a *unidirectional laminate*. After curing, the laminate would have a very high strength and stiffness when loaded in

the direction of the strands, called the *longitudinal* direction. However, the resulting product would have a very low strength and stiffness in the direction perpendicular to the fiber direction, called the *transverse* direction. If any off-axis loads are encountered, the part may fail or deform significantly. Table 2-18 gives sample data for a unidirectional laminated, carbon/epoxy composite.

To overcome the lack of off-axis strength and stiffness, laminated structures should be made with a variety of orientations of the layers. One popular arrangement is shown in Figure 2-25. Naming the longitudinal direction of the surface layer the 0° ply, this structure is referred to as

$$0^\circ, 90^\circ, +45^\circ, -45^\circ, -45^\circ, +45^\circ, 90^\circ, 0^\circ$$

The symmetry and the balance of this type of layering technique result in more nearly uniform properties in two directions. The term *quasi-isotropic* is sometimes used to describe such a structure. Note that the properties perpendicular to the faces of the layered structure (through the thickness) are still quite low because fibers do not extend in that direction. Also, the strength and the stiffness in the primary directions are somewhat lower than if the plies were aligned in the same direction. Table 2-18 also shows sample data for a quasi-isotropic laminate compared with one having unidirectional fibers in the same matrix.

Laminate Orientation Code. The example shown earlier of how an eight-ply laminated composite structure, shown in Figure 2-25, is produced is convenient but that method becomes complicated and extensive for structures made from a large number of plies. The *laminate orientation code* seeks to simplify and shorten the designation for a given fabricated composite structure. Note in the previous example that the eight plies are symmetrical about the midplane between plies four and five. This is typical for many practical designs of composite structures and it is unnecessary to repeat the names of all plies. At times, two or more adjacent plies of the same orientation are used and it is simpler to indicate the number of identical plies. Some of the conventions followed in the code are:

1. The code for a given structure is enclosed within square brackets [].
2. The naming of the plies starts at the bottom, the ply that is in contact with the form tool.

TABLE 2-18 Examples of the Effect of Laminate Construction on Strength and Stiffness

Laminate type	Tensile strength				Modulus of elasticity			
	Longitudinal		Transverse		Longitudinal		Transverse	
	ksi	MPa	ksi	MPa	10^6 psi	GPa	10^6 psi	GPa
Unidirectional	200	1380	5	34	21	145	1.6	11
Quasi-isotropic	80	552	80	552	8	55	8	55

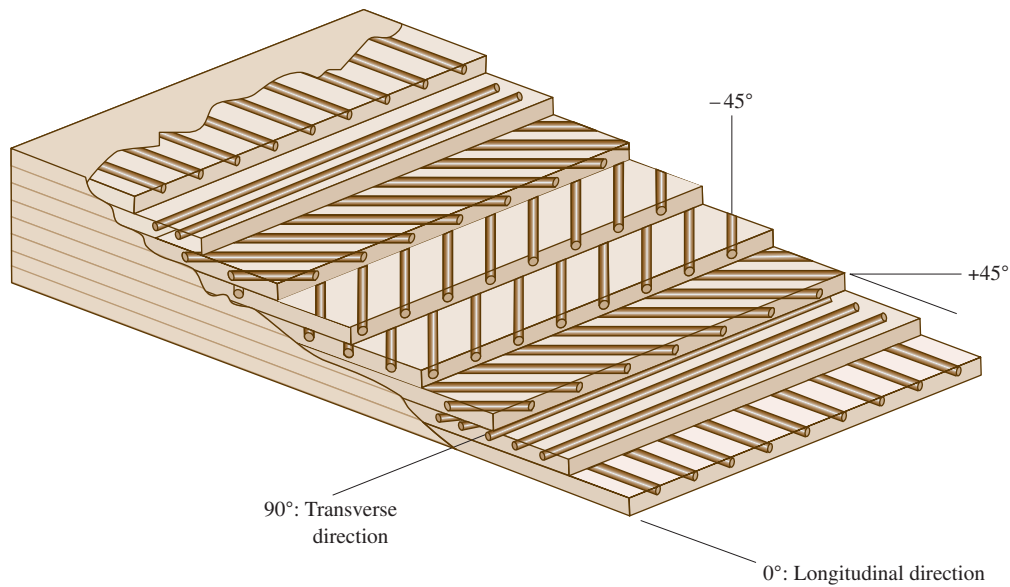


FIGURE 2-25 Multilayer, laminated, composite construction designed to produce quasi-isotropic properties

3. The degree symbol ($^{\circ}$) is not shown in the code.
4. Successive plies are separated by a forward slash (/).
5. No spaces are used.
6. When + and - plies of the same angle are adjacent, the $\pm$ symbol is used.
7. The + sign is not applied to a single ply.
8. The 0° direction is defined in the part specifications and is typically either the primary axis of the completed component or the direction of the largest applied load. Often plies with a larger number of reinforcement fibers are aligned with the 0° direction.
9. When two or more plies have the same orientation of reinforcement fibers, a subscript indicates the number of plies.
10. For symmetrical structures, only half of the plies are defined and the subscript s is added to the final bracket.
11. For a structure having an odd number of plies, but which is otherwise symmetrical, the orientation of the central ply is written with a horizontal bar over it.

Some examples are:

- (a) Symmetrical, as in Figure 2-25:
 $\blacksquare 0^{\circ}, 90^{\circ}, +45^{\circ}, -45^{\circ}, -45^{\circ}, +45^{\circ}, 90^{\circ}, 0^{\circ}$
 $\rightarrow [0/90/\pm 45]_s$
- (b) Multiple plies at same orientation:
 $\blacksquare 0^{\circ}, 0^{\circ}, 90^{\circ}, 90^{\circ}, +45^{\circ}, -45^{\circ}, -45^{\circ}, +45^{\circ}, 90^{\circ}, 90^{\circ}, 0^{\circ}, 0^{\circ}$
 $\rightarrow [0_2/90_2/\pm 45]_s$
- (c) Odd number of plies:
 $\blacksquare 0^{\circ}, 90^{\circ}, +45^{\circ}, -45^{\circ}, 90^{\circ}, -45^{\circ}, +45^{\circ}, 90^{\circ}, 0^{\circ}$
 $\rightarrow [0/90/\pm 45/\overline{90}]_s$
- (d) Nonsymmetrical:
 $\blacksquare 90^{\circ}, 90^{\circ}, +30^{\circ}, +30^{\circ}, -30^{\circ}, -30^{\circ}, 0^{\circ}, 0^{\circ}$
 $\rightarrow [90_2/30_2/-30_2/0_2]$

- (e) Repeating sets of plies with the same orientation:
 $\blacksquare 0^{\circ}, 90^{\circ}, +45^{\circ}, 0^{\circ}, 90^{\circ}, +45^{\circ}, 0^{\circ}, 90^{\circ}, +45^{\circ}$
 $\rightarrow [0/90/45]_3$

Predicting Composite Properties

The following discussion summarizes some of the important variables needed to define the properties of a composite. The subscript c refers to the composite, m refers to the matrix, and f refers to the fibers. The strength and the stiffness of a composite material depend on the elastic properties of the fiber and matrix components. But another parameter is the relative volume of the composite composed of fibers, V_f , and that composed of the matrix material, V_m . That is,

V_f = volume fraction of fiber in the composite

V_m = volume fraction of matrix in the composite

Note that for a unit volume, $V_f + V_m = 1$; thus, $V_m = 1 - V_f$.

We will use an ideal case to illustrate the way in which the strength and the stiffness of a composite can be predicted. Consider a composite with unidirectional, continuous fibers aligned in the direction of the applied load. The fibers are typically much stronger and stiffer than the matrix material. Furthermore, the matrix will be able to undergo a larger strain before fracture than the fibers can. Figure 2-26 shows these phenomena on a plot of stress versus strain for the fibers and the matrix. We will use the following notation for key parameters from Figure 2-26:

s_{uf} = ultimate strength of fiber

ϵ_{uf} = strain in the fiber corresponding to its ultimate strength

σ'_m = stress in the matrix at the same strain as ϵ_{uf}

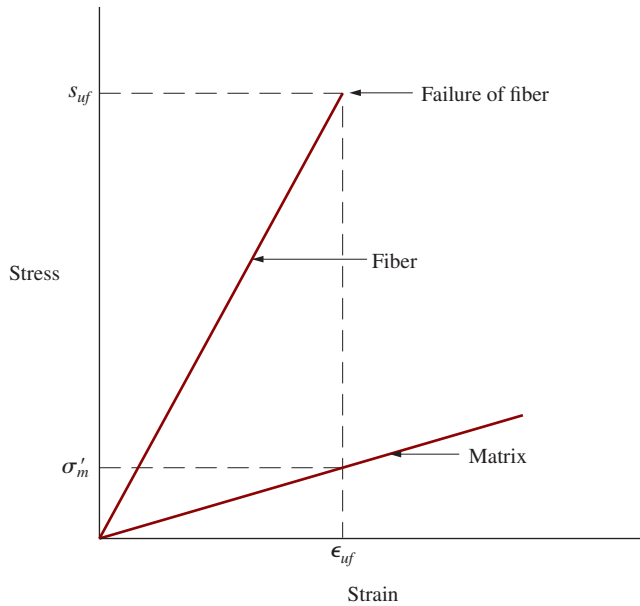


FIGURE 2-26 Stress versus strain for fiber and matrix materials

The ultimate strength of the composite, s_{uc} , is at some intermediate value between s_{uf} and σ'_m , depending on the volume fraction of fiber and matrix in the composite. That is,

➤ **Rule of Mixtures for Ultimate Strength**

$$s_{uc} = s_{uf} V_f + \sigma'_m V_m \quad (2-10)$$

At any lower level of stress, the relationship among the overall stress in the composite, the stress in the fibers, and the stress in the matrix follows a similar pattern:

➤ **Rule of Mixtures for Stress in a Composite**

$$\sigma_c = \sigma_f V_f + \sigma_m V_m \quad (2-11)$$

Figure 2-27 illustrates this relationship on a stress-strain diagram.

Both sides of Equation (2-11) can be divided by the strain at which these stresses occur. Since for each material, $\sigma/\epsilon = E$, the modulus of elasticity for the composite can be shown as

➤ **Rule of Mixtures for Modulus of Elasticity**

$$E_c = E_f V_f + E_m V_m \quad (2-12)$$

The density of a composite can be computed in a similar fashion:

➤ **Rule of Mixtures for Density of a Composite**

$$\rho_c = \rho_f V_f + \rho_m V_m \quad (2-13)$$

As mentioned previously (Section 2-2), density is defined as mass per unit volume. A related property, *specific weight*, is defined as weight per unit volume and is

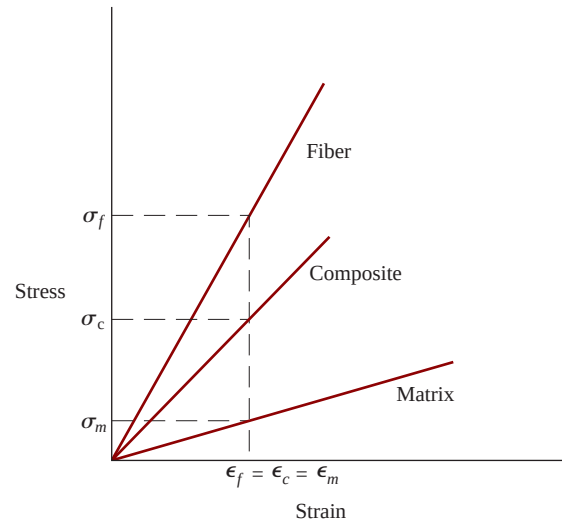


FIGURE 2-27 Relationship among stresses and strains for a composite and its fiber and matrix materials

denoted by the symbol γ (the Greek letter gamma). The relationship between density and specific weight is simply $\gamma = \rho g$, where g is the acceleration due to gravity. Multiplying each term in Equation (2-13) by g gives the formula for the specific weight of a composite:

➤ **Rule of Mixtures for Specific Weight of a Composite**

$$\gamma_c = \gamma_f V_f + \gamma_m V_m \quad (2-14)$$

The forms of Equations (2-10) through (2-14) are examples of the *rules of mixtures*.

Table 2-19 lists example values for the properties of some matrix and filler materials. Remember that wide variations can occur in such properties, depending on the exact formulation and the condition of the materials.

Design Guidelines for Members Made from Composites

The most important difference between designing with metals and designing with composites is that metals are typically taken to be homogeneous with isotropic strength and stiffness properties, whereas composites are decidedly *not* homogeneous or isotropic.

The failure modes of composite materials are complex. Tensile failure when the load is in-line with continuous fibers occurs when the individual fibers break. If the composite is made with shorter, chopped fibers, failure occurs when the fibers are pulled free from the matrix. Tensile failure when the load is perpendicular to continuous fibers occurs when the matrix itself fractures. If the fibers are in a woven form, or if a mat having shorter, randomly oriented fibers is used, other failure modes, such as fiber breakage or pullout, prevail. Such composites would have more nearly equal properties in any direction, or, as shown in Figure 2-25, multilayer laminate construction can be used.

TABLE 2-19 Example Properties of Matrix and Reinforcement Materials

	Tensile strength		Tensile modulus		Specific weight	
	ksi	MPa	10 ⁶ psi	GPa	lb/in ³	kN/m ³
Matrix materials:						
Polyester	10	69	0.40	2.76	0.047	12.7
Epoxy	18	124	0.56	3.86	0.047	12.7
Aluminum	45	310	10.0	69	0.100	27.1
Titanium	170	1170	16.5	114	0.160	43.4
Reinforcement materials:						
S-glass	600	4140	12.5	86.2	0.09	24.4
Carbon-PAN	470	3240	33.5	231	0.064	17.4
Carbon-PAN (high-strength)	820	5650	40	276	0.065	17.7
Carbon (high-modulus)	325	2200	100	690	0.078	21.2
Aramid	500	3450	19.0	131	0.052	14.1

Example Problem 2-2

Compute the expected properties of ultimate tensile strength, modulus of elasticity, and specific weight of a composite made from unidirectional strands of carbon-PAN fibers in an epoxy matrix. The volume fraction of fibers is 30%. Use data from Table 2-19.

Solution

Objective Compute the expected values of s_{uc} , E_c , and γ_c for the composite.

Given Matrix-epoxy: $s_{um} = 18$ ksi; $E_m = 0.56 \times 10^6$ psi; $\gamma_m = 0.047$ lb/in³.
 Fiber-carbon-PAN: $s_{uf} = 470$ ksi; $E_f = 33.5 \times 10^6$ psi; $\gamma_f = 0.064$ lb/in³.
 Volume fraction of fiber: $V_f = 0.30$, and $V_m = 1.0 - 0.30 = 0.70$.

Analysis and Results The ultimate tensile strength, s_{uc} , is computed from Equation (2-10):

$$s_{uc} = s_{uf}V_f + \sigma'_mV_m$$

To find σ'_m , we first find the strain at which the fibers would fail at s_{uf} . Assume that the fibers are linearly elastic to failure. Then

$$\epsilon_f = s_{uf}/E_f = (470 \times 10^3 \text{ psi})/(33.5 \times 10^6 \text{ psi}) = 0.014$$

At this same strain, the stress in the matrix is

$$\sigma'_m = E_m\epsilon = (0.56 \times 10^6 \text{ psi})(0.014) = 7840 \text{ psi}$$

Then, in Equation (2-10),

$$s_{uc} = (470\,000 \text{ psi})(0.30) + (7840 \text{ psi})(0.70) = 146\,500 \text{ psi}$$

The modulus of elasticity computed from Equation (2-12):

$$E_c = E_fV_f + E_mV_m = (33.5 \times 10^6)(0.30) + (0.56 \times 10^6)(0.70)$$

$$E_c = 10.4 \times 10^6 \text{ psi}$$

The specific weight is computed from Equation (2-14):

$$\gamma_c = \gamma_fV_f + \gamma_mV_m = (0.064)(0.30) + (0.047)(0.70) = 0.052 \text{ lb/in}^3$$

Summary of Results

$$s_{uc} = 146\,500 \text{ psi}$$

$$E_c = 10.4 \times 10^6 \text{ psi}$$

$$\gamma_c = 0.052 \text{ lb/in}^3$$

Comment Note that the resulting properties for the composite are intermediate between those for the fibers and the matrix.

Thus, an important design guideline to produce optimum strength is as follows:

Align the fibers with the direction of the load.

Another important failure mode is *interlaminar shear*, in which the plies of a multilayer composite separate under the action of shearing forces. The following is another design guideline:

Avoid or minimize shear loading, if possible.

Connections to composite materials are sometimes difficult to accomplish and provide places where fractures or fatigue failure could initiate. The manner of forming composites often allows the integration of several components into one part. Brackets, ribs, flanges, and the like, can be molded in along with the basic form of the part. The design guideline, then, is the following:

Combine several components into an integral structure.

When high panel stiffness is desired to resist flexure, as in beams or in broad panels such as floors, the designer can take advantage of the fact that the most effective material is near the outside surfaces of the panel or beam shape. Placing the high-strength fibers on these outer layers while filling the core of the shape with a light, yet rigid, material produces an efficient design in terms of weight for a given strength and stiffness. See Internet sites 25 and 26. Figure 2–28 illustrates some examples of such designs. Another design guideline follows:

Use light core material covered with strong composite layers.

Because most composites use a polymeric material for the matrix, the temperatures that they can withstand are limited. Both strength and stiffness decrease as temperature increases. The polyimides provide better high-temperature properties [up to 600°F (318°C)] than most other polymer matrix materials. Epoxies are typically limited to from 250°F to 350°F (122°C–178°C). Any application above room temperature should be checked with material suppliers. The following is a design guideline:

Avoid high temperatures.

When high temperatures [above 350°F (178°C)] must be encountered, matrix materials other than polymers should be considered, leading to this guideline:

Consider using ceramic matrix composites (CMCs) or metal matrix composites (MMCs) when high temperatures are encountered.

When using sandwich structures such as those illustrated in Figure 2–28, it is essential to have strong adhesive bonding between the various cover layers and with the core material or honeycomb to inhibit interlaminar shear or peeling. Some choices for adhesives for composites are polyester, epoxy, cyanoacrylate, polyurethane, silicone, and anaerobic adhesives (similar to thread-locking compounds). In addition, the surfaces of materials to be joined may be treated to enhance adhesion and the resulting strength of the bond.

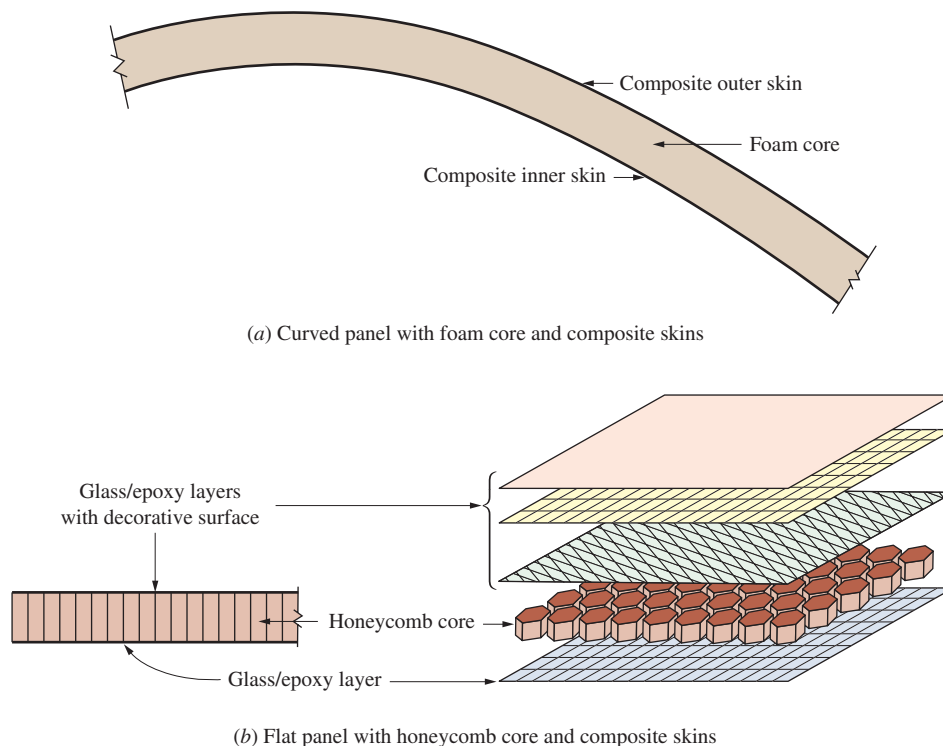


FIGURE 2–28 Laminated panels with lightweight cores

Select carefully adhesives used to bond core materials with composite surface layers and prepare properly the contact surfaces.

When a structure requires wide panels, sometimes called plates, the resulting stiffness is a function of the thickness of the wall (i.e., the number of plies in the composite) and the overall geometry. A balance can be sought by adding stiffeners to the panel to break the wide surface into a series of smaller areas. The stiffeners can be molded into the panel or applied separately with adhesives or mechanical fasteners. Three popular shapes for stiffeners are the *hat section*, *channel*, and *Z-section* that can be made of composite material using the pultrusion process or roll-formed from metal sheet. See Figure 2–29.

Consider using stiffeners on large panels to achieve adequate rigidity with minimum weight.

As described earlier in this section, many different fabrication techniques are used for composite materials. The shape can dictate a part's manufacturing technique. This is a good reason to implement the principles of concurrent engineering and adopt another design guideline:

Involve manufacturing considerations early in the design.

References 33 and 35 present broad and in-depth coverage of the design and manufacture of composite structures with emphasis on the applications of the principles. Reference 36 is one of the landmark documents defining analytical methods of design and analysis of composites.

The complete analysis of composite structures requires detailed knowledge of the mechanical and physical properties of the composite materials at the layer stage, at the finished laminate stage, during curing, at joints, at fastening points, and around complex geometric forms. Because of the anisotropic nature of composites, data for strength, modulus of elasticity in tension and shear, and Poisson's ratio in each of the three dimensions, x , y , and z are required for reliable analysis. Furthermore, many of these properties are nonlinear and dependent on detailed knowledge of loading patterns and magnitudes, and environmental conditions (temperature, moisture exposure, chemical exposure, and so forth). Dynamic response and fatigue resistance should also be analyzed.

Some of the material property data can be acquired from suppliers such as those listed in Internet sites 23–26

and 31–32. Databases and research reports from organizations such as those listed in Internet sites 19–22 and 33 are good sources of information on material properties and processing methods.

Analysis of composites structures inherently requires computer-aided design and analysis tools, including finite-element analysis, because of the special nature of the three-dimensional, nonlinear analysis required for reliable results. Several commercially available software products are tailored to composite analysis, such as those mentioned in Internet site 34.

Nanotechnology Applications in Materials. The world of materials science, engineering, and technology is rapidly changing with the commercialization of *nanotechnology*, the practice of applying tiny particles of matter. The term *nanotechnology* is derived from the measure of length as small as 1×10^{-9} meters (1.0 *nanometer* or 1.0 nm). While arbitrary, some consider a nanomaterial particle to be any with a characteristic dimension from 1 to 100 nanometers. Applying this technology is having major influence on developments in physics, biology, and chemistry in fields such as materials science, electronics, biotechnology, food processing, environmental technology, medical technology, sensors, coatings for surface protection, tire reinforcements, solar cells, highly efficient batteries and energy storage devices, adhesives, and so-called *smart materials* that change their characteristics in response to electrical stimuli, temperature, stress, strain, and so forth. See Reference 21.

One type of nanomaterial that is involved in much materials research and development is the carbon nanotube (CNT), very fine layers of graphite oriented in cylindrical form with diameters of only 1 to 3 nm and lengths from fine whiskers, platelets, and powders up to fibers 20 microns (20 μm) long. Reported strengths of CNTs range to 20 times that of high-alloy steels with approximately one-sixth the weight. Thus its strength to weight ratio is approximately 120 times higher. CNTs also have excellent electrical conductivity allowing their use in a variety of electrical and electronic systems. Another form of nanomaterial is the set of organic nanoparticles, called *dendrimers*. Some of the emerging applications for nanomaterials include:

- Creation of a series of *nanocomposites* (NCs), in which CNTs or other nanoparticles are blended with common materials to enhance their strength, stiffness, or toughness. Examples are:

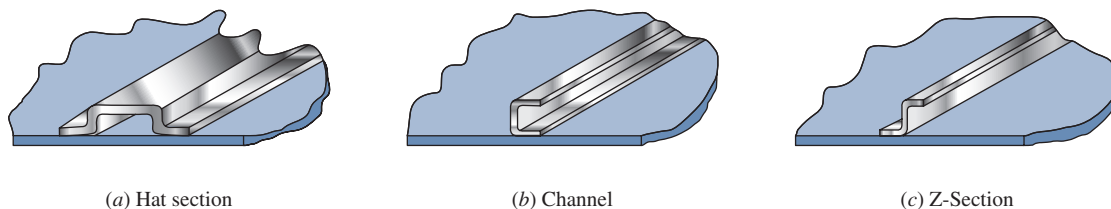


FIGURE 2–29 Panel stiffening techniques

- Adding CNTs to CMCs (ceramic matrix composites) enhances the toughness of the matrix while retaining the other desirable properties of high hardness, wear resistance, and high temperature resistance. They are called CMNCs.
- Adding CNTs to PMCs (polymer matrix composites) produce PMNCs that have enhanced matrix strength and stiffness and higher overall capabilities of the composite. An issue is the adherence of the polymers to the surface of the carbon nanotubes and the homogeneous distribution of the nanoparticles in the matrix.
- Adding CNTs to a MMCs (metal matrix composite) also enhances the strength and stiffness of the composite, which becomes a MMNC.
- Blending CNTs with common metal powders (iron, copper, magnesium, aluminum, and others) prior to sintering enhances the performance of powder metal products.
- Internet sites 35 describes high performance materials that use CNTs to enhance polymer matrix composites. PMNCs are produced as prepregs along with woven fabrics and unidirectional fibers of carbon, E-glass, S-glass, aramid, and others. Applications are in aerospace, marine, sports equipment, and motor sports.
- Internet sites 36 describes multiple applications in nanotechnology, nanomaterials, and nanoelectronics. A series of nanotools and nanodevices are produced to aid in the manipulation of matter at the nano and atomic regime for fabrication, metrology, lithography, chemical vapor deposition, 3D printing, and nanofluidics—the study of the behavior of fluids in nanoscale applications.
- Internet sites 37 describes a process for applying surface coatings of nanoparticles of aluminum oxide to provide enhanced scratch resistance of hard surfaces, plastics, architectural, and automotive applications.
- Internet sites 38 describes a process of applying patented steel alloys with microstructural grain sizes in the 10 to 100 nm size. A thermal spray technique is used to apply the coatings using atomized powder, cored wire, or electrode forms to produce coating thicknesses in the 5 to 30 mil (0.13–0.75 mm) range. Also, weld overlays in thicknesses from 1/8 in to 1/2 in (3–12 mm) are produced. Applications include coating shovel buckets, hydraulic cylinder rods, inside surfaces of piping systems that carry abrasive materials, storage bins, conveyor components, and similar products to reduce significantly wear, corrosion, and erosion.

2-18 MATERIALS SELECTION

One of the most important tasks for a designer is the specification of the material from which any individual component of a product is to be made. The decision must consider a huge number of factors, many of which have been discussed in this chapter.

The process of material selection must commence with a clear understanding of the functions and design requirements for the product and the individual component. Refer to Section 1–4 for a discussion of these concepts. Then, the designer should consider the interrelationships among the following:

- The functions of the component.
- The component's shape.
- The material from which the component is to be made.
- The manufacturing process used to produce the component.

Overall requirements for the performance of the component must be detailed. This includes, for example:

- The nature of the forces applied to the component.
- The types and magnitudes of stresses created by the applied forces.
- The allowable deformation of the component at critical points.
- Interfaces with other components of the product.
- The environment in which the component is to operate.
- Physical size and weight of the component.
- Aesthetics expected for the component and the overall product.
- Cost targets for the product as a whole and this component in particular.
- Anticipated manufacturing processes available.

A much more detailed list may be made with more knowledge of specific conditions.

From the results of the exercises described previously, you should develop a list of key material properties that are important. Examples often include:

1. Strength as indicated by ultimate tensile strength, yield strength, compressive strength, fatigue strength, shear strength, and others.
2. Stiffness as indicated by the tensile modulus of elasticity, shear modulus of elasticity, or flexural modulus.
3. Weight and mass as indicated by specific weight or density.
4. Ductility as indicated by the percent elongation.
5. Toughness as indicated by the impact energy (Izod, Charpy, etc.).
6. Creep performance data.
7. Corrosion resistance and compatibility with the environment.
8. Cost for the material.
9. Cost to process the material.

A list of candidate materials should then be created using your knowledge of the behavior of several material types, successful similar applications, and emerging materials technologies. A rational decision analysis should be applied to determine the most suitable types of materials from the list of candidates. This could take the form of a matrix in which data for the properties just listed for each candidate material are entered and ranked. An analysis of the complete set of data will aid in making the final decision.

Decision Analysis Process. An example of how decision analysis may be used to aid in deciding which material to specify in a given application is shown in Figure 2–30. The process described here is relatively simple and is sometimes called the *criteria rating process*. The decision analysis chart, sometimes called a matrix, is at the top, listing five candidate materials being considered for the production of a set of 300 wheels for use on special material handling carts. The left column lists five criteria

for evaluation. Observe also the three sets of supporting information below the matrix:

- Notes about the proposed materials:** Describes pertinent mechanical properties (from the appendix) and qualitative evaluations of the materials' corrosion resistance, cost of purchasing the materials, and costs to process the materials. With more information available, actual costs could be listed for a more detailed analysis. Note that the ratings for the processing costs for both magnesium and zinc are affected by the proposed use of die casting that entails relatively high cost for die design and fabrication for only 300 parts to be produced. The low rating for processing titanium is based on the relative difficulty of machining titanium compared with steel or aluminum.
- Application:** Gives more detail about the use of the wheels and the environment in which the carts will be operated, including the quantity to be produced, weather, and other environmental issues,

MATERIAL SELECTION—DECISION ANALYSIS CHART EXAMPLE

Desired property	Candidate materials				
	Steel	Aluminum	Magnesium	Zinc	Titanium
	Rating [1 to 10 scale]				
Yield strength (High is better)	9	6	3	7	10
Modulus of elasticity (High is better)	10	4	2	6	8
Ductility (High is better)	10	8	7	4	5
Density (Low is better)	3	8	10	5	6
Corrosion resistance (High is better)	4	10	7	7	10
Material cost (Low is better)	10	7	7	7	2
Processing cost (Low is better)	10	10	5	5	5
Totals:	56	53	41	41	46
Ranking:	1	2	4	4	3

<i>Notes about the proposed materials:</i>	s_y (MPa)	E (GPa)	% Elong.	Density (kg/m ³)	Corr. res.	Matl cost	Proc. cost
1. Steel: Wrought SAE 3140 OQT 1000	920	207	17	7680	Low	Low	Low
2. Aluminum wrought 6061-T6	276	69	12	2774	High	Moderate	Low
3. Magnesium: Die-cast AM-50	120	45	10	1770	Moderate	Moderate	High
4. Zinc: Die-cast ZA-12	320	82.7	5.5	6000	Moderate	Moderate	High
5. Titanium: Wrought Ti-6Al-4V Quenched and aged at 1000°F	1030	114	7	4432	High	Very high	High

Application: Wheel hub for four-wheel cart used in material handling within a factory environment

- Quantity: 300 pieces
- Exposed to weather in a variety of climates throughout North America
- Wheel hub; light shock loading; moderate level of load
- Must maintain good fit with solid tire made from urethane polymer

Desirable characteristics:

- High strength to survive factory conditions
- High ductility to provide high impact strength and toughness
- High stiffness to exhibit rigidity during operation
- Low weight and mass to achieve low inertia and effort to accelerate cart
- Low overall cost of acquisition and service

FIGURE 2–30 Example of a decision analysis technique to aid decision making for materials selection

general nature of the loading in regard to its level and whether shock loading is expected, and the relationship with the mating tire. This is an abbreviated list and additional detail would be expected such as actual loads to be carried, proposed size, mounting details, and so forth.

3. **Desirable characteristics:** Support the criteria for judging the proposed materials.

In the matrix, each material is rated against each desired property on a scale of 1 to 10. Admittedly, these ratings are subjective, but they are based on the data given and the nature of the application. Then the seven ratings for each material are summed and the total is listed. Finally, the sums for the five candidate materials are ranked from highest (rank = 1) to lowest.

Note that the decision analysis matrix is not a decision-making device; rather it is an aid to decision making.

It is still the responsibility of the designer to make the final decision. However, the results suggest that either steel or aluminum would perform well in this application relative to the other three suggested materials. More careful examination of the comparative ratings of steel and aluminum is recommended and perhaps additional data collection can be done. Questions can be asked such as:

- Are all the criteria of equal weight? If not, weighting factors can be applied to the list of criteria. This is commonly done.
- How can the lower corrosion resistance of the steel be managed: painting, plating, or other secondary processes?
- How do those processes affect the final overall cost for the steel wheels?
- Is the relatively low rating for modulus of elasticity for aluminum an important factor? Does it affect the final design for the wheels, perhaps by requiring thicker sections for critical features? How does this affect the final overall cost for the aluminum wheels?
- Is the lighter projected weight of the aluminum due to lower density compared with steel a strong advantage? Is this judgment firm?
- Should more detailed designs of the wheels be pursued for both steel and aluminum before committing to a final decision?

We leave the process at this point, pending answers to questions such as these, but the value of using the decision-analysis process should be evident to compare the performance of a variety of proposed materials using several parameters. See also the discussion in Chapter 1 in Sections 1–4 and 1–5 along with Chapter 1 Reference 25 and Internet Site 18.

More comprehensive materials selection processes are described in References 3–6, 25, 26, 29, and 30.

Internet sites 1, 2, 14–17, and 33 provide vast amounts of data for material properties that can be used in combination with the appendix to generate suitable candidate materials.

Other Considerations for Materials Selection. Most of the discussion in this book deals with metallic materials that are common choices for machine elements such as gears, bearings, shafts, and springs. For design in general, a wider list of types of materials from which a designer may choose for a particular application should be considered. Refer back to earlier sections of this chapter for additional information. Consider the following list of six primary classes of materials with a few examples for each listed as well:

- **Metals** Steel, aluminum, copper alloys, titanium, magnesium, cast iron, zinc
- **Polymers** PA, ABS, PC, acrylic, PVC, acetal, polyurethane, PET, polyester
- **Ceramics** Silicon nitride, zirconia, silicon carbide, alumina
- **Glasses** Silica glass, borosilicate glass, soda glass
- **Elastomers** Natural rubber, butyl rubber, neoprene, isoprene, silicones
- **Hybrids** Composites, foams, sandwich structures, honeycomb structures

The sheer number of different materials from which to choose makes material selection a daunting task. Specialized approaches, as described in References 3–6, offer significant guidance in the selection process. Furthermore, computer software packages are available to permit rapid searching and sorting based on numerous parameters to produce refined lists of candidate materials with quantitative data about their performance, cost, producibility, or other important criteria. Two examples of selection aids help to illustrate the application of this method. See Internet sites 39 for more information about one set of software packages.

When designing a machine element or a structure, both strength and stiffness must be considered and performance requirements in each area must be met. Some designs are *strength limited* (e.g., *yield strength* or *tensile strength*) while others are *stiffness limited* (e.g., *tensile modulus of elasticity* or *shear modulus of elasticity*). In addition, most design projects seek minimal or optimal weight and mass because performance of the overall system tends to be improved and there is often a strong relationship between mass and cost. For these reasons, designers should look for materials that have both high strength and low mass and also high stiffness and low mass. The material's mass is represented by density.

A good approach to meeting this design goal is to consider the *strength to density ratio* and the *stiffness to density ratio* of candidate materials. The software

mentioned in Internet sites 39 allows the determination of both ratios for virtually any material in its database and the graphic display of *strength versus density* and *stiffness versus density* charts. Figures 2–31 and 2–32 show the basic concept with general orders of magnitude indicated for comparison. The better performing materials would tend to have both high strength and low density but trade-offs are inevitable. The shaded areas show the approximate range of data for materials within a given class or type of material.

This is but a small example of the capabilities of industry-oriented material selection software packages. Additional searches and graphic charts can display almost any combination of properties such as:

- Cost of the material and its processing.
- Other mechanical properties: elongation, hardness, fatigue strength, and fracture toughness.
- Thermal properties: conductivity, specific heat, thermal expansion, and maximum service temperature.

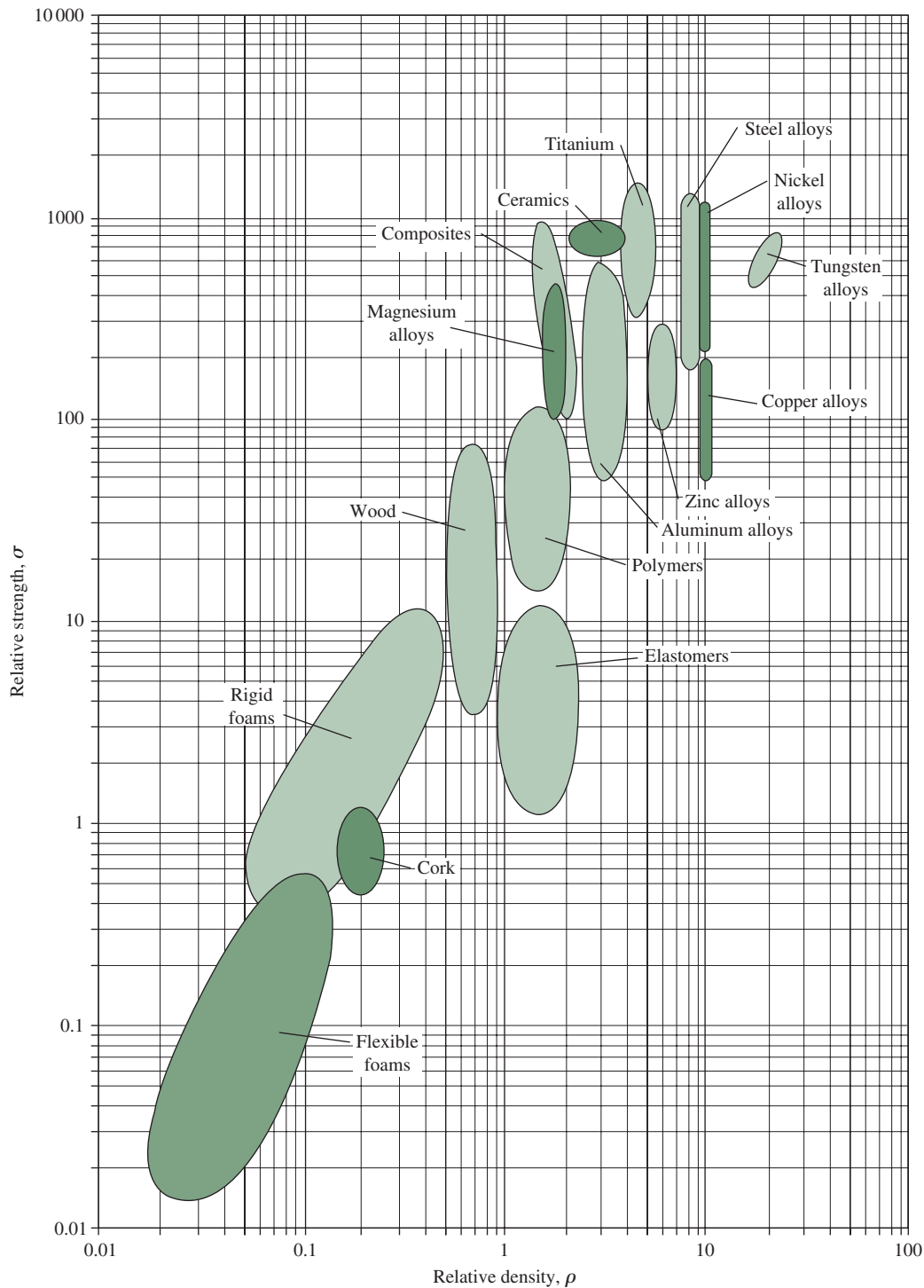


FIGURE 2–31 Relative Strength versus Relative Density

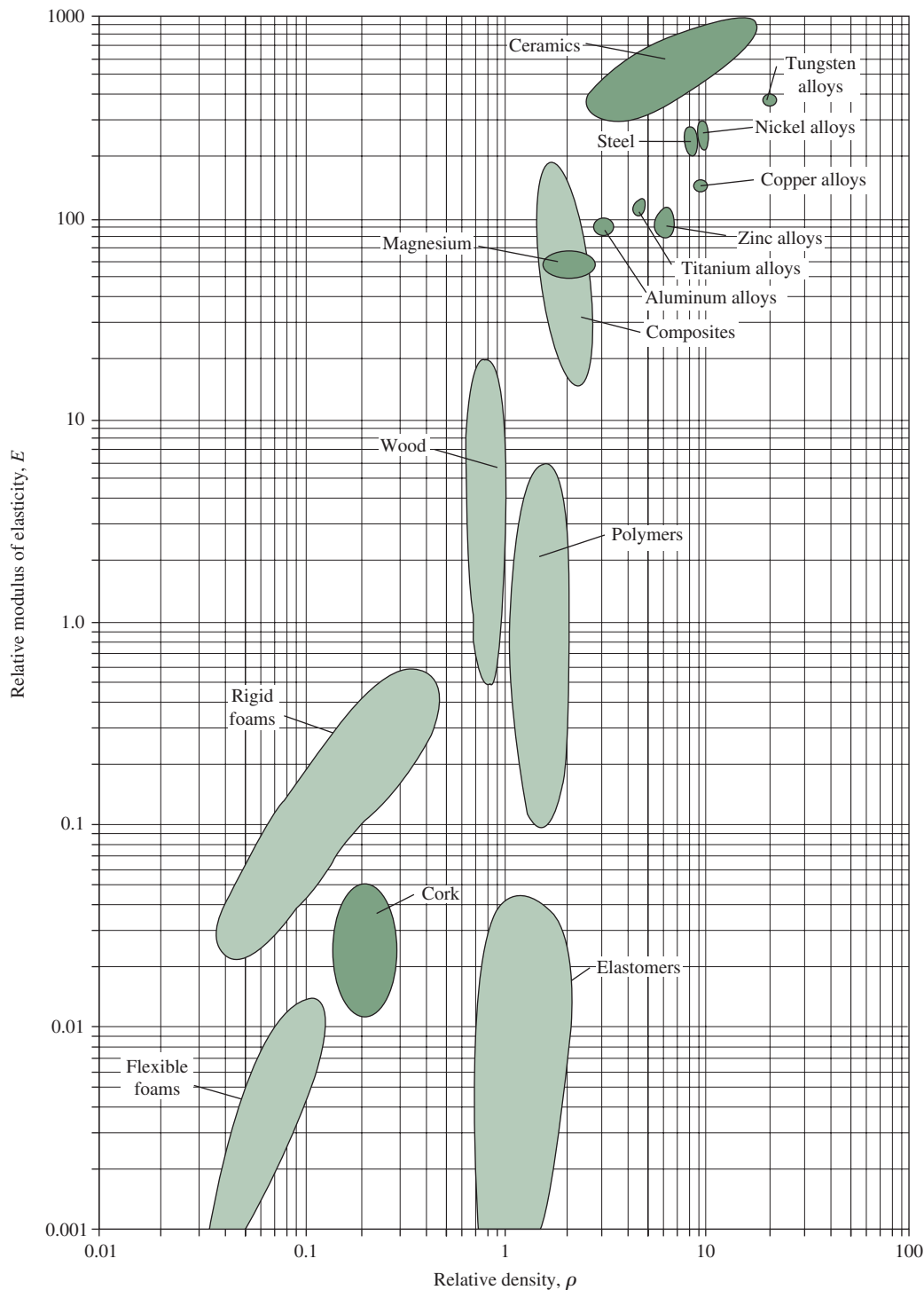


FIGURE 2-32 Relative Modulus of Elasticity versus Relative Density

- **Optical properties:** transparent or opaque?
- **Corrosion and related properties:** resistance to acids, alkalis, freshwater, seawater, organic solvents, and ultraviolet light (UV); flammability; and oxidation.
- **Electrical properties:** conductor or insulator?

Extensive tree-structures of processes appropriate to each material, within the families of *shaping*, *joining*, and *surfacing/internal structure modification*, are also part of the databases. Examples are:

- **Shaping:** Casting, machining, molding, rapid prototyping, and others.
- **Joining:** Welding, fastening, adhesives, soldering, brazing, and others.
- **Surfacing/internal structure modification:** Heat-treating, induction hardening, flame hardening, carburizing, coating, anodizing, polishing, painting, plating, and others.

Specialized searches and analyses: In addition to general searches, special modules of some software help

to focus on certain attributes and applications of materials. Among them are:

- Materials and the Environment—Eco Audits (See Reference 5 and Internet site 39)
 - Energy required to produce a material, energy to transport a product made from a particular material, and energy needed during use.
 - Emissions, disposal issues.
 - Product life cycle cost.
- Aerospace materials.
- Materials for medical devices.
- Metals and alloys.
- Polymeric materials.

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7. **Alcoa, Inc.** A producer of aluminum and fabricated products. Website can be searched for properties of specific alloys.
8. **Copper Development Association**. Provides a large database of properties of wrought and cast copper, copper alloys, brasses, and bronzes. Allows searching for appropriate alloys for typical industrial uses based on several performance characteristics. Select the *Resources* tab and then *Properties*.
9. **Metal Powder Industries Federation**. The international trade association representing the powder metal producers. Standards and publications related to the design and production of products using powder metals.
10. **INTERZINC**. A market development and technology transfer group dedicated to increasing awareness of zinc casting alloys. The site contains numerous references to aid designers in using zinc effectively, such as alloy selection, zinc casting processes, design guides, and case histories. The organization sponsors the Zinc Challenge competition for college and university students to design novel products using zinc die-casting processes.
11. **Association for Iron & Steel Technology (AIST)**. An organization that advances the technical development, production, processing, and application of iron and steel.
12. **American Iron and Steel Institute**. AISI serves as the voice of the North American steel industry, promotes the use of steel, and has played a role in the development and application of steels and steelmaking technology.
13. **Techstreet**. A part of Thompson Reuters Corporation, it provides engineers, librarians, and technical professionals with efficient, convenient, reliable access to mission-critical industry codes and standards from over 350 sources including AGMA, ASM, ANSI, ASME, ASTM, DOD, DIN, ISO, MIL, SAE, and SME.
14. **DuPont Engineering Plastics**. Information on DuPont plastics and their properties and processing. Database by type of plastic or application.
15. **PolymerPlace.com**. A web-based portal for information on the development of products and applications from plastics materials and processes.
16. **Plastics Technology Online**. The online resource for the *Plastics Technology* magazine. Information about plastics materials and processes.
17. **PLASPEC Global**. Affiliated with *Plastics Technology Online*. Provides a materials selection database, articles, and information about plastics, injection molding, extrusion, blow molding, tooling, and auxiliary equipment. Search for PLASPEC Global on the *Plastics Technology Online* website.
18. **Society of Plastics Engineers**. SPE promotes scientific and engineering knowledge and education about plastics and polymers worldwide. Site includes an online *Plastics Encyclopedia*.
19. **Composites World**. Online site for *High Performance Magazine* and *Composites Technology Magazine*. Articles on current practices in composites technology.

INTERNET SITES RELATED TO DESIGN PROPERTIES OF MATERIALS

The sites listed here provide additional details about the subject of this chapter, beyond what is included here. Most, but not all, sites are mentioned in the text. You should consider perusing this list to find sites that may help you with questions about materials whenever they arise.

1. **AZoM.com (The A to Z of Materials)**. Materials information resource for the design community. No cost, databases for metals, ceramics, polymers, and composites. Can also search by keyword, application, or industry type.
2. **Matweb**. Database of material properties for many metals, plastics, ceramics, and other engineering materials.
3. **ASM International**. The society for materials engineers and scientists, a worldwide network dedicated to advancing industry, technology, and applications of metals and other materials.
4. **SAE International**. The engineering society for advancing mobility for the benefit of mankind. A resource for technical information and standards used in designing self-propelled vehicles of all kinds. Site includes access to SAE and AMS (Aerospace Materials Standards) standards on metals, polymers, composites, and many other materials, along with components and subsystems of vehicles such as steering, suspension, engines, and many others.
5. **ASTM International**. Formerly known as the American Society for Testing and Materials. Develops and sells standards for material properties, testing procedures, and numerous other technical standards.

- with emphasis on engineering, design, and manufacturing solutions. A service of Gardner Business Media, Inc.
20. **National Center for Advanced Materials Performance (NCAMP).** Part of the National Institute for Aviation Research (NIAR) in cooperation with National Aeronautics and Space Agency (NASA) and the Federal Aviation Administration, NCAMP promotes validation and quality assurance of composite materials to be applied in the commercial and military aviation industry. The center is based at Wichita State University.
 21. **Center of Excellence for Composites & Advanced Materials (CECAM).** An academic team of universities led by Wichita State University and the NIAR (see Internet site 20) to perform research, testing, certification, and technology transfer in composites technology; to promote coordination and cooperation among the FAA, aircraft manufacturers, materials suppliers, and airline companies; and the education of the aircraft manufacturing and maintenance workforces.
 22. **Center for Advanced Materials in Transport Aircraft Structures (AMTAS).** An FAA supported center of excellence based at the University of Washington that addresses the engineering and science issues associated with safety, regulation, and product certification of advanced materials and structures. It also establishes engineering standards and develops an advanced materials and structures knowledge base.
 23. **A&P Technology.** A producer of precision braided textiles for use in the composites industry, including braided carbon fiber and fiberglass biaxial sleeveings, biaxial tapes, unidirectional tapes, and bias braided fabrics.
 24. **Cytec Solvay Group.** A provider of technologically advanced materials for composites used in commercial aerospace, military aircraft, automotive, marine, and high-performance industrial markets. Products include prepreg systems, woven fabrics, tape, roving, tooling, carbon fiber reinforcements, and thermoplastics.
 25. **Triumph Group, Inc.** Producer of composite systems used for floor panels, employing light-weight cores with fiberglass or carbon graphite skins.
 26. **General Plastics Manufacturing Company.** Producer of high-performance *Last-A-Foam*[®] cellular solid polyurethane and polyisocyanurate systems for use in light-weight cores for composite structures, other industrial applications, and graphics products.
 27. **Steel Founders' Society of America.** A trade association representing steel foundries that promotes the use of cast steel products and disseminates information about steel castings, materials, and casting processes. The site provides general overviews of castings and working with foundries.
 28. **Special Metals Corporation.** Producer of nickel-based alloys with brand names INCONEL, INCOLOY, NIMONIC, UDIMET, MONEL, and NILO.
 29. **Allegheny Technologies, Inc. (ATI).** Producer of alloys of titanium, nickel, cobalt, zirconium and other metals for aerospace, electrical energy, medical, automotive, food equipment, construction, mining and other industries.
 30. **Haynes International, Inc.** Producer of nickel- and cobalt-based alloys with brand names HASTELLOY, HAYNES, and ULTIMET.
 31. **Hexcel Corporation.** Producer of carbon fibers and matrix products for the composites industry, including prepregs, core materials, fabrics, laminates, sheet molding compounds, and systems for film infusion technology, resin transfer molding, and resin film infusion.
 32. **IDI Composites International.** A producer of sheet molding compounds and bulk molding compounds.
 33. **CINDAS LLC.** Developer of databases of materials properties: Aerospace Structural Metals Database (ASMD), Damage Tolerant Design Handbook (DTDH), Structural Alloys Handbook (SAH), Microelectronics Packaging Materials Database (MPMD), and Thermophysical Properties of Matter Database (TPMD).
 34. **Think Composites, Inc.** A consulting firm specializing in design and analysis of structures made from advanced composites. Develops and markets software and provides education and training programs in this field. Software products include GENLAM (strength and stiffness of laminates), LAMRANK (ranking of the performance of laminates), MIC-MAC (micro- and macromechanics analysis of composites), and 3DBEAM (design and analysis of three-dimensional beams).
 35. **Zyvex Technologies.** Producer of advanced high-performance materials using nanotechnology for aerospace, automotive, industrial, marine, and sports applications.
 36. **Sigma-Aldrich.** Producer of nanomaterials for the materials development market. Included are carbon nanotubes, dendrimers, nanocomposites, and others. The site includes many articles on nanomaterials and related topics.
 37. **Nanophase Technologies Corporation.** Producer of a family of nanomaterial technologies for automotive, electronics, plastics, textiles, exterior coatings, and other industries.
 38. **Nanosteel Company.** Producer of NanoSteel alloys that are applied as coatings or overlays to surfaces to enhance their hardness and resistance to wear, erosion, and corrosion. Also provides powders for additive manufacturing and 3D printing.
 39. **Granta Material Intelligence.** Producer and marketer of numerous products and services related to the effective management of materials property data and software to support materials selection in industry and academia. Included is the CES EduPack materials selection software using concepts developed by Professor Michael F. Ashby.
 40. **WorldAutoSteel.** The automotive group of the World Steel Association advances and communicates steel's unique ability to meet the automotive industry's needs and challenges in a sustainable and environmentally responsible way. Publisher of *Advanced High Strength Steel Application Guidelines* (AHSS Guidelines), available for download from the site.
 41. **Olympus Corporation.** Producer of equipment for non-destructive velocity measurements in materials to enable calculations for Young's modulus and Poisson's ratio.

PROBLEMS

1. Define *ultimate tensile strength*.
2. Define *yield point*.
3. Define *yield strength* and tell how it is measured.

4. What types of materials would have a yield point?
5. What is the difference between proportional limit and elastic limit?
6. Define *Hooke's law*.
7. What property of a material is a measure of its stiffness?
8. What property of a material is a measure of its ductility?
9. If a material is reported to have a percent elongation in a 2.00-in gage length of 2%, is it ductile?
10. Define *Poisson's ratio*.
11. If a material has a tensile modulus of elasticity of 114 GPa and a Poisson's ratio of 0.33, what is its modulus of elasticity in shear?
12. A material is reported to have a Brinell hardness of 525. What is its approximate hardness on the Rockwell C scale?
13. A steel is reported to have a Brinell hardness of 450. What is its approximate tensile strength?
For Problems 14–17, describe what is wrong with each statement.
14. "After annealing, the steel bracket had a Brinell hardness of 750."
15. "The hardness of that steel shaft is HRB 120."
16. "The hardness of that bronze casting is HRC 12."
17. "Based on the fact that this aluminum plate has a hardness of HB 150, its approximate tensile strength is 75 ksi."
18. Name two tests used to measure impact energy.
19. What are the principal constituents in steels?
20. What are the principal alloying elements in SAE 4340 steel?
21. How much carbon is in SAE 4340 steel?
22. What is the typical carbon content of a low-carbon steel? Of a medium-carbon steel? Of a high-carbon steel?
23. How much carbon does a bearing steel typically contain?
24. What is the main difference between SAE 1213 steel and SAE 12L13 steel?
25. Name four materials that are commonly used for shafts.
26. Name four materials that are typically used for gears.
27. Describe the properties desirable for the auger blades of a post hole digger, and suggest a suitable material.
28. Appendix 3 lists SAE 5160 OQT 1000. Describe the basic composition of this material, how it was processed, and its properties in relation to other steels listed in that table.
29. If a shovel blade is made from SAE 1040 steel, would you recommend flame hardening to give its edge a surface hardness of HRC 40? Explain.
30. Describe the differences between through-hardening and carburizing.
31. Describe the process of induction hardening.
32. Name 10 steels used for carburizing. What is their approximate carbon content prior to carburizing?
33. What types of stainless steels are nonmagnetic?
34. What is the principal alloying element that gives a stainless steel corrosion resistance?
35. Of what material is a typical wide-flange beam made?
36. With regard to structural steels, what does the term *HSLA* mean? What strengths are available in HSLA steel?
37. Name three types of cast iron.
38. Describe the following cast iron materials according to type, tensile strength, yield strength, ductility, and stiffness:
ASTM A48, Grade 30
ASTM A536, Grade 100-70-03
ASTM A47, Grade 32510
ASTM A220, Grade 70003
39. Describe the process of making parts from powdered metals.
40. What properties are typical for parts made from Zamak 3 zinc casting alloy?
41. What are the typical uses for Group D tool steels?
42. What does the suffix *O* in aluminum 6061-O represent?
43. What does the suffix *H* in aluminum 3003-H14 represent?
44. What does the suffix *T* in aluminum 6061-T6 represent?
45. Name the aluminum alloy and condition that has the highest strength of those listed in Appendix 9.
46. Which is one of the most versatile aluminum alloys for mechanical and structural uses?
47. Name three typical uses for titanium alloys.
48. What is the principal constituent of bronze?
49. Describe the bronze having the UNS designation C86200.
50. Name two typical uses for bronze in machine design.
51. Describe the difference between thermosetting plastics and thermoplastics.
52. Suggest a suitable plastic material for each of the following uses:
(a) Gears
(b) Football helmets
(c) Transparent shield
(d) Structural housing
(e) Pipe
(f) Wheels
(g) Electrical switch-gear, structural part
53. Name eight factors over which the designer has control when specifying a composite material.
54. Define the term *composite*.
55. Name four base resins often used for composite materials.
56. Name four types of reinforcement fibers used for composite materials.
57. Name three types of composite materials used for sporting equipment, such as tennis rackets, golf clubs, and skis.
58. Name three types of composite materials used for aircraft and aerospace structures.
59. What base resin and reinforcement are typically used for sheet-molding compound (SMC)?
60. For what applications are sheet-molding compounds used?
61. Describe six forms in which reinforcing fibers are produced.
62. Describe *wet processing* of composite materials.
63. Describe *preimpregnated materials*.
64. Describe the production processing of sheet-molding compounds.

65. Describe *pultrusion*, and list four shapes produced by this process.
66. Describe *filament winding* and four types of products made by this process.
67. Define the term *specific strength* as it is applied to structural materials.
68. Define the term *specific stiffness* as it is applied to structural materials.
69. Discuss the advantages of composite materials relative to metals with regard to specific strength and specific stiffness.
70. Compare the specific strength of SAE 1020 hot-rolled steel with that of SAE 5160 OQT 700 steel, the two aluminum alloys 6061-T6 and 7075-T6, and titanium Ti-6Al-4V.
71. Compare the specific stiffness of SAE 1020 hot-rolled steel with that of SAE 5160 OQT 700 steel, the two aluminum alloys 6061-T6 and 7075-T6, and titanium Ti-6Al-4V.
72. Compare the specific strengths of each of the five composite materials shown in Figure 2–23 with that of SAE 1020 hot-rolled steel.
73. Compare the specific stiffness of each of the five composite materials shown in Figure 2–23 with that of SAE 1020 hot-rolled steel.
74. Describe the general construction of a composite material identified as $[0/+30/-30/90]$.
75. List and discuss six design guidelines for the application of composite materials.
76. Why is it desirable to form a composite material in layers or plies with the angle of orientation of the different plies in different directions?
77. Why is it desirable to form a composite structural element with relatively thin skins of the stronger composite material over a core of light foam?
78. Describe why concurrent engineering and early manufacturing involvement are important when you are designing parts made from composite materials.
79. Describe a PMC material.
80. Describe a MMC material.
81. Describe a CMC material.

Problems 82–90. For composites made with the given matrix material and reinforcement fiber, which of the three classifications of composites mentioned in Questions 79, 80, and 81 is it in?

- | <u>Matrix material</u> | <u>Reinforcement material</u> |
|------------------------|-------------------------------|
| 82. PEEK | Carbon |
| 83. SiC | LAS |
| 84. C | C |
| 85. Mg | SiC |
| 86. PPS | C |
| 87. SiC | CAS |
| 88. Al | B |
| 89. PI | glass |
| 90. epoxy | aramid |
91. Recommend a type of composite and materials for a high-performance application such as an aircraft structural component.
 92. Recommend a type of composite and materials and a processing method for a high-volume application with low per-part cost.
 93. Name the predominant choice of composite type and

materials for low-cost applications.

Problems 94–96. For the given specification for a laminate-type composite, describe the layers.

94. $[0/\pm 45/\pm 30]_s$
95. $[90_3/\pm 45/0_2]_s$
96. $[0_4/90_2/\pm 30/0_2/90/\pm 60/0_2]$
97. Describe a nanometer.
98. Describe a CNT.
99. Describe how CNTs are used in a MMNC and what benefits can be expected compared with a MMC.
100. Describe how CNTs are used in a CMNC and what benefits can be expected compared with a CMC.

SUPPLEMENTARY PROBLEMS

1. List the approximate value of Poisson's ratio for the following materials: (a) carbon and alloy steel; (b) aluminum; (c) lead; (d) gray cast iron; (e) concrete; (f) elastomers.
2. Describe and compare the methods for measuring flexural strength and flexural modulus for rigid plastics using the 3-point bending method and the 4-point bending method.
3. List five kinds of wear.
4. Name four types of carbon and alloy steel commonly used for commercially available shapes.
5. Name two types of stainless steel commonly used for commercially available shapes.
6. Name four types of aluminum alloys commonly used for commercially available shapes.
7. Name three U.S. organizations whose names are commonly used in designations for steels used in machine design.
8. Name the U.S. organization chiefly responsible for assigning aluminum alloy designations.
9. A U.S. designer specifies SAE 4140 steel for a machine component. Name the alloy designation if the same component were to be produced in: (a) Germany; (b) the United Kingdom; (c) the European Union; (d) China; (e) Japan.
10. Repeat Problems 9 for SAE 1045 steel.
11. Repeat Problems 9 for SAE 430 stainless steel.
12. Repeat Problems 9, parts a–c, for aluminum alloy 7075.
13. List four liquid media used as quenchants during heat-treating operations.
14. Describe the process of shot peening and the benefits of using it.
15. List four carbon and alloy casting steels used in general applications.
16. List four carbon and alloy casting steels used for parts that must sustain pressurized fluids.
17. Describe the cast material called *CADI* and list four typical applications.
18. Describe the cast material called *white iron* and list four typical applications.
19. Describe the *press and sinter* procedure for producing machine components from powder metallurgy technology.
20. Describe the *isostatic pressing* procedure for producing machine components from powder metallurgy technology.
21. Describe the *metal injection molding* procedure for producing machine components from powder metallurgy technology.

22. Describe the *powder forging* procedure for producing machine components from powder metallurgy technology.
23. List four carbon or alloy powdered steel designations and give their tensile strength rating in the heat-treated condition.
24. List powdered metal designations for the following types of metals and give their tensile strength rating: (a) nickel silver; (b) bronze; (c) copper; (d) aluminum.
25. State the typical limit on part size for powder metal parts?
26. Name at least two aluminum casting alloys in each of the series: 200, 300, 400, 500, 700, and 800.
27. Name three general-purpose aluminum forging alloys.
28. Name the most widely used zinc casting alloy and give its typical tensile strength.
29. Name the three widely used ZA zinc alloys and describe the significance of the A in the designation name. For each named alloy, give its typical tensile strength.
30. Give two major reasons for specifying nickel-based alloys for a machine component.
31. Specify at least one brass or bronze alloy for the following applications: (a) bearings; (b) gears; (c) screw machine parts; (d) marine hardware; (e) pump housings; (f) aircraft parts.
32. Describe the conditions for copper alloys designated by: (a) H04; (b) H02; (c) H01; d) H08.
33. Describe what the temper designation TD means.
34. Using Figure 2-18, describe the effect of the degree of cold work between 10% and 40% on the properties of brass alloys.
35. List the six general classifications of materials that designers may specify for machine components.
36. Give three examples of hybrid materials.
37. Using Figure 2-31, rank the following general classifications of materials with regard to their relative strength: (a) natural materials (wood, rubber); (b) ceramics; (c) polymers; (d) metals; (e) cork; (f) foams; (g) composites; (h) elastomers.
38. Using Figure 2-32, rank the following general classifications of materials with regard to their relative stiffness as indicated by its Young's modulus: (a) natural materials (wood, rubber); (b) ceramics; (c) polymers; (d) metals; (e) cork; (f) foams; (g) composites; (h) elastomers.
39. Using Figure 2-32, rank the following general classifications of materials with regard to their relative density: (a) natural materials (wood, rubber); (b) ceramics;

(c) polymers; (d) metals; (e) cork; (f) foams; (g) composites; (h) elastomers.

40. Compare the three lists generated for Problems 37, 38, and 39.

INTERNET-BASED ASSIGNMENTS

1. Use the Matweb website to determine at least three appropriate materials for a shaft design. An alloy steel is preferred with a minimum yield strength of 150 ksi (1035 MPa) and a good ductility as represented by an elongation of 10% or greater.
2. Use the Matweb website to determine at least three appropriate plastic materials for use as a cam. The materials should have good strength properties and a high toughness.
3. Use the DuPont Plastics website to determine at least three appropriate plastic materials for use as a cam. The materials should have good strength properties and a high toughness.
4. Use the DuPont Plastics website to determine at least three appropriate plastic materials for use as a housing for an industrial product. Moderate strength, high rigidity, and high toughness are required.
5. Use the Alcoa website to determine at least three appropriate aluminum alloys for a mechanical component that requires moderate strength, good machinability, and good corrosion resistance.
6. Use the INTERZINC website to determine at least three appropriate zinc casting alloys for a structural component that requires good strength and that is recommended for die casting.
7. Use the Copper Development Association website to recommend at least three copper alloys for a wormgear. Good strength and ductility are desirable along with good wear properties.
8. Use the Copper Development Association website to recommend at least three copper alloys for a bearing application. Moderate strength and good friction and wear properties are required.
9. Locate the description of the ASTM Standard A992 structural steel that is commonly used for rolled-steel beam shapes. Determine how to acquire a copy of the standard.

STRESS AND DEFORMATION ANALYSIS

The Big Picture

You Are the Designer

- 3-1 Objectives of This Chapter
- 3-2 Philosophy of a Safe Design
- 3-3 Representing Stresses on a Stress Element
- 3-4 Normal Stresses Due to Direct Axial Load
- 3-5 Deformation under Direct Axial Load
- 3-6 Shear Stress Due to Direct Shear Load
- 3-7 Torsional Load—Torque, Rotational Speed, and Power
- 3-8 Shear Stress Due to Torsional Load
- 3-9 Torsional Deformation
- 3-10 Torsion in Members Having Non-Circular Cross Sections
- 3-11 Torsion in Closed Thin-Walled Tubes
- 3-12 Torsion in Open Thin-Walled Tubes
- 3-13 Shear Stress Due to Bending
- 3-14 Shear Stress Due to Bending—Special Shear Stress Formulas
- 3-15 Normal Stress Due to Bending
- 3-16 Beams with Concentrated Bending Moments
- 3-17 Flexural Center for Beam Bending
- 3-18 Beam Deflections
- 3-19 Equations for Deflected Beam Shape
- 3-20 Curved Beams
- 3-21 Superposition Principle
- 3-22 Stress Concentrations
- 3-23 Notch Sensitivity and Strength Reduction Factor

THE BIG PICTURE

Stress and Deformation Analysis

Discussion Map

- As a designer, you are responsible for ensuring the safety of the components and systems you design.
- You must apply your prior knowledge of the principles of strength of materials.

Discover

*How could consumer products and machines fail?
Describe some product failures you have seen.*

This chapter presents a brief review of the fundamentals of stress analysis. It will help you design products that do not fail, and it will prepare you for other topics later in this book.

A designer is responsible for ensuring the safety of the components and systems that he or she designs. Many factors affect safety, but one of the most critical aspects of design safety is that the level of stress to

which a machine component is subjected must be safe under reasonably foreseeable conditions. This principle implies, of course, that nothing actually breaks. Safety may also be compromised if components are

permitted to deflect excessively, even though nothing breaks.

You have already studied the principles of strength of materials to learn the fundamentals of stress analysis. Thus, at this point, you should be competent to analyze load-carrying members for stress and deflection due to direct tensile and compressive loads, direct shear, torsional shear, and bending.

Think, now, about consumer products and machines with which you are familiar, and try to explain how they *could fail*. Of course, we do not expect them to fail, because most such products are well designed. But some do fail. Can you recall any? How did they fail? What were the operating conditions when they failed? What was the material of the components that failed? Can you visualize and describe the kinds of loads that were placed on the components that failed? Were they subjected to bending, tension, compression, shear, or torsion? Could

there have been more than one type of stress acting at the same time? Are there evidences of accidental overloads? Should such loads have been anticipated by the designer? Could the failure be due to the manufacture of the product rather than its design?

Talk about product and machine failures with your associates and your instructor. Consider parts of your car, home appliances, lawn maintenance equipment, or equipment where you have worked. If possible, bring failed components to the meetings with your associates, and discuss the components and their failure.

Most of this book emphasizes developing special methods to analyze and design machine elements. These methods are all based on the fundamentals of stress analysis, and it is assumed that you have completed a course in strength of materials. This chapter presents a brief review of the fundamentals. (See References 2–4.)

YOU ARE THE DESIGNER

You are the designer of a utility crane that might be used in an automotive repair facility, in a manufacturing plant, or on a mobile unit such as a truck bed. Its function is to raise heavy loads.

A schematic layout of one possible configuration of the crane is shown in Figure 3–1. It is comprised of four primary load-carrying members, labeled 1, 2, 3, and 4. These members are connected to each other with pin-type joints at A, B, C, D, E, and F. The load is

applied to the end of the horizontal boom, member 3. Anchor points for the crane are provided at joints A and B that carry the loads from the crane to a rigid structure. Note that this is a simplified view of the crane showing only the primary structural components and the forces in the plane of the applied load. The crane would also need stabilizing members in the plane perpendicular to the drawing.

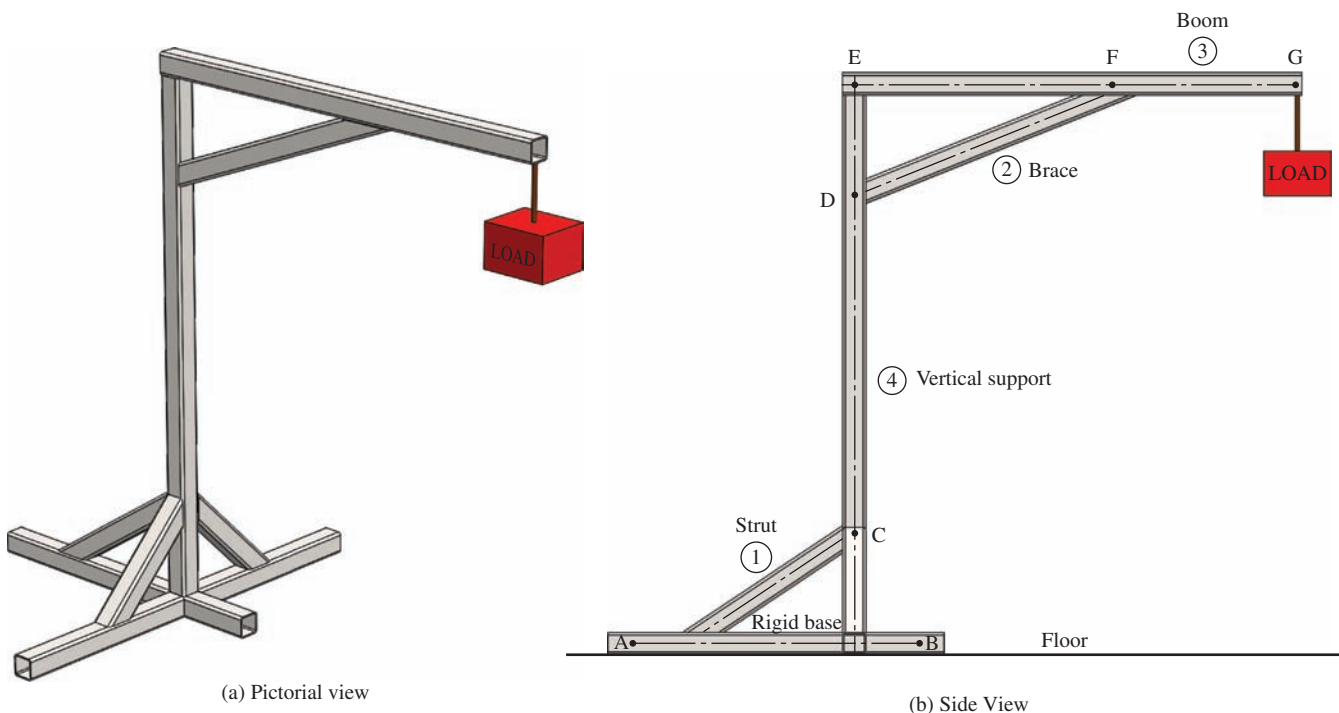


FIGURE 3–1 Schematic layout of a crane

You will need to analyze the kinds of forces that are exerted on each of the load-carrying members before you can design them. This calls for the use of the principles of statics in which you should have already gained competence. The following discussion provides a review of some of the key principles you will need in this course.

Your work as a designer proceeds as follows:

1. Analyze the forces that are exerted on each load-carrying member using the principles of statics.
2. Identify the kinds of stresses that each member is subjected to by the applied forces.
3. Propose the general shape of each load-carrying member and the material from which each is to be made.
4. Complete the stress analysis for each member to determine its final dimensions.

Let's work through steps 1 and 2 now as a review of statics. You will improve your ability to do steps 3 and 4 as you perform several practice problems in this chapter and in Chapters 4 and 5 by reviewing strength of materials and adding competencies that build on that foundation.

Force Analysis

One approach to the force analysis is outlined here.

1. Consider the entire crane structure as a free-body with the applied force acting at point G and the reactions acting at support points A and B. See Figure 3-2, which shows these forces and important dimensions of the crane structure.

2. Break the structure apart so that each member is represented as a free-body diagram, showing all forces acting at each joint. See the result in Figure 3-3.

3. Analyze the magnitudes and directions of all forces.

Comments are given here to summarize the methods used in the static analysis and to report results. You should work through the details of the analysis yourself or with colleagues to ensure that you can perform such calculations. All of the forces are directly proportional to the applied force F . We will show the results with an assumed value of $F = 10.0$ kN (approximately 2250 lb).

Step 1: The pin joints at A and B can provide support in any direction. We show the x and y components of the reactions in Figure 3-2. Then, proceed as follows:

1. Sum moments about B to find
 $R_{Ay} = 2.667 F = 26.67$ kN
2. Sum forces in the vertical direction to find
 $R_{By} = 3.667 F = 36.67$ kN.

At this point, we need to recognize that the strut AC is pin-connected at each end and carries loads only at its ends. Therefore, it is a *two-force member*, and the direction of the total force, R_A , acts along the member itself. Then R_{Ay} and R_{Ax} are the rectangular components of R_A as shown in the lower left of Figure 3-2. We can then say that

$$\tan(33.7^\circ) = R_{Ay}/R_{Ax}$$

and then

$$R_{Ax} = R_{Ay}/\tan(33.7^\circ) = 26.67 \text{ kN}/\tan(33.7^\circ) = 40.0 \text{ kN}$$

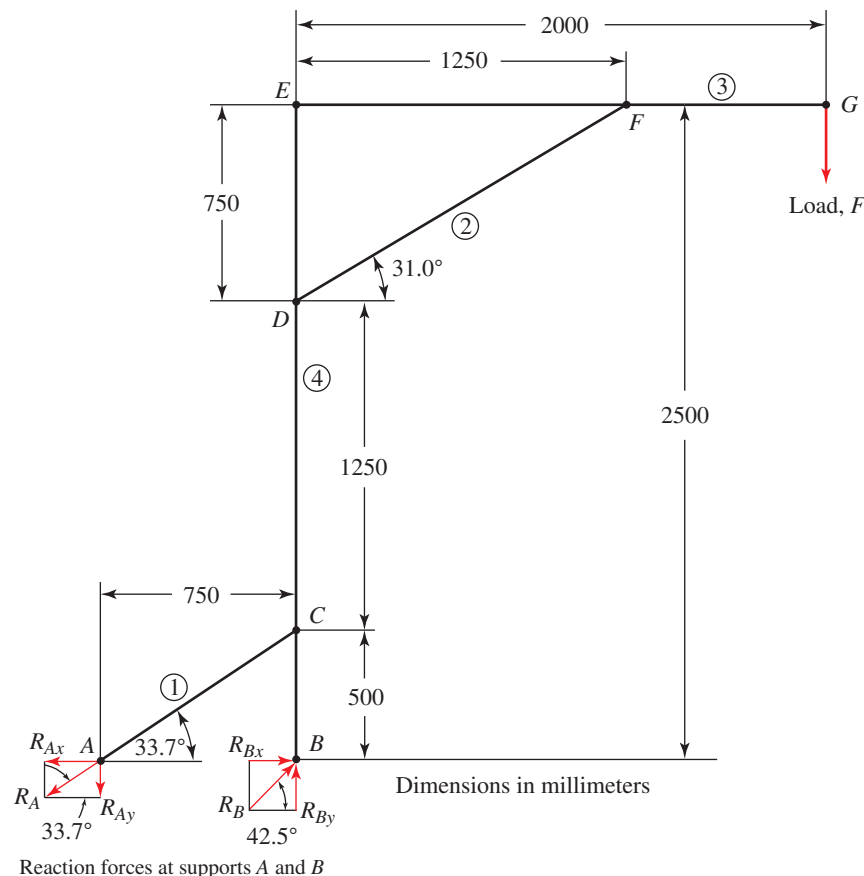


FIGURE 3-2 Free-body diagram of complete crane structure

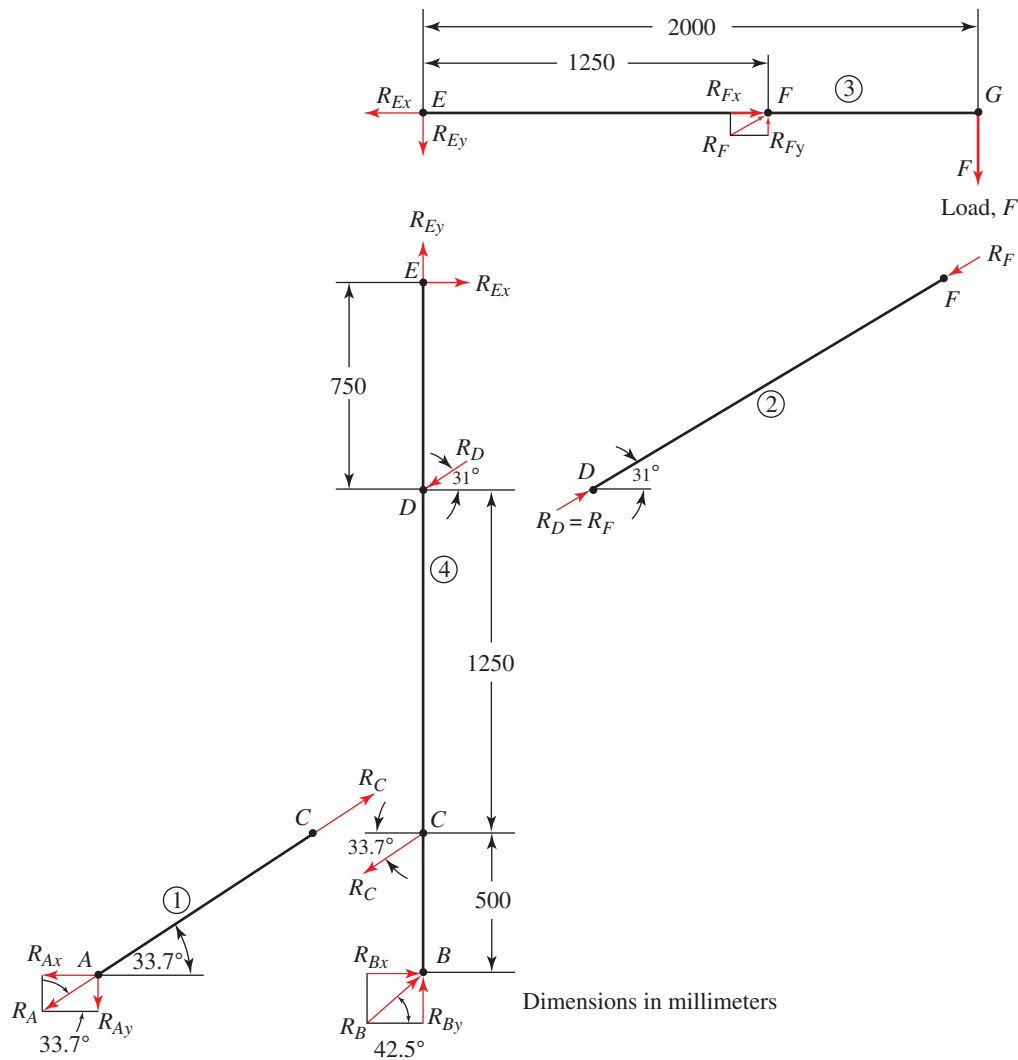


FIGURE 3-3 Free-body diagrams of each component of the crane

The total force, R_A , can be computed from the Pythagorean theorem,

$$R_A = \sqrt{R_{Ax}^2 + R_{Ay}^2} = \sqrt{(40.0)^2 + (26.67)^2} = 48.07 \text{ kN}$$

This force acts along the strut AC , at an angle of 33.7° above the horizontal, and it is the force that tends to shear the pin in joint A . The force at C on the strut AC is also 48.07 kN acting upward to the right to balance R_A on the two-force member as shown in Figure 3-3. Member AC is therefore in pure tension.

We can now use the sum of the forces in the horizontal direction on the entire structure to show that $R_{Ax} = R_{Bx} = 40.0$ kN.

The resultant of R_{Bx} and R_{By} is 54.3 kN acting at an angle of 42.5° above the horizontal, and it is the total shearing force on the pin in joint B . See the diagram in the lower right of Figure 3-2.

Step 2: The set of free-body diagrams is shown in Figure 3-3.

Step 3: Now consider the free-body diagrams of all of the members in Figure 3-3. We have already discussed member 1, recognizing it as a two-force member in tension carrying forces R_A and R_C equal to 48.07 kN. The reaction to R_C acts on the vertical member 4.

Now note that member 2 is also a two-force member, but it is in compression rather than tension. Therefore, we know that the forces on points D and F are equal and that they act in line with member 2, 31.0° with respect to the horizontal. The reactions to

these forces, then, act at point D on the vertical support, member 4, and at point F on the horizontal boom, member 3. We can find the value of R_F by considering the free-body diagram of member 3. You should be able to verify the following results using the methods already demonstrated.

$$R_{Fy} = 1.600 F = (1.600)(10.0 \text{ kN}) = 16.00 \text{ kN}$$

$$R_{Fx} = 2.667 F = (2.667)(10.0 \text{ kN}) = 26.67 \text{ kN}$$

$$R_F = 3.110 F = (3.110)(10.0 \text{ kN}) = 31.10 \text{ kN}$$

$$R_{Ey} = 0.600 F = (0.600)(10.0 \text{ kN}) = 6.00 \text{ kN}$$

$$R_{Ex} = 2.667 F = (2.667)(10.0 \text{ kN}) = 26.67 \text{ kN}$$

$$R_E = 2.733 F = (2.733)(10.0 \text{ kN}) = 27.33 \text{ kN}$$

Now all forces on the vertical member 4 are known from earlier analyses using the principle of action-reaction at each joint.

Types of Stresses on Each Member

Consider again the free-body diagrams in Figure 3-3 to visualize the kinds of stresses that are created in each member. This will lead to the use of particular kinds of stress analysis as the design process is completed. Members 3 and 4 carry forces perpendicular to their long axes and, therefore, they act as beams in bending. Figure 3-4 shows these members with the additional shearing force and bending moment diagrams. You should have learned to prepare such

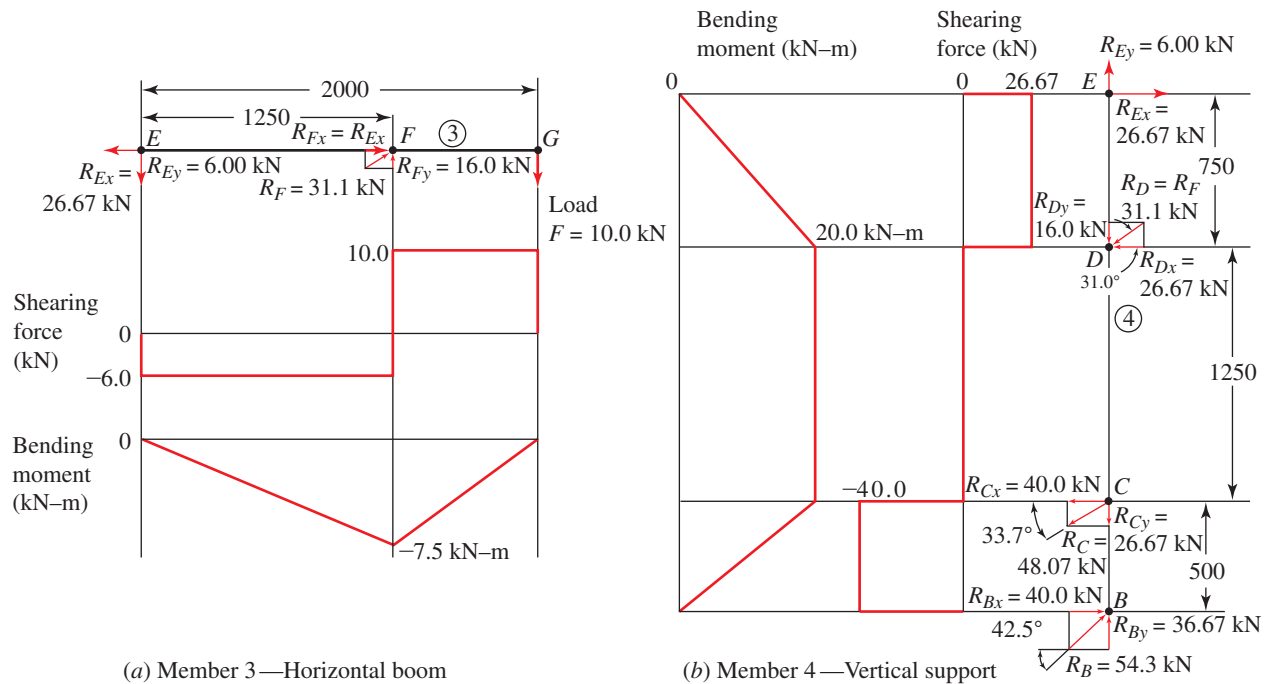


FIGURE 3-4 Shearing force and bending moment diagrams for members 3 and 4

diagrams in the prerequisite study of strength of materials. The following is a summary of the kinds of stresses in each member.

Member 1: The strut is in pure tension.

Member 2: The brace is in pure compression. Column buckling should be checked.

Member 3: The boom acts as a beam in bending. The right end between F and G is subjected to bending stress and vertical shear stress. Between E and F there is bending and shear combined with an axial tensile stress.

Member 4: The vertical support experiences a complex set of stresses depending on the segment being considered as described here.

Between E and D: Combined bending stress, vertical shear stress, and axial tension.

Between D and C: Combined bending stress and axial compression.

Between C and B: Combined bending stress, vertical shear stress, and axial compression.

Pin Joints: The connections between members at each joint must be designed to resist the total reaction force acting at each, computed in the earlier analysis. In general, each connection will likely include a cylindrical pin connecting two parts. The pin will typically be in direct shear. ■

3-1 OBJECTIVES OF THIS CHAPTER

After completing this chapter, you will:

1. Have reviewed the principles of stress and deformation analysis for several kinds of stresses, including the following:

Normal stresses due to direct tension and compression forces

Shear stress due to direct shear force

Shear stress due to torsional load for both circular and non-circular sections

Shear stress in beams due to bending

Normal stress in beams due to bending

2. Be able to interpret the nature of the stress at a point by drawing the *stress element* at any point in a load-carrying member for a variety of types of loads.

3. Have reviewed the importance of the *flexural center* of a beam cross section with regard to the alignment of loads on beams.
4. Have reviewed beam-deflection formulas.
5. Be able to analyze beam-loading patterns that produce abrupt changes in the magnitude of the bending moment in the beam.
6. Be able to use the principle of superposition to analyze machine elements that are subjected to loading patterns that produce combined stresses.
7. Be able to properly apply stress concentration factors in stress analyses.

3-2 PHILOSOPHY OF A SAFE DESIGN

In this book, every design approach will ensure that the stress level is below yield in ductile materials, automatically ensuring that the part will not break under a static

load. For brittle materials, we will ensure that the stress levels are well below the ultimate tensile strength. We will also analyze deflection where it is critical to safety or performance of a part.

Two other failure modes that apply to machine members are fatigue and wear. *Fatigue* is the response of a part subjected to repeated loads (see Chapter 5). *Wear* is discussed within the chapters devoted to the machine elements, such as gears, bearings, and chains, for which it is a major concern.

3-3 REPRESENTING STRESSES ON A STRESS ELEMENT

One major goal of stress analysis is to determine *the point* within a load-carrying member that is subjected to the highest stress level. You should develop the ability to visualize a *stress element*, a single, infinitesimally small cube from the member in a highly stressed area, and to show vectors that represent the kind of stresses that exist on that element. The orientation of the stress element is critical, and it must be aligned with specified axes on the member, typically called x , y , and z .

Figure 3-5 shows three examples of stress elements with two basic fundamental kinds of stress: Normal (tensile and compressive) and shear. Both the complete three-dimensional cube and the simplified, two-dimensional square forms for the stress elements are shown. The square is one face of the cube in a selected plane. The

sides of the square represent the projections of the faces of the cube that are perpendicular to the selected plane. It is recommended that you visualize the cube form first and then represent a square stress element showing stresses on a particular plane of interest in a given problem. In some problems with more general states of stress, two or three square stress elements may be required to depict the complete stress condition.

Tensile and compressive stresses, called *normal stresses*, are shown acting perpendicular to opposite faces of the stress element. Tensile stresses tend to pull on the element, whereas compressive stresses tend to crush it.

Shear stresses are created by direct shear, vertical shear in beams, or torsion. In each case, the action on an element subjected to shear is a tendency to *cut* the element by exerting a stress downward on one face while simultaneously exerting a stress upward on the opposite, parallel face. This action is that of a simple pair of shears or scissors. But note that if only one pair of shear stresses acts on a stress element, it will not be in equilibrium. Rather, it will tend to spin because the pair of shear stresses forms a couple. To produce equilibrium, a second pair of shear stresses on the other two faces of the element must exist, acting in a direction that opposes the first pair.

In summary, shear stresses on an element will always be shown as two pairs of equal stresses acting on (parallel to) the four sides of the element. Figure 3-5(c) shows an example.

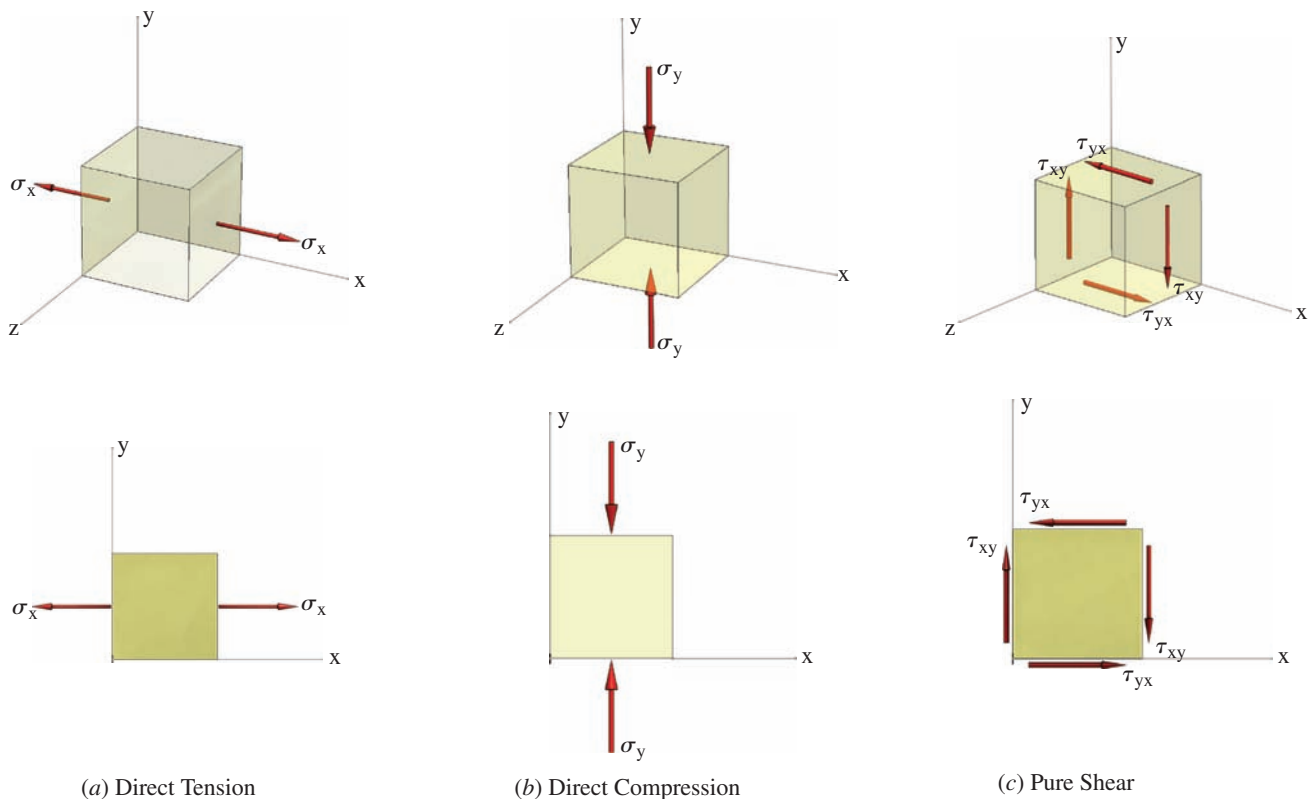


FIGURE 3-5 Stress elements for normal and shear stresses

Sign Convention for Shear Stresses

This book adopts the following convention:

Positive shear stresses tend to rotate the element in a clockwise direction.

Negative shear stresses tend to rotate the element in a counterclockwise direction.

A double subscript notation is used to denote shear stresses in a plane. For example, in Figure 3-5(c), drawn for the x - y plane, the pair of shear stresses, τ_{xy} , indicates a shear stress acting on the element face that is perpendicular to the x -axis and parallel to the y -axis. Then τ_{yx} acts on the face that is perpendicular to the y -axis and parallel to the x -axis. In this example, τ_{xy} is positive and τ_{yx} is negative.

3-4 NORMAL STRESSES DUE TO DIRECT AXIAL LOAD

Stress can be defined as the internal resistance offered by a unit area of a material to an externally applied load. *Normal stresses* (σ) are either *tensile* (positive) or *compressive* (negative).

For a load-carrying member in which the external load is uniformly distributed across the cross-sectional area of the member, the magnitude of the stress can be calculated from the direct stress formula:

Direct Tensile or Compressive Stress

$$\sigma = \text{force/area} = F/A \quad (3-1)$$

The units for stress are always *force per unit area*, as is evident from Equation (3-1). Common units in the U.S. Customary system and the SI metric system follow.

U.S. Customary Units

$$\text{lb/in}^2 = \text{psi}$$

$$\text{kips/in}^2 = \text{ksi}$$

$$\text{Note: } 1.0 \text{ kip} = 1000 \text{ lb}$$

$$1.0 \text{ ksi} = 1000 \text{ psi}$$

SI Metric Units

$$\text{N/m}^2 = \text{pascal} = \text{Pa}$$

$$\text{N/mm}^2 = \text{megapascal}$$

$$= 10^6 \text{ Pa} = \text{MPa}$$

The conditions on the use of Equation (3-1) are as follows:

1. The load-carrying member must be straight.
2. The line of action of the load must pass through the centroid of the cross section of the member.
3. The member must be of uniform cross section near where the stress is being computed.
4. The material must be homogeneous and isotropic.
5. In the case of compression members, the member must be short to prevent buckling. The conditions under which buckling is expected are discussed in Chapter 6.

Example Problem 3-1

A tensile force of 9500 N is applied to a 12-mm-diameter round bar, as shown in Figure 3-6. Compute the direct tensile stress in the bar.

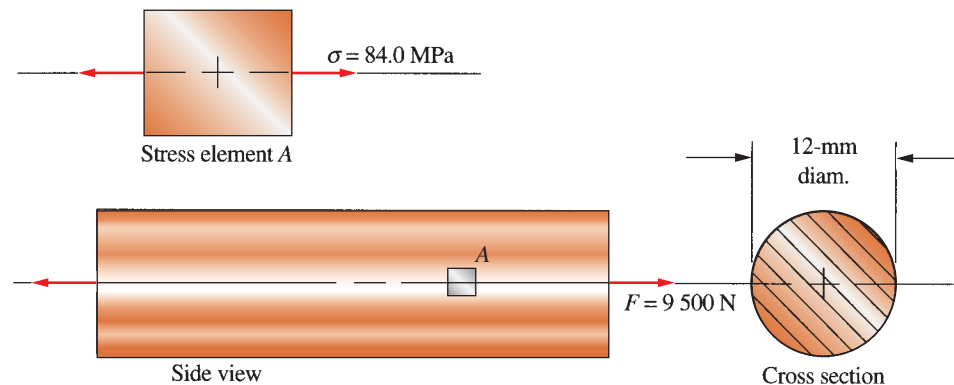


FIGURE 3-6 Tensile stress in a round bar

Solution

Objective Compute the tensile stress in the round bar.

Given Force = $F = 9500 \text{ N}$; diameter = $D = 12 \text{ mm}$.

Analysis Use the direct tensile stress formula, Equation (3-1): $\sigma = F/A$. Compute the cross-sectional area from $A = \pi D^2/4$.

Results $A = \pi D^2/4 = \pi(12 \text{ mm})^2/4 = 113 \text{ mm}^2$
 $\sigma = F/A = (9500 \text{ N})/(113 \text{ mm}^2) = 84.0 \text{ N/mm}^2 = 84.0 \text{ MPa}$

Comment The results are shown on stress element A in Figure 3-6, which can be taken to be anywhere within the bar because, ideally, the stress is uniform on any cross section. The cube form of the element is as shown in Figure 3-5 (a).

Example Problem 3-2

For the round bar subjected to the tensile load shown in Figure 3-6, compute the total deformation if the original length of the bar is 3600 mm. The bar is made from a steel having a modulus of elasticity of 207 GPa.

Solution

Objective Compute the deformation of the bar.

Given Force = $F = 9500$ N; diameter = $D = 12$ mm.
Length = $L = 3600$ mm; $E = 207$ GPa

Analysis From Example Problem 3-1, we found that $\sigma = 84.0$ MPa. Use Equation (3-3).

Results $\delta = \frac{\sigma L}{E} = \frac{(84.0 \times 10^6 \text{ N/m}^2)(3600 \text{ mm})}{(207 \times 10^9 \text{ N/m}^2)} = 1.46 \text{ mm}$

3-5 DEFORMATION UNDER DIRECT AXIAL LOAD

The following formula computes the stretch due to a direct axial tensile load or the shortening due to a direct axial compressive load:

$$\delta = FL/EA \quad (3-2)$$

⇒ **Deformation Due to Direct Axial Load**

where δ = total deformation of the member carrying the axial load

F = direct axial load

L = original total length of the member

E = modulus of elasticity of the material

A = cross-sectional area of the member

Noting that $\sigma = F/A$, we can also compute the deformation from

$$\delta = \sigma L/E \quad (3-3)$$

3-6 SHEAR STRESS DUE TO DIRECT SHEAR LOAD

Direct shear stress occurs when the applied force tends to cut through the member as scissors or shears do or when a punch and a die are used to punch a slug of material from a sheet. Another important example of direct shear in machine design is the tendency for a key to be sheared off at the section between the shaft and the hub of a machine element when transmitting torque. Figure 3-7 shows the action.

The method of computing direct shear stress is similar to that used for computing direct tensile stress because the applied force is assumed to be uniformly distributed across the cross section of the part that is resisting the force. But the kind of stress is *shear stress* rather than *normal stress*. The symbol used for shear stress is the

Greek letter tau (τ). The formula for direct shear stress can thus be written

⇒ **Direct Shear Stress**

$$\tau = \text{shearing force/area in shear} = F/A_s \quad (3-4)$$

This stress is more properly called the *average shearing stress*, but we will make the simplifying assumption that the stress is uniformly distributed across the shear area.

3-7 TORSIONAL LOAD—TORQUE, ROTATIONAL SPEED, AND POWER

The relationship among the power (P), the rotational speed (n), and the torque (T) in a shaft is described by the equation

⇒ **Power–Torque–Speed Relationship**

$$T = P/n \quad (3-5)$$

In SI units, power is expressed in the unit of *watt* (W) or its equivalent, *newton meter per second* ($\text{N} \cdot \text{m/s}$), and the rotational speed is in *radians per second* (rad/s).

In the U.S. Customary Unit System, power is typically expressed as *horsepower*, equal to $550 \text{ ft} \cdot \text{lb/s}$. The typical unit for rotational speed is rpm, or revolutions per minute. But the most convenient unit for torque is the pound-inch ($\text{lb} \cdot \text{in}$). Considering all of these quantities and making the necessary conversions of units, we use the following formula to compute the torque (in $\text{lb} \cdot \text{in}$) in a shaft carrying a certain power P (in hp) while rotating at a speed of n rpm.

⇒ **P – T – n Relationship for U.S. Customary Units**

$$T = 63\,000 P/n \quad (3-6)$$

The resulting torque will be in $\text{lb} \cdot \text{in}$. You should verify the value of the constant, 63 000.

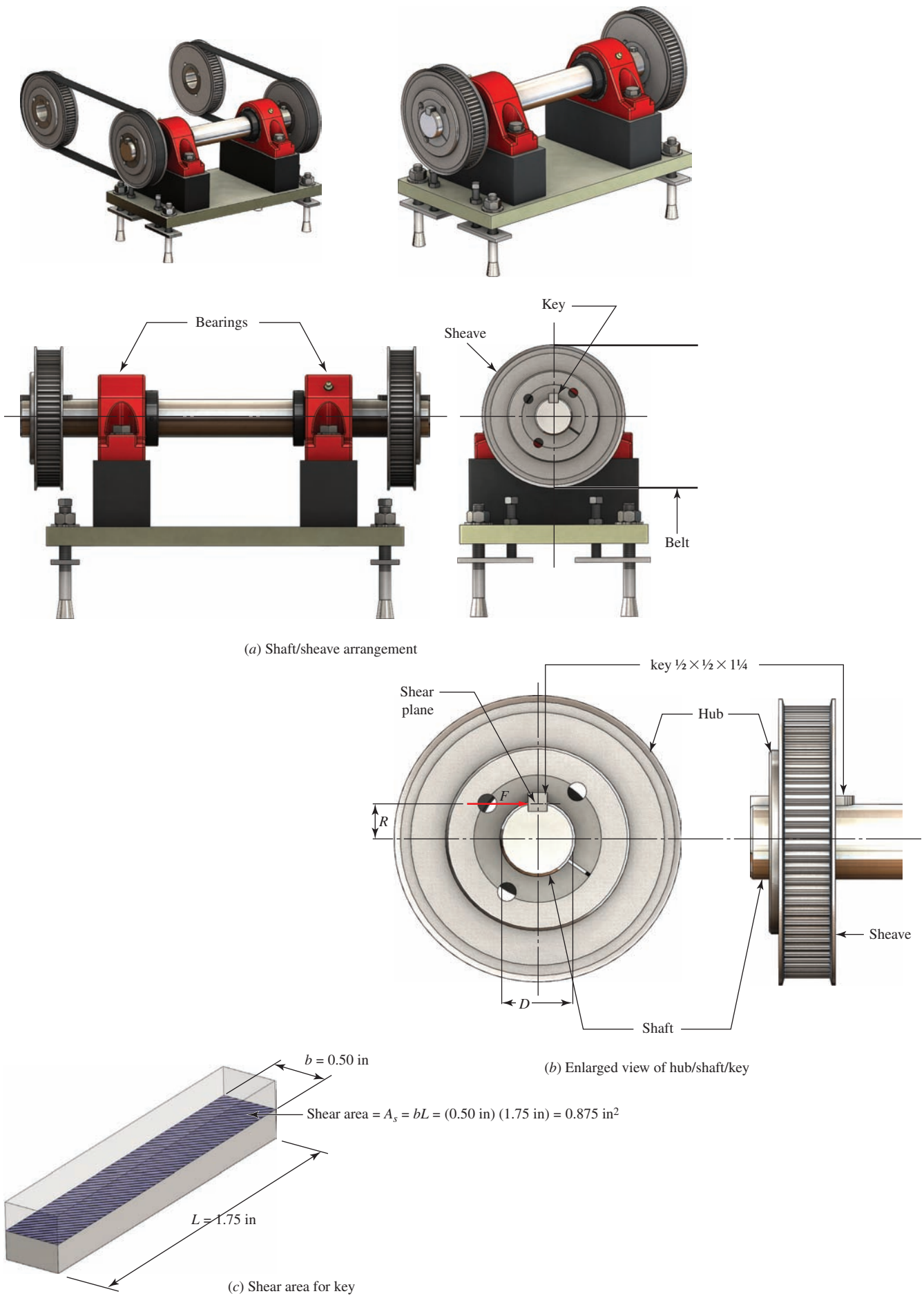


FIGURE 3-7 Direct shear on a key

Example Problem 3–3

Figure 3–7 shows a shaft carrying two sprockets for synchronous belt drives that are keyed to the shaft. Figure 3–7 (b) shows that a force F is transmitted from the shaft to the hub of the sprocket through a square key. The shaft has a diameter of 2.25 in and transmits a torque of 14 063 lb·in. The key has a square cross section, 0.50 in on a side, and a length of 1.75 in. Compute the force on the key and the shear stress caused by this force.

Solution

Objective	Compute the force on the key and the shear stress.
Given	Layout of shaft, key, and hub shown in Figure 3–7. Torque = $T = 14\,063\text{ lb}\cdot\text{in}$; key dimensions = $0.5\text{ in} \times 0.5\text{ in} \times 1.75\text{ in}$. Shaft diameter = $D = 2.25\text{ in}$; radius = $R = D/2 = 1.125\text{ in}$.
Analysis	Torque $T = \text{force } F \times \text{radius } R$. Then $F = T/R$. Use equation (3–4) to compute shearing stress: $\tau = F/A_s$. Shear area is the cross section of the key at the interface between the shaft and the hub: $A_s = bL$.
Results	$F = T/R = (14\,063\text{ lb}\cdot\text{in})/(1.125\text{ in}) = 12\,500\text{ lb}$ $A_s = bL = (0.50\text{ in})(1.75\text{ in}) = 0.875\text{ in}^2$ $\tau = F/A_s = (12\,500\text{ lb})/(0.875\text{ in}^2) = 14\,300\text{ lb/in}^2$
Comment	This level of shearing stress will be uniform on all parts of the cross section of the key.

Example Problem 3–4

Compute the torque on a shaft transmitting 750 W of power while rotating at 183 rad/s. (*Note:* This is equivalent to the output of a 1.0-hp, 4-pole electric motor, operating at its rated speed of 1750 rpm. See Chapter 21.)

Solution

Objective	Compute the torque T on the shaft.
Given	Power = $P = 750\text{ W} = 750\text{ N}\cdot\text{m/s}$. Rotational speed = $n = 183\text{ rad/s}$.
Analysis	Use Equation (3–5).
Results	$T = P/n = (750\text{ N}\cdot\text{m/s})/(183\text{ rad/s})$ $T = 4.10\text{ N}\cdot\text{m/rad} = 4.10\text{ N}\cdot\text{m}$
Comments	In such calculations, the unit of $\text{N}\cdot\text{m/rad}$ is dimensionally correct, and some advocate its use. Most, however, consider the radian to be dimensionless, and thus torque is expressed in $\text{N}\cdot\text{m}$ or other familiar units of force times distance.

Example Problem 3–5

Compute the torque on a shaft transmitting 1.0 hp while rotating at 1750 rpm. Note that these conditions are approximately the same as those for which the torque was computed in Example Problem 3–4 using SI units.

Solution

Objective	Compute the torque on the shaft.
Given	$P = 1.0\text{ hp}$; $n = 1750\text{ rpm}$.
Analysis	Use Equation (3–6).
Results	$T = 63\,000\text{ P}/n = [63\,000(1.0)]/1750 = 36.0\text{ lb}\cdot\text{in}$

3–8 SHEAR STRESS DUE TO TORSIONAL LOAD

When a *torque*, or twisting moment, is applied to a member, it tends to deform by twisting, causing a rotation of one part of the member relative to another. Such twisting causes a shear stress in the member. For a small element

of the member, the nature of the stress is the same as that experienced under direct shear stress. However, in *torsional shear*, the distribution of stress is not uniform across the cross section.

The most frequent case of torsional shear in machine design is that of a round circular shaft transmitting power. Chapter 12 covers shaft design.

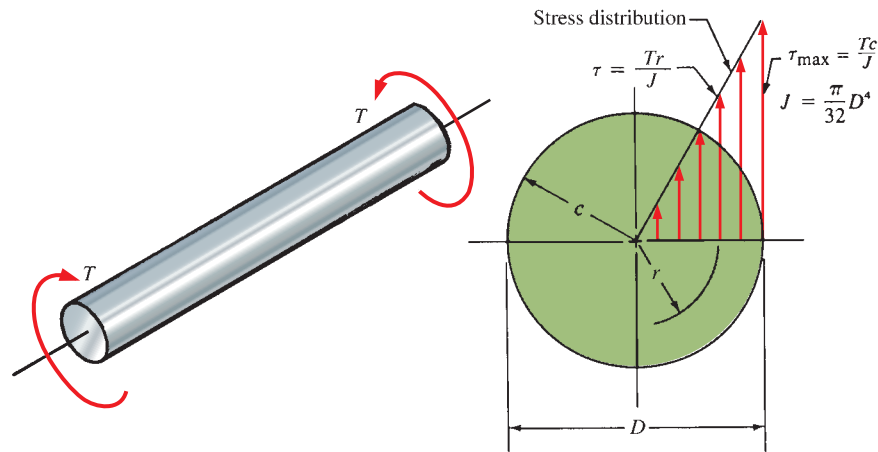


FIGURE 3-8 Stress distribution in a solid shaft

Torsional Shear Stress Formula

When subjected to a torque, the outer surface of a solid round shaft experiences the greatest shearing strain and therefore the largest torsional shear stress. See Figure 3-8. The value of the maximum torsional shear stress is found from

Maximum Torsional Shear Stress in a Circular Shaft

$$\tau_{\max} = Tc/J \quad (3-7)$$

where c = radius of the shaft to its outside surface
 J = polar moment of inertia

See Appendix 1 for formulas for J .

If it is desired to compute the torsional shear stress at some point inside the shaft, the more general formula is used:

General Formula for Torsional Shear Stress

$$\tau = Tr/J \quad (3-8)$$

where r = radial distance from the center of the shaft to the point of interest

Figure 3-8 shows graphically that this equation is based on the linear variation of the torsional shear stress from zero at the center of the shaft to the maximum value at the outer surface.

Equations (3-7) and (3-8) apply also to hollow shafts (Figure 3-9 shows the distribution of shear stress). Again note that the maximum shear stress occurs at the outer surface. Also note that the entire cross section carries a relatively high stress level. As a result, the hollow shaft is more efficient. Notice that the material near the center of the solid shaft is not highly stressed.

For design, it is convenient to define the *polar section modulus*, Z_p :

Polar Section Modulus

$$Z_p = J/c \quad (3-9)$$

Then the equation for the maximum torsional shear stress is

$$\tau_{\max} = T/Z_p \quad (3-10)$$

Formulas for the polar section modulus are also given in Appendix 1. This form of the torsional shear stress equation is useful for design problems because the polar section modulus is the only term related to the geometry of the cross section.

Example Problem 3-6

Compute the maximum torsional shear stress in a shaft having a diameter of 10 mm when it carries a torque of $4.10 \text{ N} \cdot \text{m}$.

Solution

Objective Compute the torsional shear stress in the shaft.

Given Torque = $T = 4.10 \text{ N} \cdot \text{m}$; shaft diameter = $D = 10 \text{ mm}$.
 c = radius of the shaft = $D/2 = 5.0 \text{ mm}$.

Analysis Use Equation (3-7) to compute the torsional shear stress: $\tau_{\max} = Tc/J$. J is the polar moment of inertia for the shaft: $J = \pi D^4/32$ (see Appendix 1).

$$\begin{aligned} \text{Results} \quad J &= \pi D^4/32 = [(\pi)(10 \text{ mm})^4]/32 = 982 \text{ mm}^4 \\ \tau_{\max} &= \frac{(4.10 \text{ N} \cdot \text{m})(5.0 \text{ mm})10^3 \text{ mm}}{982 \text{ mm}^4 \text{ m}} = 20.9 \text{ N/mm}^2 = 20.9 \text{ MPa} \end{aligned}$$

Comment The maximum torsional shear stress occurs at the outside surface of the shaft around its entire circumference.

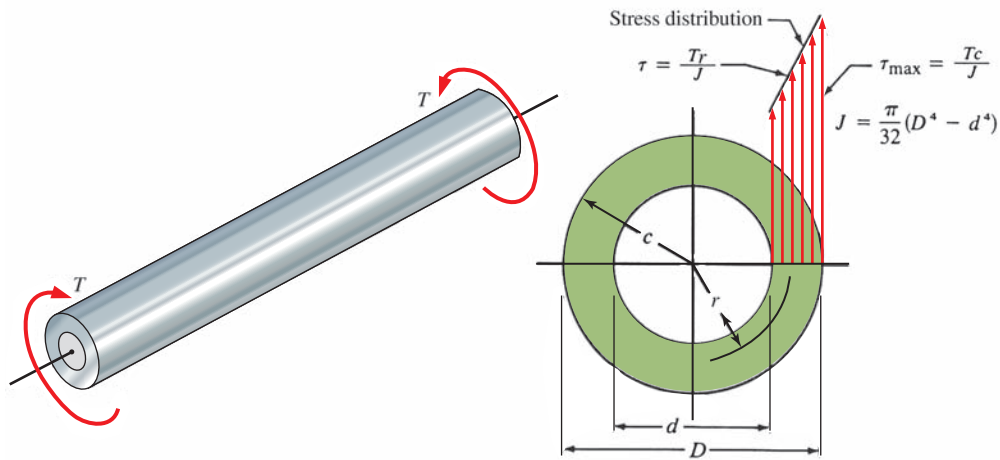


FIGURE 3-9 Stress distribution in a hollow shaft

3-9 TORSIONAL DEFORMATION

When a shaft is subjected to a torque, it undergoes a twisting in which one cross section is rotated relative to other cross sections in the shaft. The angle of twist is computed from

where θ = angle of twist (radians)

L = length of the shaft over which the angle of twist is being computed

G = modulus of elasticity of the shaft material in *shear*

⇒ Torsional Deformation

$$\theta = TL/GJ \quad (3-11)$$

Example Problem 3-7

Compute the angle of twist of a 10-mm-diameter shaft carrying 4.10 N·m of torque if it is 250 mm long and made of steel with $G = 80$ GPa. Express the result in both radians and degrees.

Solution

Objective Compute the angle of twist in the shaft.

Given Torque = $T = 4.10$ N·m; length = $L = 250$ mm.
Shaft diameter = $D = 10$ mm; $G = 80$ GPa.

Analysis Use Equation (3-11). For consistency, let $T = 4.10 \times 10^3$ N·mm and $G = 80 \times 10^3$ N/mm². From Example Problem 3-6, $J = 982$ mm⁴.

Results

$$\theta = \frac{TL}{GJ} = \frac{(4.10 \times 10^3 \text{ N} \cdot \text{mm})(250 \text{ mm})}{(80 \times 10^3 \text{ N/mm}^2)(982 \text{ mm}^4)} = 0.013 \text{ rad}$$

Using $\pi \text{ rad} = 180^\circ$,

$$\theta = (0.013 \text{ rad})(180^\circ/\pi \text{ rad}) = 0.75^\circ$$

Comment Over the length of 250 mm, the shaft twists 0.75°.

3-10 TORSION IN MEMBERS HAVING NON-CIRCULAR CROSS SECTIONS

The behavior of members having noncircular cross sections when subjected to torsion is radically different from that for members having circular cross sections. However, the factors of most use in machine design are the maximum stress and the total angle of twist for such

members. The formulas for these factors can be expressed in similar forms to the formulas used for members of circular cross section (solid and hollow round shafts).

The following two formulas can be used:

⇒ Torsional Shear Stress

$$\tau_{\max} = T/Q \quad (3-12)$$

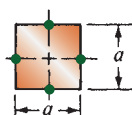
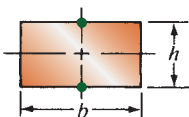
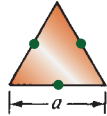
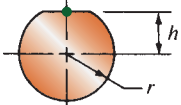
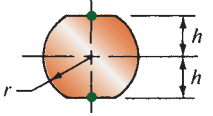
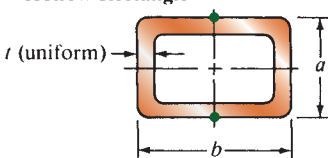
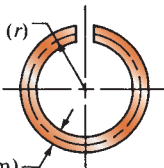
Cross-sectional shape	$K =$ for use in $\theta = TL/GK$ $Q =$ for use in $\tau = T/Q$	Colored dot (●) denotes location of $\tau_{\max}$																					
Square 	$K = 0.141a^4$ $Q = 0.208a^3$	$\tau_{\max}$ at midpoint of each side																					
Rectangle 	$K = bh^3\left[\frac{1}{3} - 0.21\frac{h}{b}\left(1 - \frac{(h/b)^4}{12}\right)\right]$ $Q = \frac{bh^2}{[3 + 1.8(h/b)]}$	(Approximate; within $\approx 5\%$) $\tau_{\max}$ at midpoint of long sides																					
Triangle (equilateral) 	$K = 0.0217a^4$ $Q = 0.050a^3$																						
Shaft with One Flat 	$K = C_1r^4$ $Q = C_2r^3$	<table><tr><td>h/r</td><td>0</td><td>0.2</td><td>0.4</td><td>0.6</td><td>0.8</td><td>1.0</td></tr><tr><td>C_1</td><td>0.30</td><td>0.51</td><td>0.78</td><td>1.06</td><td>1.37</td><td>1.57</td></tr><tr><td>C_2</td><td>0.35</td><td>0.51</td><td>0.70</td><td>0.92</td><td>1.18</td><td>1.57</td></tr></table>	h/r	0	0.2	0.4	0.6	0.8	1.0	C_1	0.30	0.51	0.78	1.06	1.37	1.57	C_2	0.35	0.51	0.70	0.92	1.18	1.57
h/r	0	0.2	0.4	0.6	0.8	1.0																	
C_1	0.30	0.51	0.78	1.06	1.37	1.57																	
C_2	0.35	0.51	0.70	0.92	1.18	1.57																	
Shaft with Two Flats 	$K = C_3r^4$ $Q = C_4r^3$	<table><tr><td>h/r</td><td>0.5</td><td>0.6</td><td>0.7</td><td>0.8</td><td>0.9</td><td>1.0</td></tr><tr><td>C_3</td><td>0.44</td><td>0.67</td><td>0.93</td><td>1.19</td><td>1.39</td><td>1.57</td></tr><tr><td>C_4</td><td>0.47</td><td>0.60</td><td>0.81</td><td>1.02</td><td>1.25</td><td>1.57</td></tr></table>	h/r	0.5	0.6	0.7	0.8	0.9	1.0	C_3	0.44	0.67	0.93	1.19	1.39	1.57	C_4	0.47	0.60	0.81	1.02	1.25	1.57
h/r	0.5	0.6	0.7	0.8	0.9	1.0																	
C_3	0.44	0.67	0.93	1.19	1.39	1.57																	
C_4	0.47	0.60	0.81	1.02	1.25	1.57																	
Hollow Rectangle 	$K = \frac{2t(a-t)^2(b-t)^2}{(a+b-2t)}$ $Q = 2t(a-t)(b-t)$ Gives average stress; good approximation of maximum stress if t is small—thin-walled tube Inner corners should have generous fillets																						
Split Tube Mean radius (r) 	$K = 2\pi r t^3/3$ $Q = \frac{4\pi^2 r^2 t^2}{(6\pi r + 1.8t)}$ t must be small—thin-walled tube																						

FIGURE 3-10 Methods for determining values for K and Q for several types of cross sections

Deflection for Noncircular Sections

$$\theta = TL/GK \quad (3-13)$$

Note that Equations (3-12) and (3-13) are similar to Equations (3-10) and (3-11), with the substitution of Q for Z_p and K for J . Refer Figure 3-10 for the methods of

determining the values for K and Q for several types of cross sections useful in machine design. These values are appropriate only if the ends of the member are free to deform. If either end is fixed, as by welding to a solid structure, the resulting stress and angular twist are quite different. (See References 1-3, 6, and 7.)

Example Problem 3-8

A 2.50-in-diameter shaft carrying a chain sprocket has one end milled in the form of a square to permit the use of a hand crank. The square is 1.75 in on a side. Compute the maximum shear stress on the square part of the shaft when a torque of 15 000 lb·in is applied.

Also, if the length of the square part is 8.00 in, compute the angle of twist over this part. The shaft material is steel with $G = 11.5 \times 10^6$ psi.

Solution

Objective Compute the maximum shear stress and the angle of twist in the shaft.

Given Torque = $T = 15\,000$ lb·in; length = $L = 8.00$ in.
The shaft is square; thus, $a = 1.75$ in.
 $G = 11.5 \times 10^6$ psi.

Analysis Figure 3–10 shows the methods for calculating the values for Q and K for use in Equations (3–12) and (3–13).

Results $Q = 0.208a^3 = (0.208)(1.75 \text{ in})^3 = 1.115 \text{ in}^3$
 $K = 0.141a^4 = (0.141)(1.75 \text{ in})^4 = 1.322 \text{ in}^4$
 Now the stress and the deflection can be computed.

$$\tau_{\max} = \frac{T}{Q} = \frac{15\,000 \text{ lb}\cdot\text{in}}{(1.115 \text{ in}^3)} = 13\,460 \text{ psi}$$

$$\theta = \frac{TL}{GK} = \frac{(15\,000 \text{ lb}\cdot\text{in})(8.00 \text{ in})}{(11.5 \times 10^6 \text{ lb/in}^2)(1.322 \text{ in}^4)} = 0.0079 \text{ rad}$$

Convert the angle of twist to degrees:

$$\theta = (0.0079 \text{ rad})(180^\circ/\pi \text{ rad}) = 0.452^\circ$$

Comments Over the length of 8.00 in, the square part of the shaft twists 0.452° . The maximum shear stress is 13 460 psi, and it occurs at the midpoint of each side as shown in Figure 3–10.

3-11 TORSION IN CLOSED, THIN-WALLED TUBES

A general approach for closed, thin-walled tubes of virtually any shape uses Equations (3–12) and (3–13) with special methods of evaluating K and Q . Figure 3–11 shows such a tube having a constant wall thickness. The values of K and Q are

$$K = 4A^2t/U \quad (3-14)$$

$$Q = 2tA \quad (3-15)$$

where A = area enclosed by the median boundary (indicated by the dashed line in Figure 3–11)

t = wall thickness (which must be uniform and thin)

U = length of the median boundary

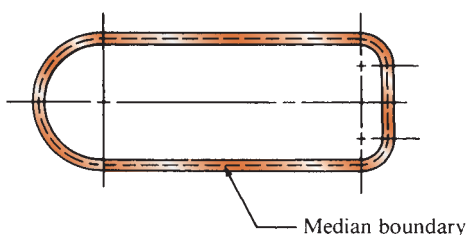


FIGURE 3-11 Closed, thin-walled tube with a constant wall thickness

The shear stress computed by this approach is the *average stress* in the tube wall. However, if the wall thickness t is small (a thin wall), the stress is nearly uniform throughout the wall, and this approach will yield a close approximation of the maximum stress. For the analysis of tubular sections having nonuniform wall thickness, see References 1–3, 6, and 7.

To design a member to resist torsion only, or torsion and bending combined, it is advisable to select hollow tubes, either round or rectangular, or some other closed shape. They possess good efficiency both in bending and in torsion.

3-12 TORSION IN OPEN, THIN-WALLED TUBES

The term *open tube* refers to a shape that appears to be tubular but is not completely closed. For example, some tubing is manufactured by starting with a thin, flat strip of steel that is roll-formed into the desired shape (circular, rectangular, square, and so on). Then the seam is welded along the entire length of the tube. It is interesting to compare the properties of the cross section of such a tube before and after it is welded. The following example problem illustrates the comparison for a particular size of circular tubing.

Example Problem 3-9

Figure 3–12 shows a tube before [Part (b)] and after [Part (a)] the seam is welded. Compare the stiffness and the strength of each shape.

Solution

Objective Compare the torsional stiffness and the strength of the closed tube of Figure 3–12(a) with those of the open-seam (split) tube shown in Figure 3–12(b).

Given The tube shapes are shown in Figure 3–12. Both have the same length, diameter, and wall thickness, and both are made from the same material.

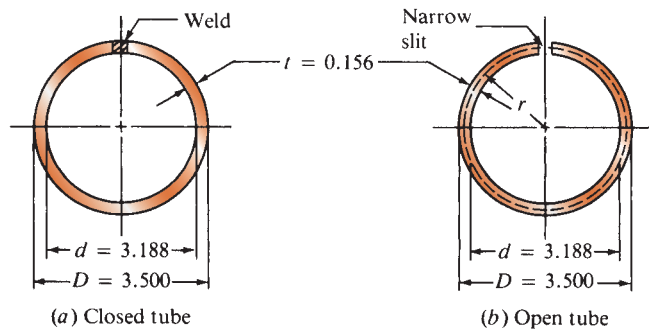


FIGURE 3-12 Comparison of closed and open tubes

Analysis Equation (3-13) gives the angle of twist for a noncircular member and shows that the angle is inversely proportional to the value of K . Similarly, Equation (3-11) shows that the angle of twist for a hollow circular tube is inversely proportional to the polar moment of inertia J . All other terms in the two equations are the same for each design. Therefore, the ratio of θ_{open} to θ_{closed} is equal to the ratio J/K . From Appendix 1, we find

$$J = \pi(D^4 - d^4)/32$$

From Figure 3-10, we find

$$K = 2\pi r t^3/3$$

Using similar logic, Equations (3-12) and (3-10) show that the maximum torsional shear stress is inversely proportional to Q and Z_p for the open and closed tubes, respectively. Then we can compare the strengths of the two forms by computing the ratio Z_p/Q . By Equation (3-9), we find that

$$Z_p = J/c = J/(D/2)$$

The equation for Q for the split tube is listed in Figure 3-10.

Results We make the comparison of torsional stiffness by computing the ratio J/K . For the closed, hollow tube,

$$J = \pi(D^4 - d^4)/32$$

$$J = \pi(3.500^4 - 3.188^4)/32 = 4.592 \text{ in}^4$$

For the open tube before the slit is welded, from Figure 3-10,

$$K = 2\pi r t^3/3$$

$$K = [(2)(\pi)(1.672)(0.156)^3]/3 = 0.0133 \text{ in}^4$$

$$\text{Ratio} = J/K = 4.592/0.0133 = 345$$

Then we make the comparison of the strengths of the two forms by computing the ratio Z_p/Q .

The value of J has already been computed to be 4.592 in^4 . Then

$$Z_p = J/c = J/(D/2) = (4.592 \text{ in}^4)/[(3.500 \text{ in})/2] = 2.624 \text{ in}^3$$

For the open tube,

$$Q = \frac{4\pi^2 r^2 t^2}{(6\pi r + 1.8t)} = \frac{4\pi^2 (1.672 \text{ in})^2 (0.156 \text{ in})^2}{[6\pi (1.672 \text{ in}) + 1.8(0.156 \text{ in})]} = 0.0845 \text{ in}^3$$

Then the strength comparison is

$$\text{Ratio} = Z_p/Q = 2.624/0.0845 = 31.1$$

Comments Thus, for a given applied torque, the slit tube would twist 345 times as much as the closed tube. The stress in the slit tube would be 31.1 times higher than in the closed tube. Also note that if the material for the tube is thin, it will likely buckle at a relatively low stress level, and the tube will collapse suddenly. This comparison shows the dramatic superiority of the closed form of a hollow section to an open form. A similar comparison could be made for shapes other than circular.

3-13 SHEAR STRESS DUE TO BENDING

A beam carrying loads transverse to its axis will experience shearing forces, denoted by V . In the analysis of beams, it is usual to compute the variation in shearing force across the entire length of the beam and to draw the *shearing force diagram*. Then the resulting vertical shearing stress can be computed from

Vertical Shearing Stress in Beams

$$\tau = VQ/It \quad (3-16)$$

where I = rectangular moment of inertia of the cross section of the beam

t = thickness of the section at the place where the shearing stress is to be computed

Q = *first moment*, with respect to the overall centroidal axis, of the area of that part of the cross section that lies away from the axis where the shearing stress is to be computed.

To calculate the value of Q , we define it by the following equation,

First Moment of the Area

$$Q = A_p \bar{y} \quad (3-17)$$

where A_p = that part of the area of the section above the place where the stress is to be computed

$\bar{y}$ = distance from the neutral axis of the section to the centroid of the area A_p

In some books or references, and in earlier editions of this book, Q was called the *statical moment*. Here we will use the term *first moment of the area*.

For most section shapes, the maximum vertical shearing stress occurs at the centroidal axis. Specifically, if the thickness is not less at a place away from the centroidal axis, then it is assured that the maximum vertical shearing stress occurs at the centroidal axis.

Figure 3-13 shows three examples of how Q is computed in typical beam cross sections. In each, the maximum vertical shearing stress occurs at the neutral axis.

Note that the vertical shearing stress is equal to the *horizontal shearing stress* because any element of material subjected to a shear stress on one face must have a shear stress of the same magnitude on the adjacent face

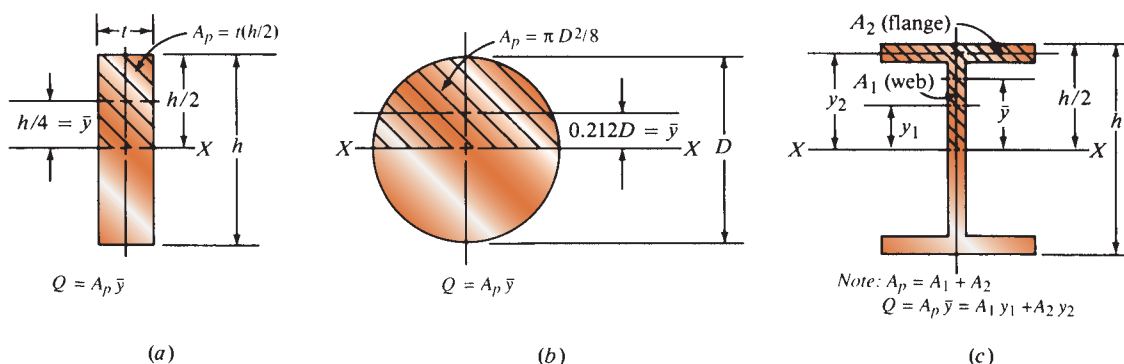


FIGURE 3-13 Illustrations of A_p and $\bar{y}$ used to compute Q for three shapes

Example Problem 3-10

Figure 3-14 shows a simply supported beam carrying two concentrated loads. The shearing force diagram is shown, along with the rectangular shape and size of the cross section of the beam. The stress distribution is parabolic, with the maximum stress occurring at the neutral axis. Use Equation (3-16) to compute the maximum shearing stress in the beam.

Solution

Objective Compute the maximum shearing stress τ in the beam in Figure 3-14.

Given The beam shape is rectangular: $h = 8.00$ in; $t = 2.00$ in.
Maximum shearing force = $V = 1000$ lb at all points between A and B.

Analysis Use Equation (3-16) to compute τ . V and t are given. From Appendix 1,

$$I = th^3/12$$

The value of the first moment of the area Q can be computed from Equation (3-17). For the rectangular cross section shown in Figure 3-13(a), $A_p = t(h/2)$ and $\bar{y} = h/4$. Then

$$Q = A_p \bar{y} = (th/2)(h/4) = th^2/8$$

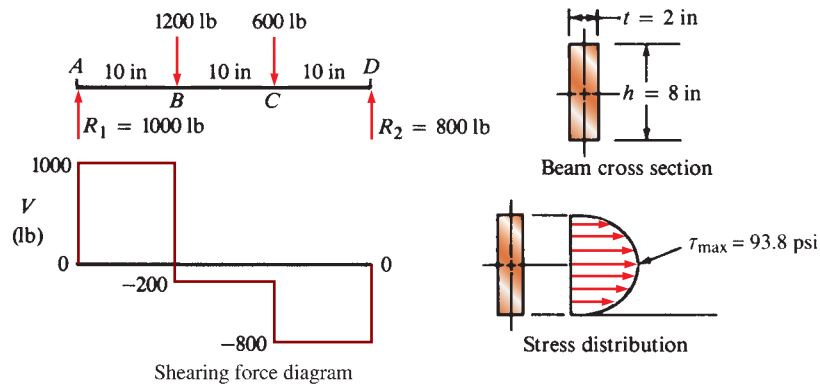


FIGURE 3-14 Shearing force diagram and vertical shearing stress for beam

Results $I = th^3/12 = (2.0 \text{ in})(8.0 \text{ in})^3/12 = 85.3 \text{ in}^4$
 $Q = A_p \bar{y} = th^2/8 = (2.0 \text{ in})(8.0 \text{ in})^2/8 = 16.0 \text{ in}^3$
 Then the maximum shearing stress is

$$\tau = \frac{VQ}{It} = \frac{(1000 \text{ lb})(16.0 \text{ in}^3)}{(85.3 \text{ in}^4)(2.0 \text{ in})} = 93.8 \text{ lb/in}^2 = 93.8 \text{ psi}$$

Comments The maximum shearing stress of 93.8 psi occurs at the neutral axis of the rectangular section as shown in Figure 3-14. The stress distribution within the cross section is generally parabolic, ending with zero shearing stress at the top and bottom surfaces. This is the nature of the shearing stress everywhere between the left support at A and the point of application of the 1200-lb load at B. The maximum shearing stress at any other point in the beam is proportional to the magnitude of the vertical shearing force at the point of interest.

for the element to be in equilibrium. Figure 3-15 shows this phenomenon.

In most beams, the magnitude of the vertical shearing stress is quite small compared with the bending stress (see the following section). For this reason, it is frequently not computed at all. Those cases where it is of importance include the following:

1. When the material of the beam has a relatively low shear strength (such as wood).
2. When the bending moment is zero or small (and thus the bending stress is small), for example, at the ends of simply supported beams and for short beams.
3. When the thickness of the section carrying the shearing force is small, as in sections made from rolled sheet, some extruded shapes, and the web of rolled structural shapes such as wide-flange beams.

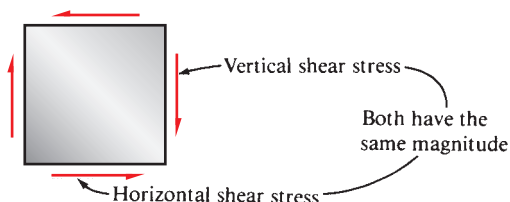


FIGURE 3-15 Shear stresses on an element

3-14 SHEAR STRESS DUE TO BENDING - SPECIAL SHEAR STRESS FORMULAS

Equation (3-16) can be cumbersome because of the need to evaluate the first moment of the area Q . Several commonly used cross sections have special, easy-to-use formulas for the maximum vertical shearing stress:

▣ $\tau_{\max}$ for Rectangle

$$\tau_{\max} = 3V/2A \text{ (exact)} \quad (3-18)$$

where A = total cross-sectional area of the beam

▣ $\tau_{\max}$ for Circle

$$\tau_{\max} = 4V/3A \text{ (exact)} \quad (3-19)$$

▣ $\tau_{\max}$ for I-Shape

$$\tau_{\max} \approx V/th \text{ (approximate: about 15% low)} \quad (3-20)$$

where t = web thickness

h = height of the web (e.g., a wide-flange beam)

▣ $\tau_{\max}$ for Thin-walled Tube

$$\tau_{\max} \approx 2V/A \text{ (approximate: a little high)} \quad (3-21)$$

In all of these cases, the maximum shearing stress occurs at the neutral axis.